Quantum Entanglement

Leonard Susskind

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Chapter 1

Quantum Entanglements, Part 1

These notes follow Leonard Susskind's opening lecture on quantum entanglement, with curation by LazyingArt LLC. The lecture begins, characteristically, with a sermon about intuition. It then turns that sermon into a program: we will not begin with wave mechanics, but with information, bits, configurations, update rules, and only then the elementary algebra that makes those update rules precise. The chapter should therefore be read in that order. We are being taught not merely a list of facts, but a way of rebuilding intuition.

1.1 Why Quantum Mechanics Feels Unnatural

Susskind begins by reminding us that our ordinary physical intuition is an evolutionary inheritance. It was formed in a world of rocks, arrows, prey, predators, and moderate velocities. Even an animal chasing another animal is already solving a problem in kinematics: it tracks relative velocity, direction, and the moment at which the sign of that relative velocity changes. So one should not underestimate how much classical mechanics is already wired into ordinary behavior.

But modern physics begins precisely where that inheritance stops being reliable. Relativity concerns velocities that no human or animal ever encountered in ordinary life. Quantum mechanics concerns electrons, uncertainty, and the breakdown of the very trajectory-based picture that ordinary experience encourages us to adopt. A particle, according to common sense, is a thing that has a position at every instant; if it has a position at every instant, then it has a trajectory; if it has a trajectory, then we compute its velocity from the differences of positions. That whole chain of intuition comes from the world of thrown stones and flying arrows. It is not a guide to the microscopic world.

So the problem is not that quantum mechanics is illogical. The problem is that it lies outside the regime that trained our brains. We therefore need abstraction before we can recover intuition. Susskind says, in effect, that physicists rewire themselves. They learn abstract mathematics first, and only afterward develop new habits of thought that make the subject feel less alien.

That sermon is not decorative. It explains the method of the lecture. This will not be a conventional first course in quantum mechanics, centered from the outset on the Schrödinger equation and the wave picture. The emphasis will be different. We are going to concentrate on the logic of quantum mechanics and the logic of quantum information. That is why the lecture pivots almost immediately to its first concrete object: the bit.

1.2 Classical Bits and the Counting of States

Physics, in this register, is information. To say something about a physical system is to provide information about it, often numerically, but very often in the form of a discrete alternative. Heads or tails, yes or no, up or down: these are all examples of the same logical structure.

Definition 1.1. A bit is a question about a system with exactly two possible answers.

Susskind insists on starting with the classical case. A classical bit may be represented by 0 and 1, although the choice of symbols is arbitrary. We could equally well use 1 and -1, or any other pair of labels. The point is only that there are two possibilities. Once that is fixed, we can write the two classical states as

$$|0\rangle, \qquad |1\rangle. \tag{1.1}$$

At this stage the Dirac notation looks like excess baggage, and Susskind says so. It adds nothing to the classical description *yet*. But it is installed here because it will be the natural notation later, when the bit becomes quantum.

For many bits we simply write a string inside the ket:

$$|001011\cdots\rangle. \tag{1.2}$$

This is a configuration of a multibit system. The notation is intentionally plain. We are not yet taking tensor products apart; we are just recording a string of binary values.

Now comes the first quantitative question: how many states does such a system have? One bit has two states. Two bits have four. Each new bit doubles the number of possibilities. Therefore an n-bit classical system has

$$N_s = 2^n \tag{1.3}$$

possible configurations, where N_s is the number of states. Inverting the relation gives

$$n = \log_2 N_s. \tag{1.4}$$

This is the point where the lecture turns the word *bit* from a label into a measure. The number of bits of information carried by a system is, by definition, the logarithm base two of the number of available states:

$$I = \log_2 N_s. \tag{1.5}$$

For systems built from binary alternatives this reproduces an integer. But the definition is broader. A die has six states, so its information content is $\log_2 6$, which is not an integer. The definition survives even when the system is not literally built from a set of binary switches.

This is a small but important beat in the lecture. Susskind first defines the bit operationally, then counts states, then inverts the relation. The movement is deliberate: we are being taught to think of a state space as something whose size already measures information.

1.3 How Ordinary Physics Becomes a Bit String

The next question is the natural one: are bits merely a convenient language for toy systems such as coins, or do they really reach ordinary physics? Susskind's answer is emphatic. Almost any physical

information can be represented by bits, at least approximately and often to arbitrarily improving approximation.

Begin with a single measured quantity, such as the temperature of a room. If we care only about finite accuracy, then we may truncate the value and write it in base two. Once that is done, the number is simply a finite string of zeros and ones. If we want more accuracy, we keep more places. If we want a finer notion of temperature, we redefine the unit and again represent the result in binary. The essential move is the same every time: finite accuracy turns a real-valued quantity into a long but finite bit string.

The same idea applies to fields. A field is something that varies from place to place. To encode it, we divide space into many small cells, order the cells, and record the field value in each cell with some chosen precision. The result is a long sequence of binary strings, one block per cell, and therefore a long sequence of bits. In that sense, a temperature field, an electric field, or a magnetic field may all be represented by binary data once we agree to discretize space and tolerate finite precision.

Susskind then gives a second example, even more directly in the language of yes-or-no questions. Impose a lattice on a room and suppose that no cell can contain more than one particle. Then each cell carries one bit:

$$\text{occupied} \longleftrightarrow 1, \qquad \text{empty} \longleftrightarrow 0. \tag{1.6}$$

A particle configuration is thus encoded by a string of occupancies. What does the string mean? It means exactly this: the first cell is occupied, the second is empty, the third is empty, the fourth is occupied, and so on. We have traded a spatial picture for a sequence of answers to yes-or-no questions.

This is one of the lecture's recurring motifs: *so far no quantum mechanics.* Nothing quantum has entered yet. We are still talking about classical information, classical bits, and classical approximation schemes. But the point is already substantial. If physical information can be encoded by bit strings, then digital computation is not an accident. That is why computers can simulate physics at all.

There is, however, a caveat, and the lecture states it explicitly. Exact classical real numbers may require infinitely many bits. Rational numbers may repeat, irrational numbers certainly do not terminate, and long-time classical tracking may demand more and more precision. But that does not disturb the practical or conceptual point being made here. It only sharpens it: a computer works because finite approximation is often enough, and because physical descriptions can be improved systematically by using more bits and finer discretization.

1.4 Configurations, Update Rules, and State Space

Up to this point the lecture has concentrated on how to *describe* a system at one time. The next pivot is from description to law. A physical theory is not only a list of possible configurations; it must also tell us how one configuration becomes another.

Definition 1.2. A configuration is the state of a system at a given instant of time.

Susskind is careful here. First comes configuration, not motion. We specify the state at one instant. Then, if we wish, we may discretize time as well as space and speak of updating from one digital

instant to the next. In that language a physical theory consists of two parts: a space of possible states and a rule that updates one state into another slightly later state.

To make this visible, the lecture introduces an abstract state space. Its points are not points in ordinary space; they are entire configurations. That distinction matters. A closed loop in this state space is not a particle running around a spatial circle. It is a system moving through a cycle of possible configurations.

For a one-bit system the state space has only two points. Then the simplest possible update laws can be written in bare form:

$$0 \mapsto 0, \qquad 1 \mapsto 1, \tag{1.7}$$

or

$$0 \mapsto 1, \qquad 1 \mapsto 0. \tag{1.8}$$

The first is the trivial fixed-point law: nothing happens. The second is the flip-flop law: the system alternates forever between the two possibilities.

The same idea extends immediately to two bits. A two-bit system has four states. One perfectly good law moves the system around a closed loop in the logical space of those four configurations. Another good law describes two separate flip-flops happening in parallel. The lecture is not trying to exhaust the possibilities. It is trying to install a picture: the laws of motion may be drawn as arrows between states.

That is why the section matters. We are being led from a string of bits to a geometry of possibilities, and from there to dynamics as a pattern of arrows. Once that picture is in place, one further question becomes unavoidable: are all arrow patterns equally good?

1.5 Reversibility, Information Loss, and Unitarity

Now the lecture reaches its conceptual center. Susskind first shows examples of acceptable update laws, and then he draws another law that is logically easy to imagine but physically defective. Two distinct states flow into a single later state. Going forward is still defined. Going backward is not.

The good diagrams all have the same structural feature. Each state has one definite next state, but also one definite previous state. Susskind phrases this as the *uniqueness of the future* and the *uniqueness of the past*. That is what reversibility means here.

1.5.1 Question & Answer

Question. Why is the merging update rule forbidden? It seems perfectly sensible as a forward rule. If the system is here, it goes there. Why is that not enough?

Answer. Because the rule destroys information about the past. If two distinct starting points both flow into the same later point, then once we arrive there we can no longer tell where we came from. The future is defined, but the past is not uniquely reconstructible. In Susskind's language, the rule loses information. At the fundamental level, where we imagine keeping track of every degree of freedom and not merely a coarse-grained summary, classical physics does not allow that.

This is the rhythm of the lecture: first a picture that looks reasonable, then the recognition that something has gone wrong, and then the sharpened principle that repairs it. The repair is not

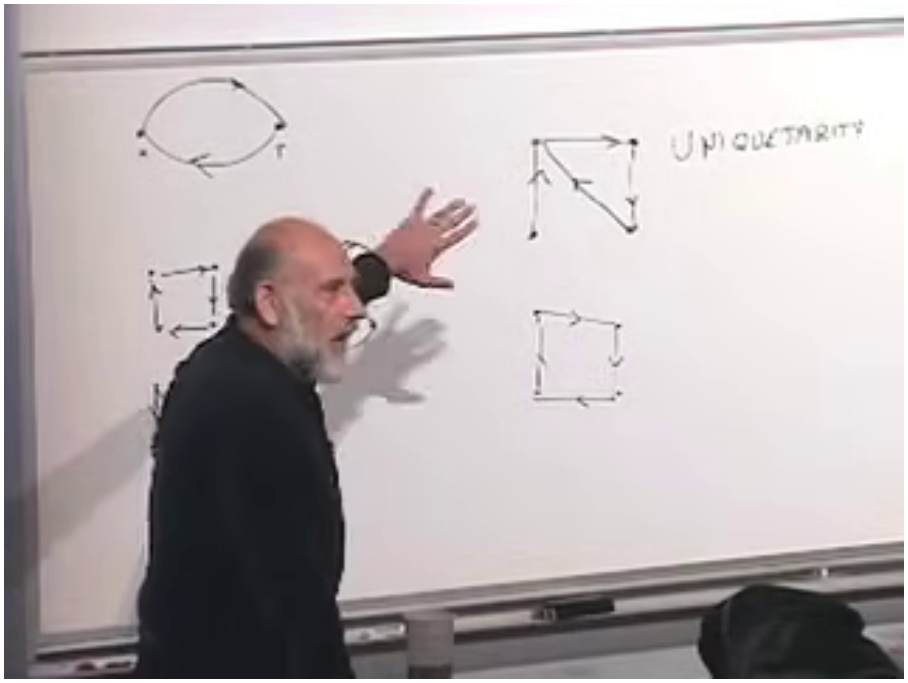


Figure 1.1: Board evidence from the discussion of reversible and irreversible update rules. The handwritten word *unitarity* appears beside the emphasized four-state sketch.

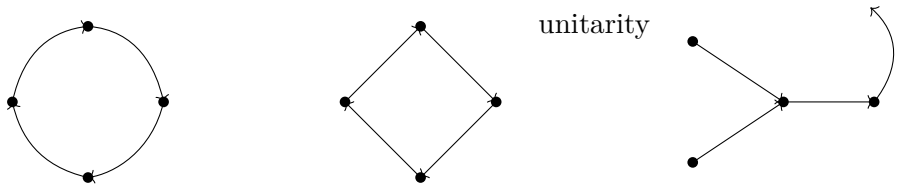


Figure 1.2: A clean schematic reconstruction of the board logic: reversible loops on the left, and a contrasting information-losing update rule on the right. The redraw stays schematic because the board does not securely label the states.

a matter of logic alone. Susskind explicitly says that such information-losing rules are logically conceivable; they are simply not the form taken by fundamental physical laws.

A second clarification follows immediately. Reversibility is not quite the same thing as exact time-reversal symmetry. A cyclic law still has an orientation: under forward evolution it runs one way around the loop, not both ways at once. What reversibility means is that there exists a reverse law taking us back uniquely. So the lecture draws a distinction between *time reversibility* and the stronger notion of exact symmetry under interchange of past and future.

A third clarification comes from thermodynamics. It often *looks* as if information is lost. Release the gas from one corner of a room and after a while the whole room looks the same no matter which corner we started from. But, Susskind argues, this is usually not fundamental loss. It is coarse-graining. We have stopped keeping track of the microscopic details. If we followed every molecule with enough precision, then in principle the information would still be there. What is lost in practice is our power to resolve it.

This is also the place where the lecture gestures beyond classical physics. Susskind jokes about

naming the property "unique-tarity," and then gives the quantum name: *unitarity*. The quantum theory will have its own formal statement of this idea, but the principle has already appeared in classical dress: past and future must both be unique.

1.6 Vectors, Rows, Columns, and Inner Products

After the break Susskind says, in effect, that he was about to jump into quantum mechanics, but he wants to stop first and lay out some elementary mathematics. The timing matters. The algebra is not an isolated appendix; it is being installed because the classical discussion has made it necessary.

A vector is introduced in the least ceremonial possible way. Susskind does not begin with axioms of a vector space. He says plainly: a vector is a finite sequence of numbers. At this point the entries are real numbers. Complex conjugation, daggers, and the full quantum notation are postponed.

A four-component vector may be written as a row,

$$A = (A_1, A_2, A_3, A_4), \tag{1.9}$$

or as a column,

$$A = \begin{pmatrix} A_1 \\ A_2 \\ A_3 \\ A_4 \end{pmatrix}. \tag{1.10}$$

The two forms contain the same information. They are simply different notations adapted to different operations. Susskind even warns us not to visualize these as arrows in ordinary space. For the present purpose they are just organized lists of numbers.

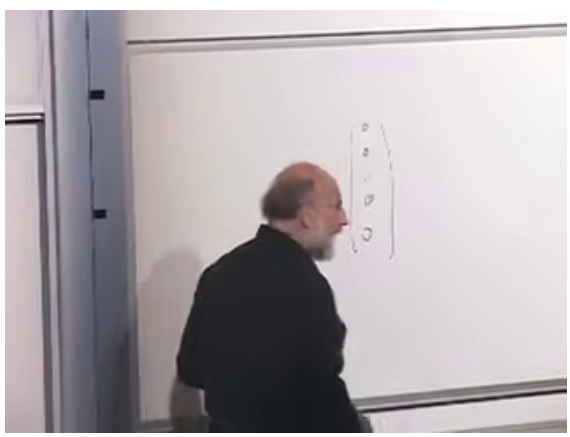


Figure 1.3: The board at the moment the lecture isolates a column vector. The entries are too small to transcribe literally, so the clean reconstruction uses generic components $A_1, \ldots, A_4$.

Now take a second vector, written as a row:

$$B = (B_1, B_2, B_3, B_4). \tag{1.11}$$

If the row and the column have the same number of entries, we may multiply them. The rule is the familiar one:

$$BA = B_1A_1 + B_2A_2 + B_3A_3 + B_4A_4. \tag{1.12}$$

This is the inner product. It is not another vector. It is a single number. In ordinary three-dimensional language it would be called the dot product, and Susskind says so, but he prefers the more general name because the objects are being treated abstractly.

The pedagogical point is simple and operational. Rows multiply columns to produce scalars. Once that is secure, matrices can be introduced with almost no extra conceptual burden.

1.7 Matrices as Update Operators

A matrix is now presented as a machine that acts on a vector and produces a new vector. That description is intentionally concrete. We are not yet being handed the formal theory of linear operators. We are being given the piece of algebra that will encode update laws.

A 4×4 matrix is written as

$$M = \begin{pmatrix} M_{11} & M_{12} & M_{13} & M_{14} \\ M_{21} & M_{22} & M_{23} & M_{24} \\ M_{31} & M_{32} & M_{33} & M_{34} \\ M_{41} & M_{42} & M_{43} & M_{44} \end{pmatrix}. \tag{1.13}$$

Its rows and columns are themselves collections of numbers, but together they define one operation. Acting on the column vector A, the matrix produces a new column vector whose entries are row-by-column inner products:

$$(MA)_i = \sum_j M_{ij} A_j. \tag{1.14}$$

Written out componentwise, the first two entries are

$$(MA)_1 = M_{11}A_1 + M_{12}A_2 + M_{13}A_3 + M_{14}A_4, \tag{1.15}$$

$$(MA)_2 = M_{21}A_1 + M_{22}A_2 + M_{23}A_3 + M_{24}A_4. \tag{1.16}$$

The later entries are obtained in exactly the same way. Susskind does two and leaves the pattern to the audience.

A worked classical example. Now the lecture turns back to the classical discussion and shows why matrices matter. Suppose the system has finitely many configurations arranged in a cycle. Susskind first tries to draw a five-state example, then abandons it and retreats to four states so that the arithmetic will fit. That change does not alter the mathematical point.

We encode the four classical states by one-hot column vectors:

$$e_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \qquad e_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \qquad e_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \qquad e_4 = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}. \tag{1.17}$$

A cycle $1 \to 2 \to 3 \to 4 \to 1$ may then be represented by the permutation matrix

$$P = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix}. \tag{1.18}$$

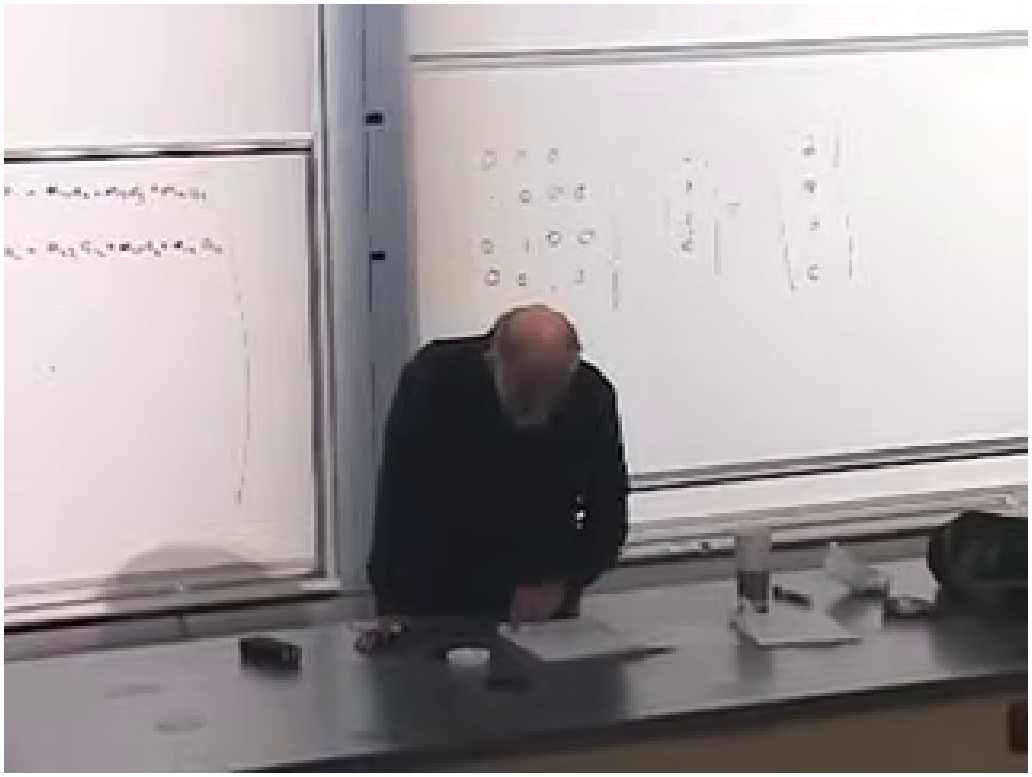


Figure 1.4: Board layout for the discussion of a matrix acting on a column vector. The frame preserves the spatial organization of the explanation more reliably than the individual micro-entries.

This is a deliberate clean convention. On the board the direction is corrected live, so the notes should not pretend that the first version written there was final. What is stable is the cyclic structure. With the present convention,

$$Pe_1 = e_2, \qquad Pe_2 = e_3, \qquad Pe_3 = e_4, \qquad Pe_4 = e_1. \tag{1.19}$$

The same matrix acts on a general column by shifting its entries cyclically:

$$P \begin{pmatrix} A \\ B \\ C \\ D \end{pmatrix} = \begin{pmatrix} D \\ A \\ B \\ C \end{pmatrix}. \tag{1.20}$$

This is the lecture's first real worked example of an update rule written algebraically. A law of motion has become a matrix.

Matrix multiplication as repeated updating. Once a matrix represents one step of evolution, the reason for multiplying matrices by matrices becomes immediate. If

$$V' = MV, \tag{1.21}$$

then a second step gives

$$V'' = MV' = M^2 V. \tag{1.22}$$

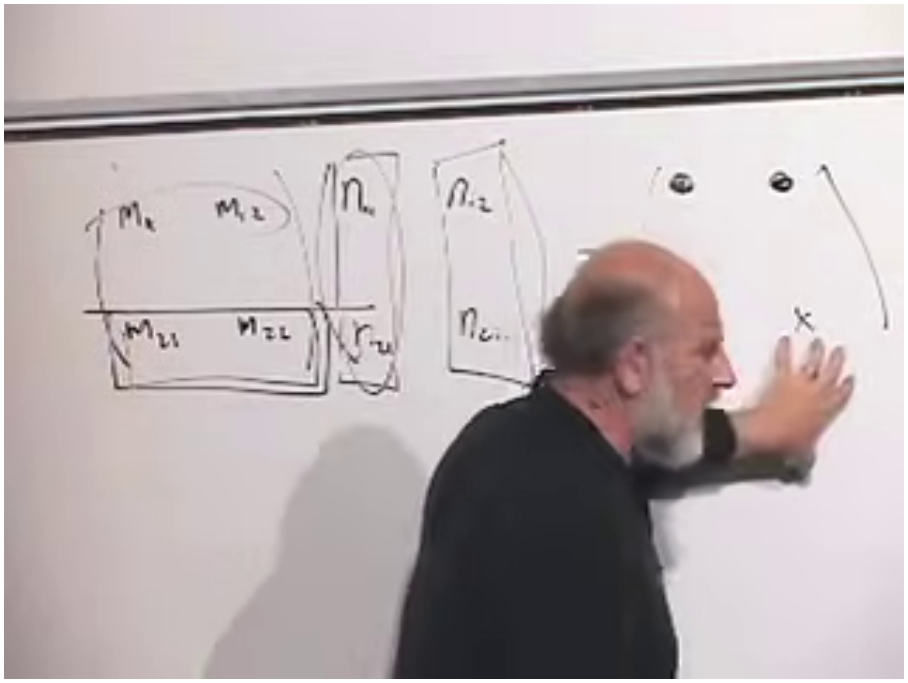


Figure 1.5: Board layout for the 2×2 matrix product. The pedagogical point is that the right-hand matrix may be read column by column, each column being acted on by the left-hand matrix.

So matrix multiplication composes updates. To evolve by two time steps, we apply the update matrix twice. To evolve by five, we multiply it into itself five times.

To make the arithmetic explicit, Susskind reduces the discussion to 2×2 matrices:

$$M = \begin{pmatrix} M_{11} & M_{12} \\ M_{21} & M_{22} \end{pmatrix}, \qquad N = \begin{pmatrix} N_{11} & N_{12} \\ N_{21} & N_{22} \end{pmatrix}. \tag{1.23}$$

Their product $C = MN$ is another matrix. The first entry is obtained by taking the first row of M and multiplying it by the first column of N:

$$C_{11} = M_{11}N_{11} + M_{12}N_{21}. \tag{1.24}$$

The next entry comes from the first row of M and the second column of N:

$$C_{12} = M_{11}N_{12} + M_{12}N_{22}. \tag{1.25}$$

The general rule is

$$C_{ij} = \sum_k M_{ik}N_{kj}. \tag{1.26}$$

The board layout in the lecture is especially revealing here. The right-hand matrix is drawn almost as two neighboring columns, because that is how Susskind wants us to think of it. Multiply the left matrix by the first column and put the result in the first column of the answer; multiply the left matrix by the second column and put the result in the second column of the answer. Matrix multiplication is thus visibly the same operation as matrix-times-vector multiplication, performed column by column.

Row vectors multiplied by matrices. The final pattern is the one obtained by multiplying a row vector by a matrix. If

$$B = (B_1, B_2, B_3, B_4), \tag{1.27}$$

then BM is another row vector, and its j-th entry is found by multiplying the original row B by the j-th column of M:

$$(BM)_j = \sum_i B_i M_{ij}. \tag{1.28}$$

This closes the elementary algebra of the lecture. A row times a column is a number. A matrix times a column is a column. A row times a matrix is a row. A matrix times a matrix is a matrix. Susskind ends by stressing that if we can do these manipulations fluently, then we already possess the basic mathematics that quantum mechanics will use over and over again.

Only at the end, in the questions, does the lecture phrase reversibility in matrix language. A reversible update should be represented by a matrix with an inverse. That is the algebraic shadow of the earlier physical statement that there must be a unique future and a unique past.

1.8 Summary

The first lecture does not yet present the new quantum rules. Instead it prepares the ground with unusual care. First we are told why abstraction is necessary: ordinary intuition is a classical inheritance, not a trustworthy guide to the microscopic world. Then the lecture reframes physics as information. From there it constructs classical bits, counts their states, and shows how surprisingly much of ordinary physics may be represented by long strings of binary data.

Only after that groundwork does the lecture turn from configuration to law. A physical system consists of possible states together with an update rule. Not every conceivable update rule is acceptable: fundamental laws must preserve the distinction between different pasts, which is why reversibility and uniqueness of past and future become central. The quantum name for the corresponding principle has already been previewed: unitarity.

Finally, after the break, the lecture installs the algebraic machinery required for the next step. Vectors, inner products, matrices, and matrix multiplication are not side remarks. They are the clean language of updating. So the lecture ends exactly where it should end: classical bits have been lifted into linear algebra, and the stage is set for qubits.

Chapter 2

Quantum Entanglements, Part 1

These notes follow Leonard Susskind's second lecture in the supplementary *Quantum Entanglement* track, in an edition curated by LazyingArt LLC. The lecture opens by leaving the world of classical bits and entering quantum mechanics at its simplest nontrivial point: a single electron spin. But it does not begin with formalism. It begins with a puzzle. The electron can be prepared as though it were pointing in many different directions, yet an up-down measurement returns only two answers. The chapter should be read in that order. We first let the classical picture build up enough force to fail, and only then do we introduce the complex vector space of states that can carry the physics.

2.1 Classical Bits, Qubits, and the Spin Example

We begin where the lecture begins: classical physics with a discrete state space. In that setting the space of states is a Boolean space, a collection of points, one point for each possible state, and the logic governing those states is ordinary classical logic. A classical bit is the simplest example. It has two possible values and nothing in between.

If we draw a two-state object with one end distinguished from the other, then "up" is one state and "down" is another. That is a classical bit in its barest form: two possibilities, no intermediate state. One can build many-bit systems out of such objects, and in classical physics that point of view goes a surprisingly long way.

Quantum mechanics forces a different notion of state, and Susskind chooses the simplest example immediately: the spin of an electron. He also warns us, at the outset, that the word *vector* is about to become dangerous.

On the one hand, there is the familiar vector in ordinary space: an arrow with magnitude and direction. To keep the distinction visible, we will call that a *spatial vector*. The electron carries such a spatial vector in the form of its magnetic moment. The magnetic moment has a fixed magnitude for every electron, but its direction may vary.

On the other hand, there is the abstract vector that belongs to a vector space of states. That vector is not an arrow in space. It is a mathematical object, and before long it will be this abstract vector, not the spatial arrow, that carries the state of the qubit.

That is already the lecture's first tension. The electron seems to come equipped with something that can point in any direction in space, but the measurement story we are heading toward will not

return a continuum of answers. We are being steered toward a conflict between classical imagery and quantum logic.

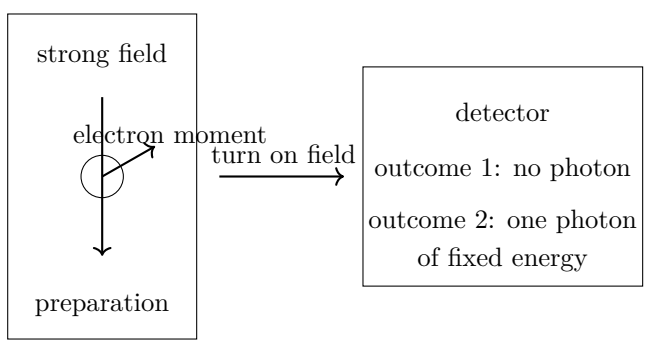
2.2 The Classical Preparation-and-Detection Story, and Why It Fails

Susskind next does something very deliberate: he tells a classical story in full. He even says that for the moment we are still thinking very classically. The point is not to endorse that story. The point is to let it fail in a way that makes the quantum alternative unavoidable.

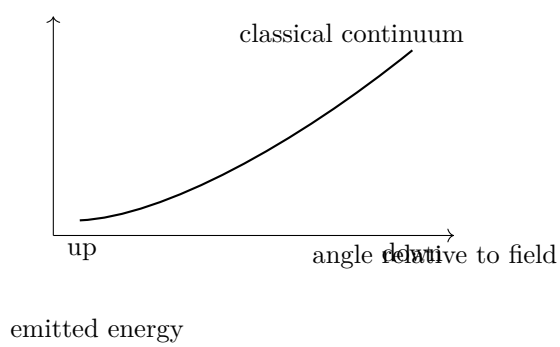
Imagine the electron as a tiny bar magnet with a north pole and a south pole. Suppose we place it in a strong external magnetic field. A classical magnet does not, in general, jump instantly into alignment. In vacuum, with no friction, it precesses around the external field. Then it radiates away its excess energy and settles into the energetically preferred orientation. That is the lecture's classical picture of *preparation*: choose the direction of the external field, wait long enough, and the electron's magnetic moment comes into equilibrium with that field.

The same setup then suggests a classical model of *detection*. Put the electron into a known field and watch how much radiation it emits as it relaxes. If the electron begins almost aligned with the field, only a little energy is emitted. If it begins farther away, more energy is emitted. If it begins anti-aligned, the emitted energy is maximal. So, classically, the emitted energy should be a continuous function of the initial angle.

The lecture's experimental staging is captured well enough by a schematic reconstruction:



In this classical picture we do *not* need an explicit formula for emitted energy as a function of angle, and the lecture never gives one. What matters is only the qualitative statement that the response is continuous.



This, however, is exactly the point at which the lecture reverses itself. Susskind says, in effect: that was the wrong story. For a real electron the detector does *not* produce a continuous output.

Once the magnetic field is turned on, only two things can happen:

- no photon is emitted;
- one photon of a definite energy is emitted, namely the energy corresponding to the transition from down to up.

Nothing in between ever appears. The detector does not report a small amount of up or a medium amount of down. It returns one of two answers.

What *does* depend on the preparation is the probability distribution over those two answers. If the electron has been prepared vertically upward and we measure it in the same up-down setup, then no photon is emitted. If it has been prepared vertically downward and we reverse the field appropriately, the photon is emitted with probability one. If it has been prepared horizontally and then measured in the vertical device, the two outcomes occur with equal probabilities:

$$P(\text{one photon}) = \frac{1}{2}, \qquad P(\text{no photon}) = \frac{1}{2}. \tag{2.1}$$

The lecture also pauses, in audience questions, over the operational meaning of this probability. A single run does not display a distribution. We must repeat the whole prepare-and-measure procedure many times. And once a given measurement has found the electron in the up state, repeating the same measurement does not recover the earlier preparation. The system has been fixed relative to that detector.

2.2.1 Question & Answer

Question. If the electron can be prepared in arbitrary directions, why does the up-down detector return only up or down?

Answer. Because the preparation direction does not create new outcomes for a fixed detector. A vertical detector always asks a binary question. The preparation changes only the probabilities of the two available answers. The mystery, then, is not that new detector values appear. The mystery is that a continuously variable preparation changes only a two-valued probability distribution.

That is the lecture's main puzzle. In one sense the electron seems to have only two measurable states. In another sense it can still be prepared in many directions. A set of two points will not be enough to describe such an object.

2.3 From the Puzzle to a Complex Vector Space of States

Now comes the lecture's central pivot. Susskind says, in substance, that we need a different mathematics of states. Not a Boolean set of two points, but a vector space.

He also tells us what the bargain is. We are going to do some abstract mathematics first and interpret it afterward. The first piece of that mathematics is the algebra of complex numbers.

We introduce the imaginary unit i by

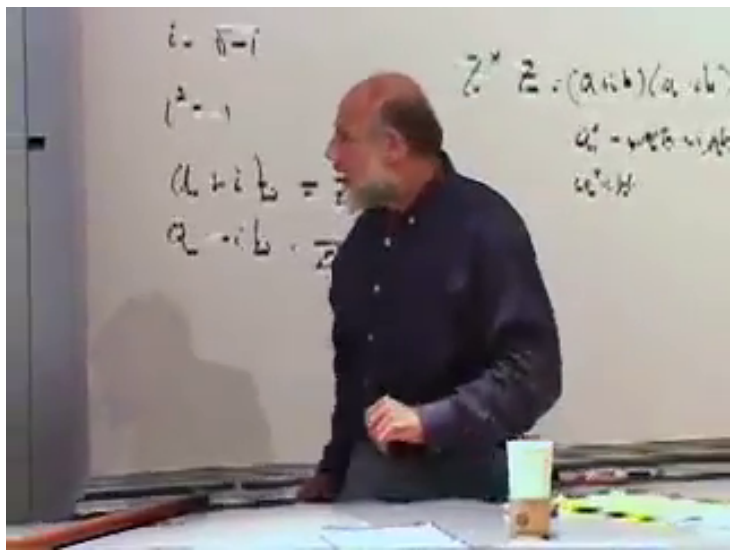
$$i^2 = -1. \tag{2.2}$$

A complex number is written

$$z = a + ib, \qquad z^* = a - ib, \tag{2.3}$$

where a and b are real and z^* denotes the complex conjugate.

The lecture's blackboard cluster for this interlude is worth keeping, even though the frame is only partial evidence for the exact line-by-line algebra.



A key example is the product of a complex number with its conjugate:

$$\begin{aligned} z^* z &= (a + ib)(a - ib) && (2.4)\\ &= a^2 - iab + iab + b^2 && (2.5)\\ &= a^2 + b^2. && (2.6) \end{aligned}$$

The cross terms cancel, and the result is real and nonnegative. That is not yet quantum mechanics, but it is the algebraic pattern quantum mechanics will use when amplitudes become probabilities.

Definition 2.1. A complex vector space, in the minimal sense used in this lecture, is a collection of objects that can be added and multiplied by complex numbers, with the result remaining in the same collection.

The lecture then gives the simplest concrete realization of this idea: column vectors with complex entries. If

$$A \leftrightarrow \begin{pmatrix} A_1 \\ A_2 \\ A_3 \end{pmatrix},$$

then multiplication by a complex number c is defined componentwise:

$$c \begin{pmatrix} A_1 \\ A_2 \\ A_3 \end{pmatrix} = \begin{pmatrix} cA_1 \\ cA_2 \\ cA_3 \end{pmatrix}, \tag{2.7}$$

and addition is defined by

$$\begin{pmatrix} A_1 \\ A_2 \\ A_3 \end{pmatrix} + \begin{pmatrix} B_1 \\ B_2 \\ B_3 \end{pmatrix} = \begin{pmatrix} A_1 + B_1 \\ A_2 + B_2 \\ A_3 + B_3 \end{pmatrix}. \tag{2.8}$$

There is no need yet to say more. The point is to establish the kind of space in which quantum states will live. For the spin of a single electron, the relevant case will be a two-dimensional complex vector space.

2.4 Column Vectors, Bras, Kets, and the Inner Product

The lecture now tightens the notation. Abstract vectors are represented by columns, and their conjugates by rows.

If

$$|a\rangle \leftrightarrow \begin{pmatrix} a_1 \\ a_2 \end{pmatrix},$$

then its conjugate is represented by

$$\langle a| \leftrightarrow \begin{pmatrix} a_1^* & a_2^* \end{pmatrix}.$$

The star is implicit in turning the ket into the bra; that is exactly the point Susskind pauses over in the lecture.

The notation $\langle b|a\rangle$ is a *bracket* split into its two halves, bra and ket. In components the rule is

$$\langle b|a\rangle = b_1^* a_1 + b_2^* a_2. \tag{2.9}$$

This is the inner product.

If we take the inner product of a vector with itself, we find

$$\langle a|a\rangle = a_1^* a_1 + a_2^* a_2. \tag{2.10}$$

Each term has the form $z^* z$, and therefore each term is real and nonnegative. So $\langle a|a\rangle$ itself is real and nonnegative.

Remark 2.2. The lecture interprets $\langle a|a\rangle$ as the square of the size of the vector. We are not yet using this abstractly; we are preparing it for a physical interpretation.

This is the point at which the mathematics begins to justify itself. Complex numbers were introduced because their conjugation law produces positive real quantities. Those are exactly the quantities that can later be interpreted as probabilities.

2.5 Up and Down as an Orthogonal Basis of States

At this point Susskind explicitly returns from the abstract vector space to the quantum bit. The detector has two possible outputs. Let us denote the corresponding states by

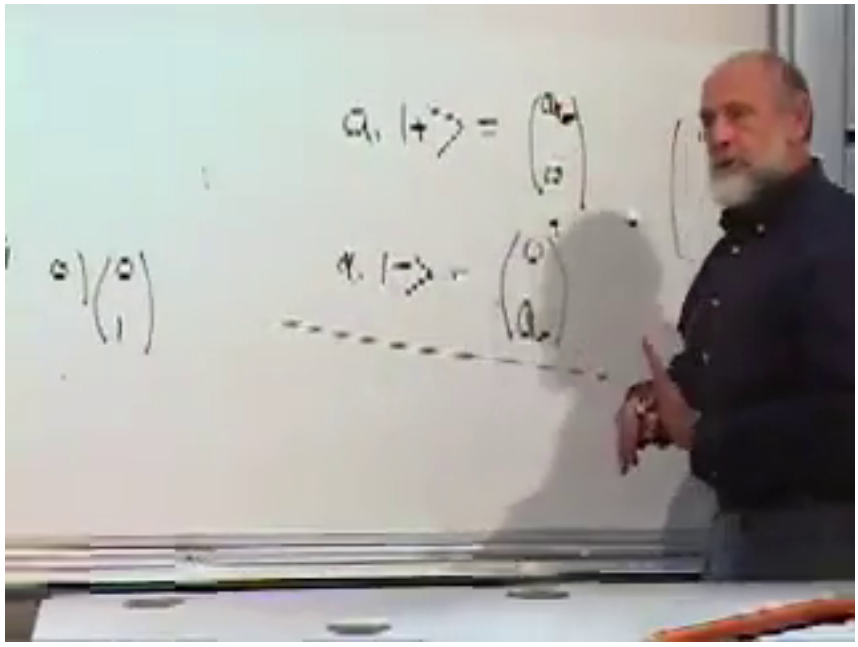
$$|+\rangle \qquad \text{and} \qquad |-\rangle.$$

These stand for the electron being found up or down in the chosen measurement basis.

Now the lecture makes the space concrete. A general state of the electron spin is written as a linear combination

$$|\psi\rangle = a_+|+\rangle + a_-|-\rangle. \tag{2.11}$$

The corresponding blackboard layout is an important piece of visual evidence because it shows exactly how the coefficient-weighted basis states are assembled into a single column vector.



Once the basis has been fixed, the canonical column-vector representatives are

$$|+\rangle \leftrightarrow \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \qquad |-\rangle \leftrightarrow \begin{pmatrix} 0 \\ 1 \end{pmatrix}. \tag{2.12}$$

Then

$$a_+|+\rangle \leftrightarrow \begin{pmatrix} a_+ \\ 0 \end{pmatrix}, \qquad a_-|-\rangle \leftrightarrow \begin{pmatrix} 0 \\ a_- \end{pmatrix},$$

and adding these two contributions gives

$$|\psi\rangle \leftrightarrow \begin{pmatrix} a_+ \\ a_- \end{pmatrix}. \tag{2.13}$$

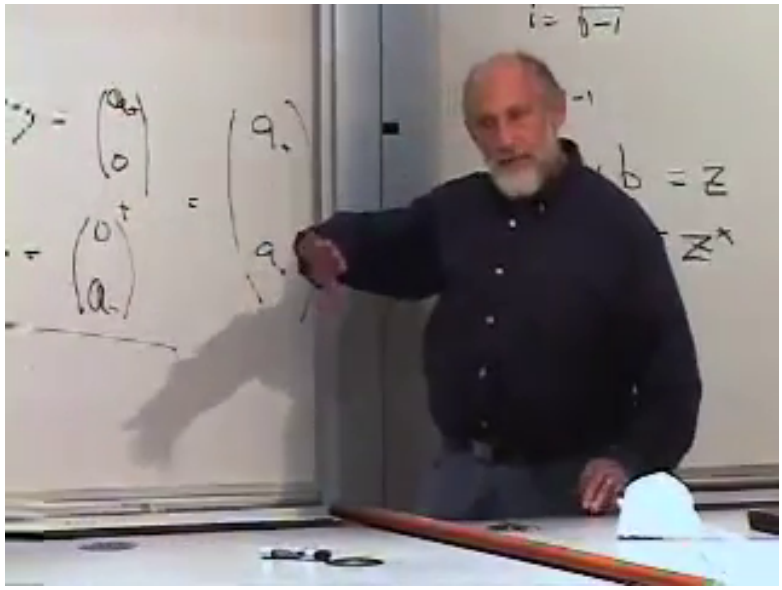
The lecture later revisits the same identification more cleanly on the board, and that cleaner basis-fixing picture should remain visible as well.

Now comes the physical interpretation. The squared magnitudes of the coefficients are the probabilities for the two detector outcomes:

$$P(+) = a_+^* a_+ = |a_+|^2, \qquad P(-) = a_-^* a_- = |a_-|^2. \tag{2.14}$$

Since these are the only two possibilities, they must add to one:

$$|a_+|^2 + |a_-|^2 = 1. \tag{2.15}$$



Equivalently,

$$\langle\psi|\psi\rangle = 1. \tag{2.16}$$

This is the normalization condition. The physical states are the normalized vectors in the vector space.

The two basis states are orthonormal:

$$\langle +|+\rangle = 1, \qquad \langle -|-\rangle = 1, \tag{2.17}$$

$$\langle +|-\rangle = 0, \qquad \langle -|+\rangle = 0. \tag{2.18}$$

The lecture explicitly checks these relations in components. For example,

$$\langle +|-\rangle = \begin{pmatrix} 1 & 0 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = 0.$$

So the up and down states correspond to orthogonal vectors in the state space.

Worked example. The lecture does not move too quickly past its first nontrivial example. It writes, in effect,

$$\begin{pmatrix} 1 \\ 1 \end{pmatrix},$$

then immediately notices that this cannot yet be a physical state because its norm-squared is 2. To repair it, we divide by $\sqrt{2}$:

$$\frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}. \tag{2.19}$$

Now the probabilities are

$$P(+) = \frac{1}{2}, \qquad P(-) = \frac{1}{2}.$$

This is the lecture's equal-weight state. It is not pure up and not pure down. It is one of the states associated with a horizontal preparation.

We now see how the lecture wants us to think. A detector aligned along a chosen axis still returns only two answers. But the state space is larger than those two detector values, because it contains all normalized linear combinations of the corresponding basis vectors.

2.6 What the State Vector Does and Does Not Encode

After the break Susskind performs a reset. More than one person is probably confused, he says, about the two notions of vector. That is exactly the right place to slow down and separate them.

A spatial vector in ordinary space has three real components. A two-component state vector,

$$\begin{pmatrix} a_+ \\ a_- \end{pmatrix},$$

has two complex components, which means four real numbers before any constraints are imposed. So the state vector is not just a disguised spatial arrow.

Of course, not all four of those real numbers are physically independent. Normalization removes one. Overall phase removes another. But the lecture does not yet build the full geometry that connects state vectors to spatial directions, and we should not force it. The right statement, and the lecture's own statement, is simpler: the two objects are related, but not simply related.

There is another important clarification here. The dimension of the state space is fixed experimentally. The up-down measurement has two possible outcomes, and that is why we use a two-dimensional complex vector space. In that sense the mathematics is not arbitrary. The experiment tells us how many basis states we need.

2.6.1 Question & Answer

Question. Is $-|+\rangle$ the same thing as $|-\rangle$?

Answer. No. The symbol $-$ in $|-\rangle$ is part of a label. It does not mean minus one times the plus state. By contrast, $-|+\rangle$ is an ordinary scalar multiple of $|+\rangle$, and it is perfectly allowed in the vector space.

This answer leads immediately to the lecture's next point. The states $|+\rangle$ and $-|+\rangle$ are not equal as vectors, but they represent the same physics, because an overall phase does not change the probabilities of any measurement. More generally, if $|\psi\rangle$ is multiplied by a complex number of magnitude one, the physical state is unchanged.

But a relative sign inside a superposition is a different matter. The states

$$a_+|+\rangle + a_-|-\rangle \qquad \text{and} \qquad a_+|+\rangle - a_-|-\rangle$$

need not represent the same physics. The lecture does not yet unpack the full meaning of relative phase, but it clearly insists on this distinction: overall phase is inessential, relative phase can matter.

2.7 From Classical Observables to Quantum Matrices

The lecture now shifts its center of gravity. Up to now we have been asking what a state is. Next we ask what a measurable quantity is.

In classical mechanics, if the possible states are discrete points labeled by n, then an observable is simply a function that assigns a real number to each state:

$$f_n.$$

If the system is not known exactly, but only through a probability distribution p_n, then the average value of the observable over many repeated experiments is

$$\langle f \rangle = \sum_n p_n f_n. \tag{2.20}$$

The lecture lingers here because it wants matrices to emerge by motivation rather than fiat. A coin provides the simplest example. Suppose heads is assigned the value $+1$ and tails the value -1. If the probabilities are p_+ and p_-, then

$$\langle f \rangle = p_+ - p_-. \tag{2.21}$$

This is the average value of the observable, not necessarily an attainable single-shot outcome.

2.7.1 Question & Answer

Question. How can the average value be zero when zero is not one of the actual outcomes?

Answer. Because the average value is not the same thing as a detector reading from a single run. If a fair coin yields $+1$ half the time and -1 half the time, then the average over many trials is zero, even though no single toss ever produces zero. The average is a summary of the distribution, not a new observable value that occurs in one trial.

The lecture then singles out a special classical observable: the indicator function that is 1 on one state and 0 on all the others. In that case the average is exactly the probability of that state. So if we know how to compute average values of all observables, we also know how to recover probabilities.

This now raises the quantum question: what mathematical objects represent observables in quantum mechanics? The answer, for the present finite-dimensional setting, is matrices.

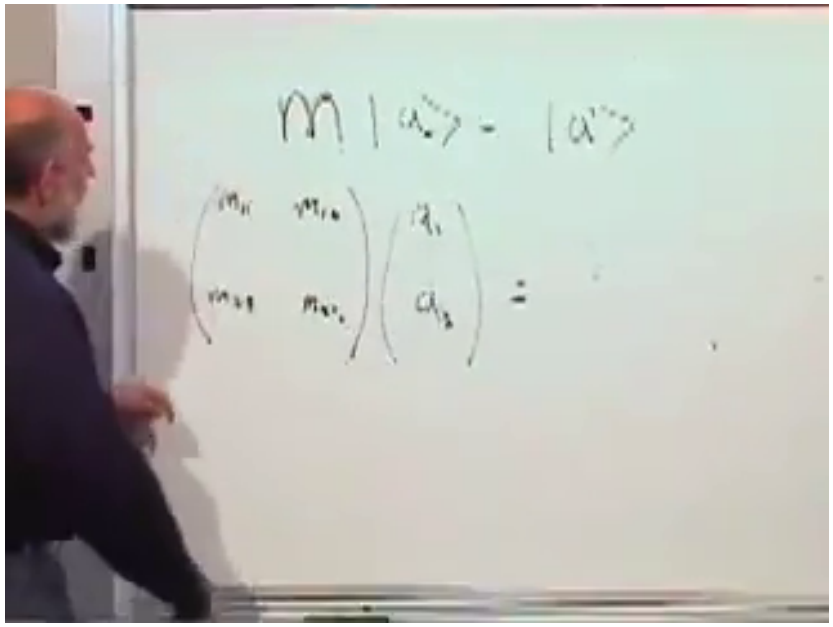
A matrix acts on a state vector and produces another vector:

$$M|a\rangle = |a'\rangle. \tag{2.22}$$

The lecture's board picture should remain in place here because it preserves the two-level transition from abstract operator language to explicit component calculation.

If

$$M = \begin{pmatrix} m_{11} & m_{12} \\ m_{21} & m_{22} \end{pmatrix}, \qquad |a\rangle = \begin{pmatrix} a_1 \\ a_2 \end{pmatrix},$$



then the component rule is

$$\begin{pmatrix} m_{11} & m_{12} \\ m_{21} & m_{22} \end{pmatrix} \begin{pmatrix} a_1 \\ a_2 \end{pmatrix} = \begin{pmatrix} m_{11}a_1 + m_{12}a_2 \\ m_{21}a_1 + m_{22}a_2 \end{pmatrix}. \tag{2.23}$$

Susskind first motivates this by going back to a very ordinary real vector space, the plane of arrows drawn on the board. In that setting matrices represent linear transformations:

- $$\begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}$$
  doubles every vector;
- $$\begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}$$
  stretches only the y-component;
- $$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$
  reflects across the diagonal $x = y$;
- $$\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$$
  rotates the plane by $90°$, in the clockwise convention used in the discussion.

The lesson is simple: matrices are the language of linear transformations of vector spaces. That is why they are the right candidates for quantum observables.

Now comes the special class singled out by the lecture.

Definition 2.3. A matrix M is Hermitian if

$$M_{ij} = M^*_{ji}. \tag{2.24}$$

This is a kind of reality condition, though it does *not* say that every entry is real. It says, first, that on the diagonal

$$M_{ii} = M^*_{ii},$$

so diagonal entries are real. It says, second, that off-diagonal entries come in complex-conjugate pairs. In the special case that all entries are real, Hermitian reduces to symmetric.

The lecture ends this part of the story with a clear declaration. In classical mechanics observables are functions on state points. In quantum mechanics observables will be represented by Hermitian operators, which at the moment means Hermitian matrices.

There is one further preview. If we want to *kick* the electron, that is not generally done by a Hermitian operator. It is done by a unitary operator. At this stage only one fact is stated: unitary operators preserve the length of vectors. Their role in dynamics and physical operations belongs to what comes next.

2.8 Summary

The lecture has carried us from a concrete spin-measurement puzzle to the threshold of the quantum formalism. We began with a classical bit and a classical magnetic-arrow picture of the electron. We let that picture generate a continuous detector response, and then watched the lecture reject it. A real electron, measured in a fixed up-down setup, yields only two outcomes. What varies with preparation is not the list of outcomes but the probability distribution over them.

That failure of the classical story forced a new mathematics of states. We introduced complex numbers, conjugation, column vectors, bras, kets, and the inner product. We chose the orthonormal basis $|+\rangle$ and $|-\rangle$, wrote a general qubit state as

$$|\psi\rangle = a_+|+\rangle + a_-|-\rangle,$$

and interpreted $|a_+|^2$ and $|a_-|^2$ as probabilities, subject to normalization.

After that, the lecture shifted from states to observables. Classical observables are functions f_n on discrete states, with averages

$$\langle f\rangle = \sum_n p_n f_n.$$

Quantum observables are more intricate: they are represented by matrices acting on state vectors, with Hermitian matrices singled out as the measurable ones. That is the mathematical entrance fee the lecture insists upon before entanglement can be discussed honestly. Once states, inner products, and Hermitian operators are in hand, the rest of the structure of quantum mechanics can be built without sleight of hand.

Chapter 3

Lecture 3: Quantum Entanglements, Part 1

We do not begin with entanglement itself. We begin where the lecture begins: with the small mathematical facts without which the subject becomes empty language. Susskind wants everyone on the same page, and so we first clear the runway with complex numbers, normalized state vectors, Hermitian observables, eigenvalues and eigenvectors, and the probability rule for spin. The guiding idea throughout is that quantum mechanics is a calculus for probabilities. These companion notes follow Leonard Susskind's lecture closely, with curation by LazyingArt LLC, and keep the two-level spin system in view at every stage.

3.1 Complex Numbers, Phases, and Conjugation

The opening review is brisk on purpose. We are not doing complex analysis for its own sake; we are fixing the notation that quantum mechanics will immediately require.

A complex number may be written in Cartesian form,

$$z = x + iy, \tag{3.1}$$

or in polar form,

$$x = r\cos\theta, \qquad y = r\sin\theta, \tag{3.2}$$

so that

$$z = r(\cos\theta + i\sin\theta) = re^{i\theta}. \tag{3.3}$$

The point of the notation $e^{i\theta}$ is that it behaves like an exponential:

$$e^{i\theta}e^{i\phi} = e^{i(\theta+\phi)}. \tag{3.4}$$

This is nothing but the angle-addition formulas in compressed form.

Complex conjugation reverses the sign of the imaginary part:

$$z^* = x - iy. \tag{3.5}$$

Applied to a pure phase,

$$(e^{i\theta})^* = e^{-i\theta}, \tag{3.6}$$

and therefore

$$e^{i\theta}e^{-i\theta} = 1. \tag{3.7}$$

More generally,

$$zz^* = x^2 + y^2. \tag{3.8}$$

So the numbers of the form $e^{i\theta}$ lie on the unit circle:

$$|e^{i\theta}|^2 = 1. \tag{3.9}$$

The lecture uses the informal phrase "unitary numbers" in passing, but the mathematically stable term is *pure phase*. Every complex number is a positive real modulus times a pure phase. That is all we need from this opening review, but we will need it at once.

3.2 States as Normalized Complex Vectors

Now comes the hard pivot. Quantum mechanics is introduced as a calculus for calculating probabilities. Probabilities of what? Probabilities for measurement outcomes. The things that are measured are observables, and the systems being measured are prepared in states.

In the simple two-level system, a general state is written as

$$|\psi\rangle = \alpha|u\rangle + \beta|d\rangle, \tag{3.10}$$

where α and β are complex numbers. In the standard basis,

$$|u\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \qquad |d\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}. \tag{3.11}$$

If we measure up or down, the probabilities are

$$P(u) = |\alpha|^2, \qquad P(d) = |\beta|^2. \tag{3.12}$$

Because the total probability must be one, the state is normalized:

$$\langle\psi|\psi\rangle = |\alpha|^2 + |\beta|^2 = 1. \tag{3.13}$$

This is the first postulate in the lecture's presentation: states are vectors in a complex vector space, and we normalize them. In a higher-dimensional example the same pattern persists. The basis vectors are columns with a single nonzero entry, and an arbitrary state is a complex linear combination of them.

At this point Susskind pauses on an important issue. The probabilities $|\alpha|^2$ and $|\beta|^2$ do not completely determine the complex amplitudes. Multiplying by a pure phase changes the number but not its magnitude:

$$|e^{i\theta}\alpha|^2 = |\alpha|^2. \tag{3.14}$$

So the amplitudes contain more information than probabilities alone. The lecture does not yet say what that extra information is for; it merely marks that there is something still to be explained.

There is another warning already in place. These state vectors are not ordinary arrows in real space. The vectors $|u\rangle$ and $|d\rangle$ are orthogonal in the space of states. Later we will also speak about spatial directions of spin and magnetic moment, and those are different objects. The distinction must be kept in mind from the beginning.

3.3 Observables, Transpose, and Hermitian Matrices

What about the quantities that we measure? The lecture now turns to observables. The outcomes of measurements are always real numbers, or at any rate can always be reduced to real numbers. That requirement is what narrows the matrix language.

Before defining the relevant class of matrices, Susskind introduces transpose and then Hermitian conjugation.

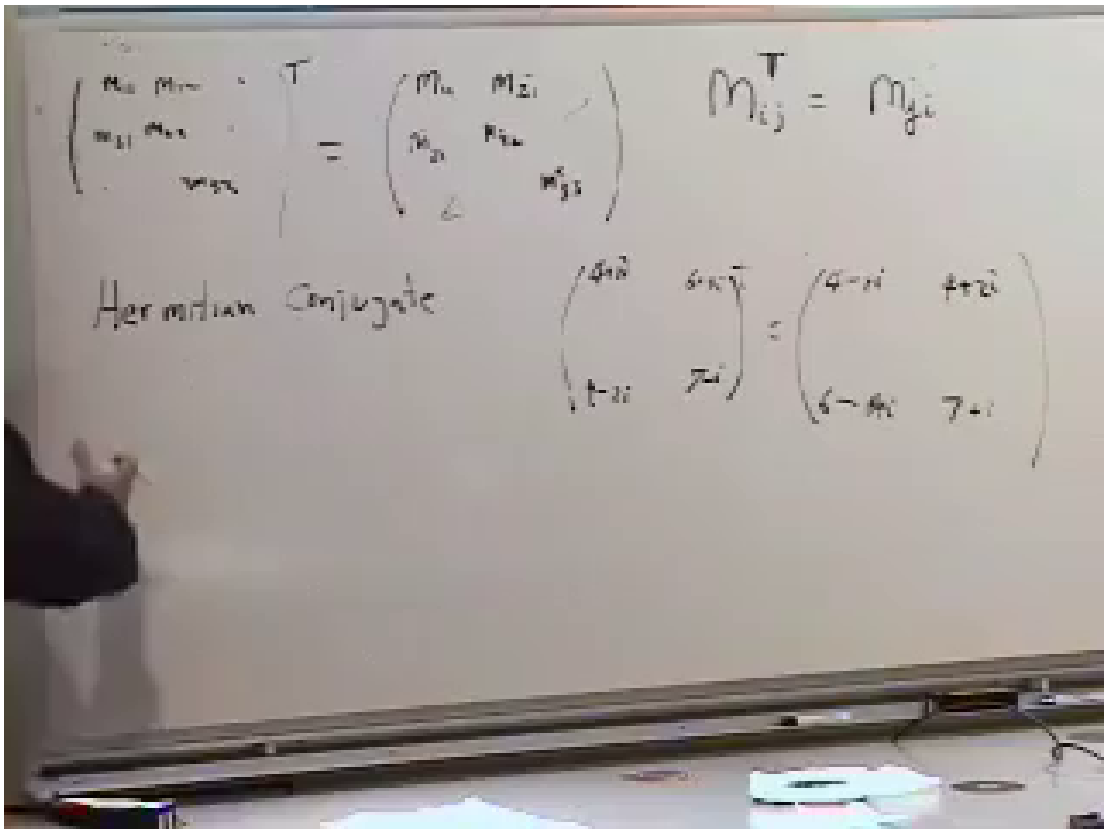


Figure 3.1: Matrix transpose and symmetry rule.

The board first displays transpose concretely. For a 3×3 matrix,

$$\begin{pmatrix} m_{11} & m_{12} & m_{13} \\ m_{21} & m_{22} & m_{23} \\ m_{31} & m_{32} & m_{33} \end{pmatrix}^T = \begin{pmatrix} m_{11} & m_{21} & m_{31} \\ m_{12} & m_{22} & m_{32} \\ m_{13} & m_{23} & m_{33} \end{pmatrix}. \tag{3.15}$$

In index notation,

$$(M^T)_{ij} = M_{ji}. \tag{3.16}$$

The effect is simply to interchange rows and columns, or if one prefers a geometric picture, to reflect across the main diagonal.

A symmetric matrix is one that is unchanged by this operation:

$$M_{ij} = M_{ji}. \tag{3.17}$$

That is the content of the board equation near the figure, and it is worth preserving because the lecture uses this simpler notion as a warm-up before turning to the complex case.

Now we add complex conjugation. The Hermitian conjugate is defined by

$$M^\dagger = (M^T)^*. \tag{3.18}$$

A Hermitian matrix is one equal to its own Hermitian conjugate:

$$M = M^\dagger. \tag{3.19}$$

In components this becomes

$$M_{ij} = (M_{ji})^*. \tag{3.20}$$

The transcript is garbled right at this point, but this is the clean condition the lecture is reaching for.

The consequences are immediate. The diagonal entries are real, because on the diagonal we have $i = j$, so the entry must equal its own complex conjugate. Off the diagonal, reflected entries are complex conjugates of one another. The right-hand board example is not sharp enough for literal transcription, so it is better to write the standard 2×2 form explicitly:

$$M = \begin{pmatrix} a & z \\ z^* & b \end{pmatrix}, \qquad a, b \text{ real}. \tag{3.21}$$

This is the matrix version of the reality condition appropriate to observables.

3.3.1 Question & Answer

Question. If observables are real, why are we using complex numbers at all?

Answer. The lecture deliberately refuses to answer this too early. Anything we literally measure can be encoded by real numbers. One may, if one wishes, package several real quantities into a complex number, but that is not the real issue. The real question is: why is quantum mechanics *driven* to complex amplitudes? Susskind postpones the answer. It belongs to time evolution and reversibility, not to the static postulates being organized here. For the moment we keep the complex vector space in place and move on.

3.4 Matrix Elements and Real Expectation Values

The lecture now writes down a few very elementary identities and immediately extracts the first genuine payoff. These are the "theorems without names" that justify the Hermitian definition.

First, the inner product satisfies

$$\langle B|A\rangle = \langle A|B\rangle^*. \tag{3.22}$$

Second, for any matrix M,

$$\langle B|M|A\rangle = \big(\langle A|M^\dagger|B\rangle\big)^*. \tag{3.23}$$

If M is Hermitian, so that $M^\dagger = M$, this becomes

$$\langle B|M|A\rangle = \big(\langle A|M|B\rangle\big)^*. \tag{3.24}$$

The quantity $\langle B|M|A\rangle$ is the matrix element of M between the states $|A\rangle$ and $|B\rangle$.

Proposition 3.1. *If M is Hermitian, then $\langle A|M|A\rangle$ is real.*

Proof. Set $B = A$ in the Hermitian matrix-element identity. Then

$$\langle A|M|A\rangle = \big(\langle A|M|A\rangle\big)^*. \tag{3.25}$$

A number equal to its own complex conjugate is real. Therefore $\langle A|M|A\rangle \in \mathbb{R}$. □

The lecture then gives this object its standard name: the *expectation value* of M in the state $|A\rangle$. The probabilistic interpretation of the name is deferred, but the mathematical point is already decisive. If an observable is to have real measurement outcomes, its average value must certainly be real. Hermitian operators have exactly that property in every state. That is the first serious reason they are the right representatives for observables.

3.5 Eigenvalues and Eigenvectors from the Simplest Examples

At this point the lecture insists that we continue doing mathematics. One can speak vaguely about entanglement and leave with a mystical feeling, but not with understanding. So the next concept, and essentially the last big mathematical concept needed for the day, is the notion of eigenvalues and eigenvectors.

Definition 3.2. A nonzero vector $|a\rangle$ is an eigenvector of M with eigenvalue λ_a if

$$M|a\rangle = \lambda_a|a\rangle. \tag{3.26}$$

For a Hermitian matrix the eigenvalue must be real.

Proposition 3.3. *If M is Hermitian and $M|a\rangle = \lambda_a|a\rangle$, then λ_a is real.*

Proof. Multiply the eigenvalue equation on the left by $\langle a|$:

$$\langle a|M|a\rangle = \lambda_a\langle a|a\rangle. \tag{3.27}$$

The left-hand side is real because M is Hermitian, and $\langle a|a\rangle$ is real and positive for a nonzero vector. Therefore

$$\lambda_a = \frac{\langle a|M|a\rangle}{\langle a|a\rangle} \tag{3.28}$$

is real. □

Now comes the lecture's first great interpretive statement. If M represents an observable, then the values one can actually measure are the eigenvalues of M. The eigenvector associated with a given eigenvalue is the state in which that value is obtained with certainty.

To make this statement feel unavoidable, Susskind begins with the easiest possible examples.

Take the diagonal matrix

$$M = \begin{pmatrix} m_{11} & 0 \\ 0 & m_{22} \end{pmatrix}. \tag{3.29}$$

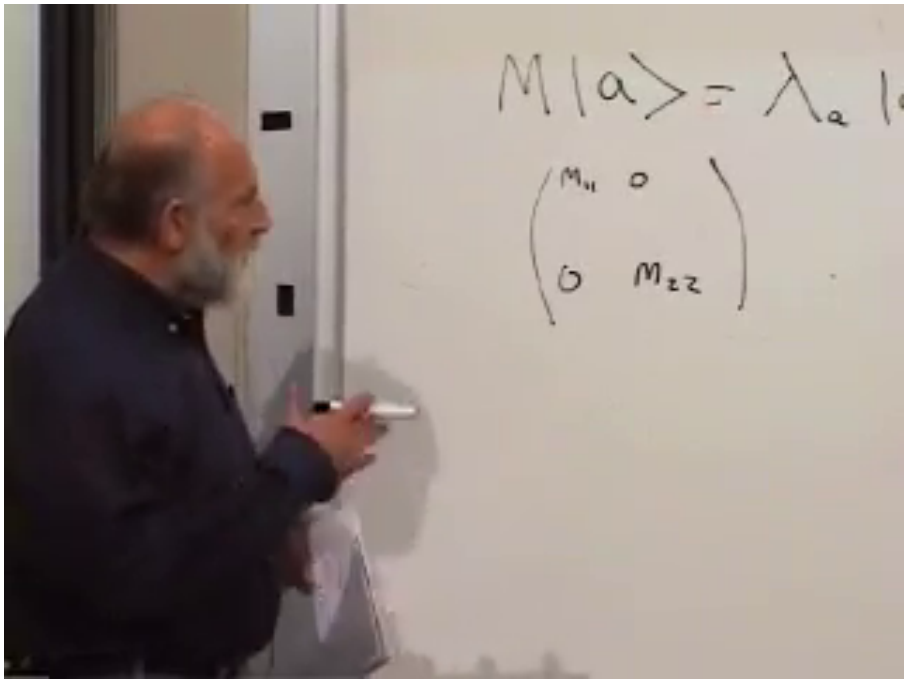


Figure 3.2: Eigenvector equation with a diagonal matrix.

The first basis vector is immediately an eigenvector:

$$\begin{pmatrix} m_{11} & 0 \\ 0 & m_{22} \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = m_{11} \begin{pmatrix} 1 \\ 0 \end{pmatrix}. \tag{3.30}$$

So $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ has eigenvalue m_{11}.

Likewise,

$$\begin{pmatrix} m_{11} & 0 \\ 0 & m_{22} \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = m_{22} \begin{pmatrix} 0 \\ 1 \end{pmatrix}. \tag{3.31}$$

So $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$ has eigenvalue m_{22}.

These are almost trivial calculations, but the lecture uses them to establish the rhythm we will need later. Matrix times vector equals number times the same vector. The number is a possible measurement value. The vector is a state of certainty for that measurement. One can already see the pattern for larger diagonal matrices: the diagonal entries are the eigenvalues, and the standard basis vectors are the eigenvectors.

3.6 Pauli Matrices and the Two-Level Spin System

Only after the diagonal examples are under control does the lecture turn to the actual two-state spin observables.

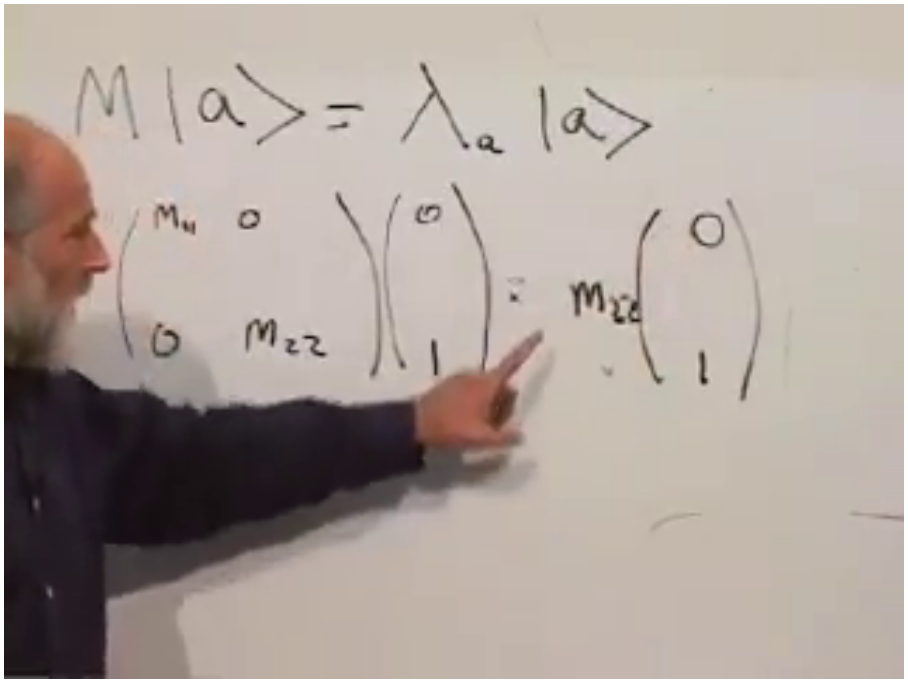


Figure 3.3: Diagonal matrix acting on the lower basis vector.

The first one is

$$\sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \tag{3.32}$$

Its eigenvectors are the familiar up and down states:

$$\sigma_3 \begin{pmatrix} 1 \\ 0 \end{pmatrix} = + \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \qquad \sigma_3 \begin{pmatrix} 0 \\ 1 \end{pmatrix} = - \begin{pmatrix} 0 \\ 1 \end{pmatrix}. \tag{3.33}$$

So the possible measured values are $+1$ and -1, and the corresponding states are precisely the states of definite spin along the 3-axis.

The second matrix is the off-diagonal example:

$$\sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}. \tag{3.34}$$

Here the lecture first writes the candidate eigenvector and only then asks what the matrix does to it. The screenshot captures that pedagogical moment.

The candidate vector

$$\begin{pmatrix} 1 \\ 1 \end{pmatrix} \tag{3.35}$$

is indeed an eigenvector, but not normalized. In quantum mechanics we normalize the state, so we use

$$\frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}. \tag{3.36}$$

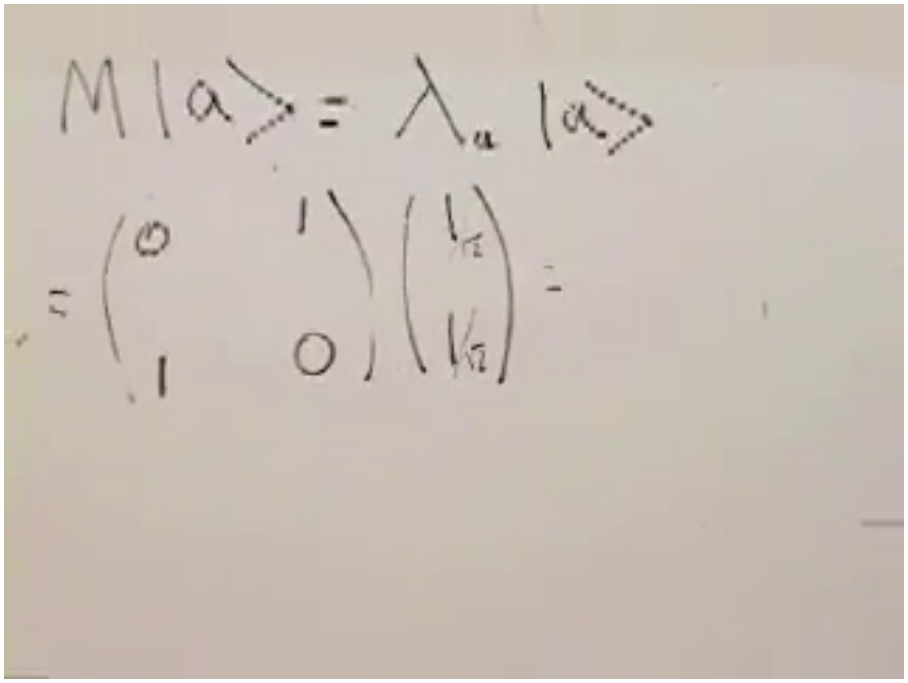


Figure 3.4: Off-diagonal matrix acting on an equal-weight vector.

The right-hand side is not yet written on the board in the frame, so we supply it from the surrounding calculation:

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}. \tag{3.37}$$

So this is an eigenvector of σ_1 with eigenvalue $+1$. The companion vector

$$\frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix} \tag{3.38}$$

satisfies

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = -\frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}. \tag{3.39}$$

The lecture pauses here to emphasize something subtle but basic. If a vector is an eigenvector, multiplying it by any nonzero scalar leaves it an eigenvector. Normalization is not part of the eigenvector condition itself. But once the vector is to represent a physical state, we normalize it.

There is also a brief geometric aside in the lecture. One can view σ_1 as interchanging the two basis directions E_1 and E_2. In that picture the diagonal directions are singled out, and that is why the eigenvectors of σ_1 point along $(1, 1)$ and $(1, -1)$.

The lecture then introduces the third Pauli matrix,

$$\sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \tag{3.40}$$

but postpones its eigenvectors for a little while. What matters immediately is that all three Pauli matrices share the same basic spectral pattern. They all square to the identity:

$$\sigma_1^2 = \sigma_2^2 = \sigma_3^2 = 1, \tag{3.41}$$

where 1 denotes the identity matrix. For example,

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}^2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}. \tag{3.42}$$

If the square of an observable is the identity, then the square of any eigenvalue must be one. Therefore the only possible measured values are

$$\lambda = \pm 1. \tag{3.43}$$

This is the recurring fact the lecture wants us to absorb: different spin observables on the same two-dimensional state space have different eigenvectors, but the same two possible outcomes.

3.7 Orthogonality, Probabilities, and Spin Along an Arbitrary Axis

Now Susskind announces a fundamental theorem. Before proving it, he says what it means. If an observable can distinguish two states by assigning them different definite outcomes, then those states must be orthogonal. In the lecture's language: orthogonal states are measurably distinct states.

Theorem 3.4. *Let M be Hermitian, and suppose*

$$M|A\rangle = \lambda_A|A\rangle, \qquad M|B\rangle = \lambda_B|B\rangle, \tag{3.44}$$

with $\lambda_A \neq \lambda_B$. Then

$$\langle B|A\rangle = 0. \tag{3.45}$$

Proof. Multiply the first equation on the left by $\langle B|$:

$$\langle B|M|A\rangle = \lambda_A\langle B|A\rangle. \tag{3.46}$$

Multiply the second equation on the left by $\langle A|$:

$$\langle A|M|B\rangle = \lambda_B\langle A|B\rangle. \tag{3.47}$$

Now complex-conjugate the second equation. Since M is Hermitian and λ_B is real,

$$\langle B|M|A\rangle = \lambda_B\langle B|A\rangle. \tag{3.48}$$

Comparing the two equations gives

$$(\lambda_A - \lambda_B)\langle B|A\rangle = 0. \tag{3.49}$$

If $\lambda_A \neq \lambda_B$, the only possibility is $\langle B|A\rangle = 0$. □

The lecture then checks this theorem on the σ_1 example. The two normalized eigenvectors are

$$\frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ 1 \end{pmatrix}, \qquad \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ -1 \end{pmatrix}, \tag{3.50}$$

and their inner product is

$$\frac{1}{\sqrt{2}}\begin{pmatrix}1 & 1\end{pmatrix}\frac{1}{\sqrt{2}}\begin{pmatrix}1\\-1\end{pmatrix} = \frac{1}{2} - \frac{1}{2} = 0. \tag{3.51}$$

So the theorem is not merely formal; it is already visible in the simplest example.

At this point the lecture supplies the promised eigenvectors of σ_2:

$$\frac{1}{\sqrt{2}}\begin{pmatrix}1\\i\end{pmatrix}, \qquad \frac{1}{\sqrt{2}}\begin{pmatrix}1\\-i\end{pmatrix}. \tag{3.52}$$

One has eigenvalue $+1$, the other eigenvalue -1. So σ_1, σ_2, and σ_3 are all similar in the sense emphasized by the lecture: they all have the same two eigenvalues, $+1$ and -1, even though their eigenvectors differ.

Now comes the probability postulate. Suppose the system is prepared in a normalized state $|B\rangle$, and suppose $|A\rangle$ is a normalized eigenvector of an observable M with eigenvalue λ_A. Then

$$P(\lambda_A|B) = |\langle A|B\rangle|^2 = \langle A|B\rangle\langle B|A\rangle. \tag{3.53}$$

This is the bridge between the spectral information and actual experimental prediction.

The lecture works examples immediately.

First prepare the electron in the $+1$ eigenstate of σ_3:

$$|B\rangle = \begin{pmatrix}1\\0\end{pmatrix}. \tag{3.54}$$

Now measure σ_1. The $+1$ eigenstate of σ_1 is

$$|A\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix}1\\1\end{pmatrix}. \tag{3.55}$$

Then

$$\langle A|B\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix}1 & 1\end{pmatrix}\begin{pmatrix}1\\0\end{pmatrix} = \frac{1}{\sqrt{2}}, \tag{3.56}$$

$$P(\sigma_1 = +1 \,|\, \sigma_3 = +1) = |\langle A|B\rangle|^2 = \frac{1}{2}. \tag{3.57}$$

If we choose instead the -1 eigenvector of σ_1,

$$\frac{1}{\sqrt{2}}\begin{pmatrix}1\\-1\end{pmatrix}, \tag{3.58}$$

we again get probability $1/2$. So a state polarized along the 3-axis gives equal chances for the two outcomes when we measure along the 1-axis.

The lecture then reverses the order of preparation and measurement, and gets the same answer. That is not an accident of wording; it reflects the symmetry of $|\langle A|B\rangle|^2$.

Finally, the lecture takes a genuinely complex example. Prepare the system in the $+1$ eigenstate of σ_1:

$$|B\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix}1\\1\end{pmatrix}, \tag{3.59}$$

and measure σ_2. For the $+1$ eigenstate of σ_2, take

$$|A\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix}1\\ i\end{pmatrix}. \tag{3.60}$$

Then

$$\langle A|B\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix}1 & -i\end{pmatrix}\frac{1}{\sqrt{2}}\begin{pmatrix}1\\ 1\end{pmatrix} = \frac{1}{2} - \frac{i}{2}, \tag{3.61}$$

$$P(\sigma_2 = +1 \,|\, \sigma_1 = +1) = \left|\frac{1}{2} - \frac{i}{2}\right|^2 = \frac{1}{2}. \tag{3.62}$$

Again the result is one-half. The lecture uses these examples to train the eye: changing the measurement axis changes the probabilities, not the set of possible values.

3.7.1 Question & Answer

Question. Is this just the cosine squared of an angle? And are perpendicular pointers the same thing as orthogonal states?

Answer. Only if one keeps the analogy under tight control. For ordinary real unit vectors in three-dimensional space, the dot product is the cosine of the angle between them. But here the inner product of two states is generally complex, and the probability is its absolute square. So the ordinary geometric picture is at best an analogy.

The more important point is the distinction between state vectors and spatial directions. A spin direction in ordinary space is a *pointer*. A state vector lives in Hilbert space. The state orthogonal to spin-up along a given axis is spin-down along that same axis; it is not spin-up along a perpendicular spatial direction. That is why the lecture repairs the vocabulary explicitly: *perpendicular* belongs to ordinary-space pointers, while *orthogonal* belongs to states.

With that clarification made, the lecture turns to the component of spin along an arbitrary spatial direction. Here we leave Hilbert-space vectors for a moment and speak classically. Let

$$\hat{n} = (n_1, n_2, n_3), \qquad n_1^2 + n_2^2 + n_3^2 = 1. \tag{3.63}$$

This is a unit pointer in ordinary three-dimensional space. Classically the component of a spin pointer σ along $\hat{n}$ is the dot product $\sigma \cdot \hat{n}$. The quantum candidate is therefore

$$\sigma \cdot \hat{n} = n_1\sigma_1 + n_2\sigma_2 + n_3\sigma_3. \tag{3.64}$$

Using the Pauli matrices,

$$\sigma \cdot \hat{n} = \begin{pmatrix} n_3 & n_1 - in_2 \\ n_1 + in_2 & -n_3 \end{pmatrix}. \tag{3.65}$$

This is Hermitian at once: the diagonal entries are real, and the off-diagonal entries are complex conjugates.

The lecture then asks whether the square of this operator is again the identity. The key point is that when we multiply matrices, order matters. So the cross terms must be written in both orders. The Pauli matrices satisfy the anticommutation relations

$$\sigma_1\sigma_2 + \sigma_2\sigma_1 = 0, \qquad \sigma_2\sigma_3 + \sigma_3\sigma_2 = 0, \qquad \sigma_1\sigma_3 + \sigma_3\sigma_1 = 0. \tag{3.66}$$

Together with $\sigma_i^2 = 1$, this gives

$$\begin{aligned}(\sigma \cdot \hat{n})^2 &= (n_1\sigma_1 + n_2\sigma_2 + n_3\sigma_3)^2 \qquad (3.67)\\ &= n_1^2\sigma_1^2 + n_2^2\sigma_2^2 + n_3^2\sigma_3^2 \\ &\quad + n_1n_2(\sigma_1\sigma_2 + \sigma_2\sigma_1) + n_2n_3(\sigma_2\sigma_3 + \sigma_3\sigma_2) + n_1n_3(\sigma_1\sigma_3 + \sigma_3\sigma_1) \\ &= (n_1^2 + n_2^2 + n_3^2)\,1 \qquad (3.68)\\ &= 1.\end{aligned}$$

Therefore any eigenvalue λ of $\sigma \cdot \hat{n}$ satisfies

$$\lambda^2 = 1, \tag{3.69}$$

and so

$$\lambda = \pm 1. \tag{3.70}$$

This is how the lecture closes. Along every axis, the possible measured values are still only $+1$ and -1. The quantum strangeness lies not in a changing spectrum, but in the changing probabilities.

3.8 Summary

The lecture has now assembled the basic language needed for the two-state quantum system. States are normalized vectors in a complex vector space. Observables are represented by Hermitian matrices, whose expectation values and eigenvalues are real. The eigenvalues are the possible measurement outcomes, the corresponding eigenvectors are the states of certainty for those outcomes, and eigenvectors with different eigenvalues are orthogonal. The probability of obtaining a given outcome is the absolute square of the relevant inner product. For spin, the Pauli matrices realize this entire structure concretely, and even along an arbitrary spatial axis the only possible outcomes remain ± 1. What varies from axis to axis is not the list of answers but the probabilities with which those answers occur.

Chapter 4

Quantum Entanglements, Part 1

These companion notes follow Leonard Susskind's fourth lecture in the *Quantum Entanglement* supplementary track and are curated for this archive by LazyingArt LLC. The lecture opens by saying, in effect, that Bell's theorem will have to wait and that entanglement itself is only just within reach. But before two spins can do anything interesting together, we still have to finish the one-spin machinery. That is the governing decision of the lecture: first complete the story of a single two-state system, and only then return to the promised subject of entanglement.

4.1 The Arbitrary-Axis Spin Operator

We resume with a single spin and an arbitrary axis in space. Susskind labels the spatial directions by $1, 2, 3$, while warning that he may lapse into x, y, z. We will keep the $1, 2, 3$ notation, with the understanding that these are just the three spatial axes.

An arbitrary direction is represented by a unit vector

$$\hat{n} = (n_1, n_2, n_3), \qquad n_1^2 + n_2^2 + n_3^2 = 1. \tag{4.1}$$

If $\sigma = (\sigma_1, \sigma_2, \sigma_3)$ denotes the spin operator, then the component of spin along $\hat{n}$ is

$$\sigma \cdot \hat{n} = n_1\sigma_1 + n_2\sigma_2 + n_3\sigma_3. \tag{4.2}$$

At this point the lecture makes an important distinction: this is the ordinary dot product in space, not the inner product of two state-vectors in Hilbert space. The n_i are just numbers, the direction cosines of the chosen axis. The σ_i, however, are matrices. Therefore $\sigma \cdot \hat{n}$ is itself an operator.

Using the Pauli matrices, we obtain the explicit form

$$\sigma \cdot \hat{n} = \begin{pmatrix} n_3 & n_1 - in_2 \\ n_1 + in_2 & -n_3 \end{pmatrix}. \tag{4.3}$$

This is the operator that answers the physical question: if we orient the measuring apparatus along $\hat{n}$, what are the possible results for the spin component along that axis?

The lecture keeps the physical interpretation close at hand. To measure $\sigma \cdot \hat{n}$, we imagine turning on a large magnetic field oriented along $\hat{n}$. The electron either emits a photon or does not. That physical process is encoded mathematically by the operator $\sigma \cdot \hat{n}$. The whole point is that the abstract operator language and the laboratory procedure are describing the same thing.

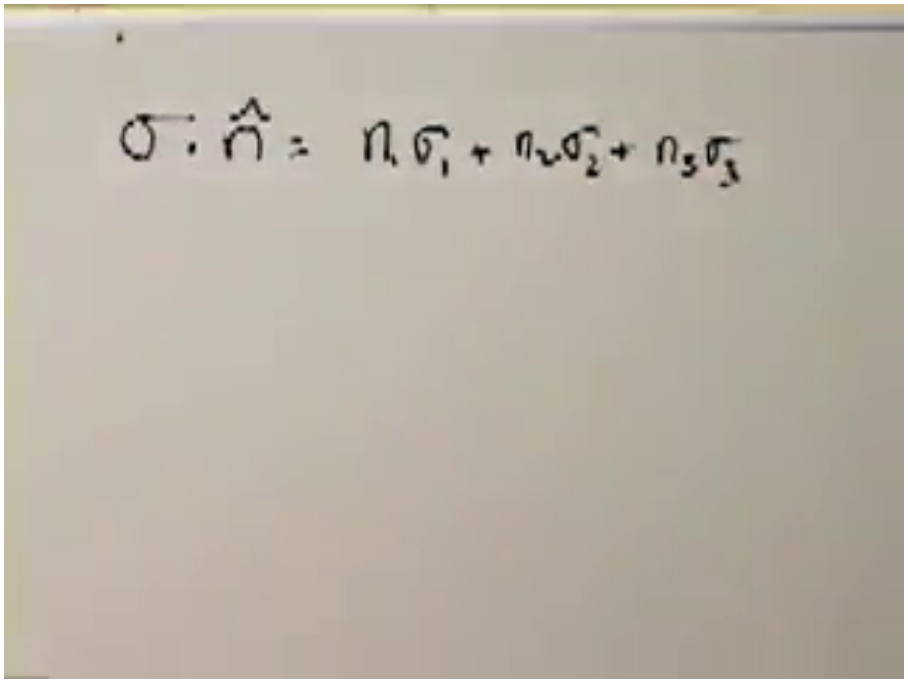


Figure 4.1: Spin component along an arbitrary direction.

4.2 Basic Properties of $\sigma \cdot \hat{n}$

It is convenient to introduce the standard shorthand

$$n_- = n_1 - in_2, \qquad n_+ = n_1 + in_2. \tag{4.4}$$

Then the matrix can be written as

$$\sigma \cdot \hat{n} = \begin{pmatrix} n_3 & n_- \\ n_+ & -n_3 \end{pmatrix}. \tag{4.5}$$

The lower shorthand line on the board is only partially visible, but the notation is clear from the surrounding lecture.

The matrix is Hermitian: the diagonal entries are real, and the off-diagonal entries are complex conjugates of one another. That is exactly what we want from an observable. The lecture then extracts a few working facts and moves on; these are tools, not a separate formalism.

Remark 4.1. Here and below, the symbol 1 means the identity operator on the two-dimensional one-spin state space.

Proposition 4.2. *For any unit vector $\hat{n}$,*

$$(\sigma \cdot \hat{n})^2 = 1. \tag{4.6}$$

Hence the eigenvalues of $\sigma \cdot \hat{n}$ are ± 1.

Proof. One may verify the square directly from the matrix and the identity $n_1^2 + n_2^2 + n_3^2 = 1$. The lecture also recalls the Pauli-matrix relation

$$\sigma_i \sigma_j = -\sigma_j \sigma_i, \qquad i \neq j, \tag{4.7}$$

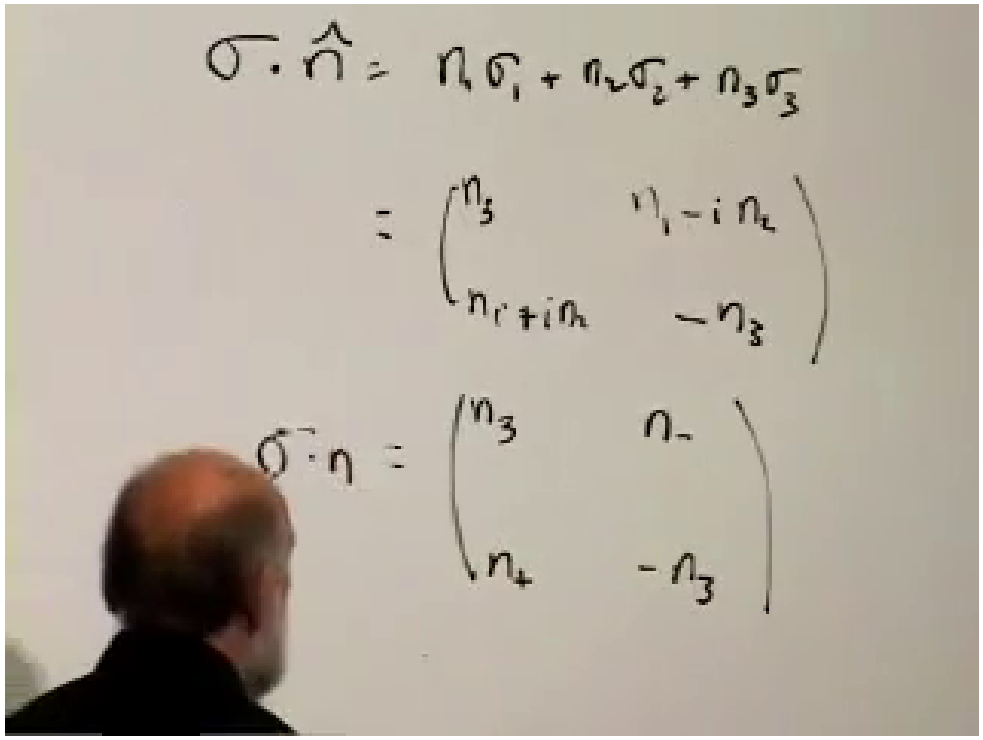


Figure 4.2: Matrix form of $\sigma \cdot \hat{n}$.

which gives the same result. If λ is an eigenvalue of $\sigma \cdot \hat{n}$, then $\lambda^2 = 1$, so $\lambda = \pm 1$. □

A second algebraic identity is introduced because it will be used repeatedly in the eigenvector calculation:

$$n_+ n_- = (n_1 + i n_2)(n_1 - i n_2) = n_1^2 + n_2^2 = 1 - n_3^2. \tag{4.8}$$

The lecture treats this exactly as it should be treated: not as a deep theorem, but as the small fact that makes the later algebra survivable.

4.3 From Preparation to Probability

Now the lecture arrives at the question it really wants to answer. Suppose we prepare the electron so that its spin is definitely up along the axis $\hat{n}$. Then we measure the spin along a different axis $\hat{m}$. What is the probability that the outcome along $\hat{m}$ is $+1$?

Before calculating anything, Susskind notes what symmetry suggests. The answer ought to depend only on the angle between $\hat{n}$ and $\hat{m}$, and it ought to be symmetric in n and m. But he does not stop there. The lecture insists that we derive the answer from the probability postulate rather than simply guess the angle-dependence and bless it afterward.

We therefore introduce the $+1$ eigenstates of the two operators:

$$\sigma \cdot \hat{n} \, |\sigma \cdot \hat{n} = 1\rangle = |\sigma \cdot \hat{n} = 1\rangle, \qquad \sigma \cdot \hat{m} \, |\sigma \cdot \hat{m} = 1\rangle = |\sigma \cdot \hat{m} = 1\rangle. \tag{4.9}$$

The equality sign inside each ket is part of the label; it is not an additional assertion. The lecture

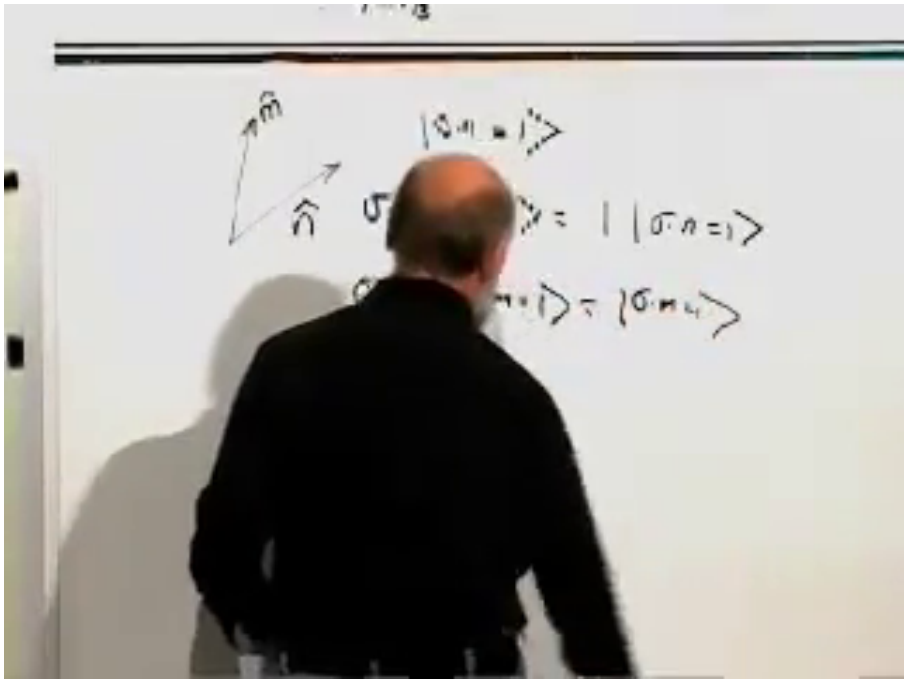


Figure 4.3: Spin-up eigenstates along $\hat{m}$ and $\hat{n}$.

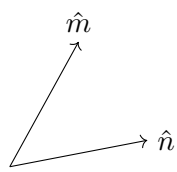


Figure 4.4: A clean reconstruction of the $\hat{n}$-$\hat{m}$ sketch that remains on the board while the eigenstate equations are written.

is explicit about this because the notation is a bit heavy. The board is partially occluded at this point, so what we write here is the cautious standard reconstruction of the intended notation.

4.3.1 Question & Answer

Question. Once the state is prepared along $\hat{n}$, what quantity do we calculate first in order to predict the outcome along $\hat{m}$?

Answer. First we calculate a probability amplitude, not a probability. The amplitude is the inner product between the prepared state and the final eigenstate whose outcome we want to test:

$$\langle \sigma \cdot \hat{m} = 1 \mid \sigma \cdot \hat{n} = 1 \rangle. \tag{4.10}$$

Only after that do we form the probability by multiplying by the complex conjugate:

$$P(\sigma \cdot \hat{m} = 1 \mid \sigma \cdot \hat{n} = 1) = \left| \langle \sigma \cdot \hat{m} = 1 \mid \sigma \cdot \hat{n} = 1 \rangle \right|^2. \tag{4.11}$$

That distinction is one of the lecture's central pedagogical beats: amplitude first, probability second.

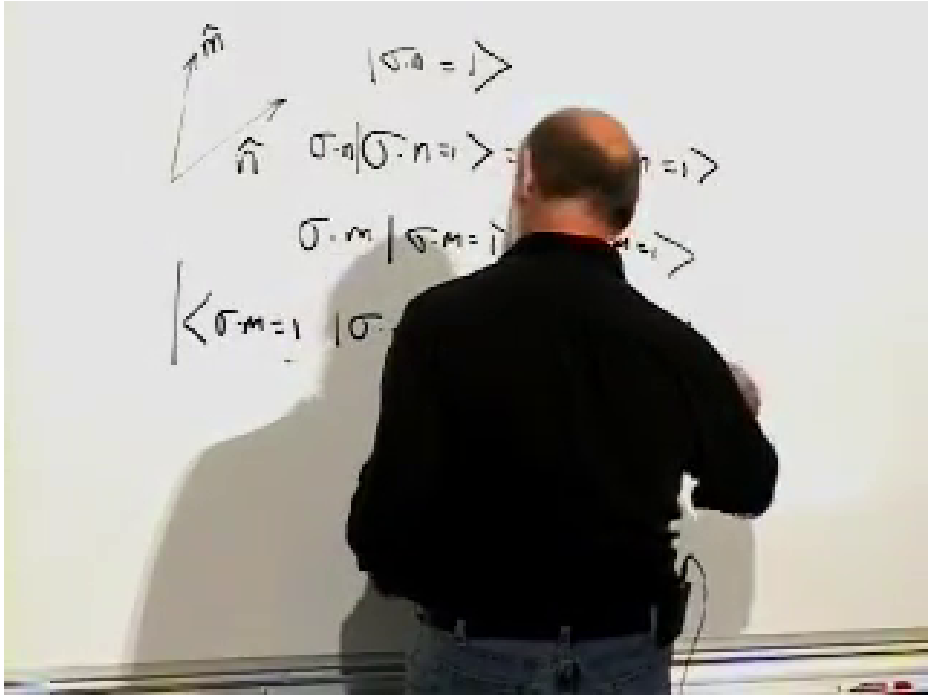


Figure 4.5: Overlap of spin-up states along $\hat{m}$ and $\hat{n}$.

The transcript is garbled around this point, but the board progression and the surrounding discussion make the intended route clear: define the relevant eigenvectors, take their overlap, then square the magnitude.

4.4 The Eigenvector Calculation and the Formula $\frac{1+\hat{n}\cdot\hat{m}}{2}$

The lecture now slows down and performs the calculation. Susskind warns the audience that it is ugly. We may clean it up a little, but we should preserve the rhythm of the argument: first solve for the eigenvectors, then normalize them, and only then take the inner product.

We seek a $+1$ eigenvector of $\sigma \cdot \hat{n}$, so we solve

$$(\sigma \cdot \hat{n})\begin{pmatrix}\alpha \\ \beta\end{pmatrix} = \begin{pmatrix}\alpha \\ \beta\end{pmatrix}. \tag{4.12}$$

At first we do *not* normalize the vector. We also keep in mind another point the lecture emphasizes: multiplying a state by an overall phase $e^{i\theta}$ changes no probability, since every physical probability is built from a quantity times its complex conjugate. Therefore, at this stage, there is no loss of physical content in choosing

$$\alpha = 1. \tag{4.13}$$

Write the lower component as z. Then

$$\begin{pmatrix} n_3 & n_- \\ n_+ & -n_3 \end{pmatrix}\begin{pmatrix}1 \\ z\end{pmatrix} = \begin{pmatrix}1 \\ z\end{pmatrix}. \tag{4.14}$$

From the top component we obtain

$$n_3 + n_- z = 1, \tag{4.15}$$

and hence

$$z = \frac{1 - n_3}{n_-}. \tag{4.16}$$

So one unnormalized +1 eigenvector is

$$\begin{pmatrix} 1 \\ \dfrac{1 - n_3}{n_-} \end{pmatrix}. \tag{4.17}$$

Now we normalize it. The corresponding bra is

$$\left(1, \frac{1 - n_3}{n_+}\right), \tag{4.18}$$

so the norm is

$$\left(1, \frac{1 - n_3}{n_+}\right) \begin{pmatrix} 1 \\ \dfrac{1 - n_3}{n_-} \end{pmatrix} = 1 + \frac{(1 - n_3)^2}{n_+ n_-} \tag{4.19}$$

$$= 1 + \frac{(1 - n_3)^2}{(1 - n_3)(1 + n_3)} \tag{4.20}$$

$$= \frac{2}{1 + n_3}. \tag{4.21}$$

Therefore a convenient normalized eigenvector is

$$|\sigma \cdot \hat{n} = 1\rangle = \sqrt{\frac{1 + n_3}{2}} \begin{pmatrix} 1 \\ \dfrac{1 - n_3}{n_-} \end{pmatrix}. \tag{4.22}$$

This is one explicit phase convention. The lecture would not want us to forget that other equivalent conventions exist; what matters physically is the ray, not the overall phase.

The corresponding bra for the +1 eigenstate of $\sigma \cdot \hat{m}$ is obtained by replacing n with m and taking the conjugate:

$$\langle \sigma \cdot \hat{m} = 1| = \left(\sqrt{\frac{1 + m_3}{2}},\ \frac{1 - m_3}{m_+}\sqrt{\frac{1 + m_3}{2}}\right). \tag{4.23}$$

The next step is the one Susskind refuses to do line by line on the board. We form the inner product

$$\langle \sigma \cdot \hat{m} = 1 \mid \sigma \cdot \hat{n} = 1\rangle \tag{4.24}$$

and then multiply by its complex conjugate. The intermediate expression is, in his words, an unholy mess. But after repeated use of identities such as $n_+ n_- = 1 - n_3^2$, everything collapses to

$$P(\sigma \cdot \hat{m} = 1 \mid \sigma \cdot \hat{n} = 1) = \frac{1 + n_1 m_1 + n_2 m_2 + n_3 m_3}{2} = \frac{1 + \hat{n} \cdot \hat{m}}{2}. \tag{4.25}$$

If θ_{nm} is the angle between the two axes, then

$$\hat{n} \cdot \hat{m} = \cos\theta_{nm}, \tag{4.26}$$

and therefore

$$P(\sigma \cdot \hat{m} = 1 \mid \sigma \cdot \hat{n} = 1) = \frac{1 + \cos\theta_{nm}}{2}. \tag{4.27}$$

Now the lecture does what it always does after a calculation: it checks the answer against obvious cases.

- If $\hat{m} = \hat{n}$, then $\cos\theta_{nm} = 1$, so $P = 1$.
- If $\hat{m} = -\hat{n}$, then $\cos\theta_{nm} = -1$, so $P = 0$.
- If $\hat{m} \perp \hat{n}$, then $\cos\theta_{nm} = 0$, so $P = \frac{1}{2}$.

So the formula passes the tests the lecture wanted: same axis gives certainty, opposite axis gives impossibility, and a right angle gives a half-half result.

Remark 4.3. After the fact, one can simplify the derivation by choosing coordinates so that $\hat{n}$ is the third axis. Then the prepared state is simply $\binom{1}{0}$, and the amplitude is immediately the upper component of the m-eigenvector. The lecture deliberately resists doing that at the start, because the point is to *derive* the rotationally invariant answer rather than assume it.

The lecture also pauses to interpret what probability means operationally. One does not measure a probability in a single shot. One either repeats the same preparation-and-measurement sequence on the same electron, re-preparing it each time along $\hat{n}$, or else prepares many identical electrons and performs the same measurement on each. In either method the fraction of $+1$ outcomes tends to the probability above.

The real surprise is now in plain view. Every measurement of spin along any chosen axis still gives only ± 1, never anything in between. Yet the theory is rotationally smooth, not because the individual outcomes are fractional, but because the *probabilities* are.

4.5 Measurement as Filter; Simultaneous Measurements

With the probability formula in hand, the lecture turns to another important point that had been hovering in the background: a measurement changes the state.

If we measure $\sigma \cdot \hat{m}$ and the outcome is $+1$, we do not merely discover something about the original state. We also leave the electron in the state $|\sigma \cdot \hat{m} = 1\rangle$. The lecture compares this to a polarizer for photons: the device both tests the polarization and prepares the outgoing polarization state. A measurement is therefore a filter.

This is the bridge to simultaneous measurement. If one measurement leaves the system in an eigenstate of the measured operator, then a second quantity can be measured simultaneously only if the same post-measurement state is an eigenstate of both observables.

4.5.1 Question & Answer

Question. When can two observables be measured simultaneously?

Answer. They must have the same eigenvectors. Otherwise the state left behind by one measurement cannot also be a definite state for the other observable.

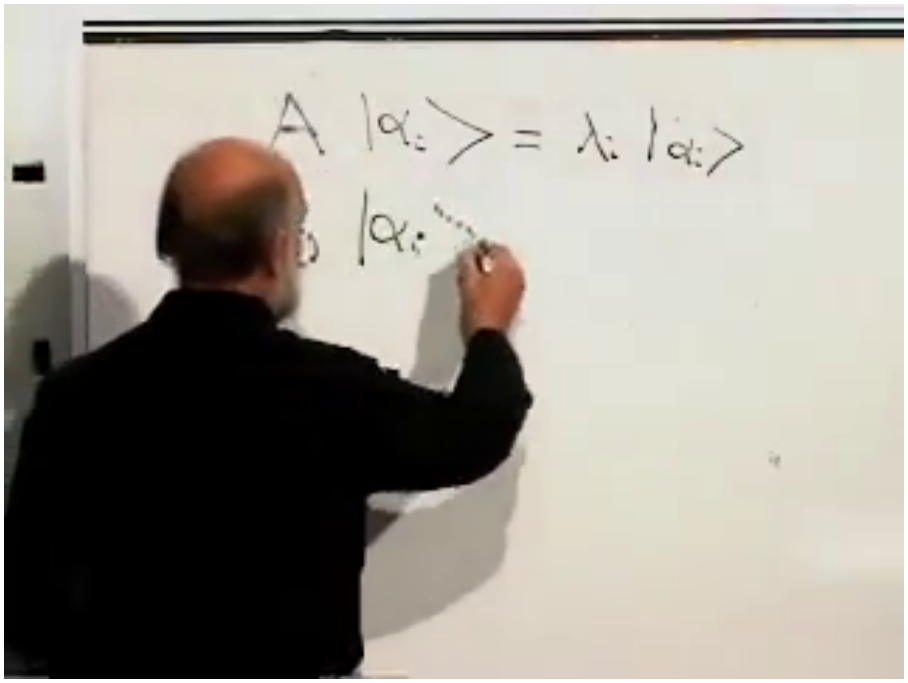


Figure 4.6: Two matrices sharing the same eigenvectors.

This is why different spin components do not admit simultaneous reproducible measurement. For $\sigma \cdot \hat{n}$ and $\sigma \cdot \hat{m}$ with $\hat{n} \neq \hat{m}$, the eigenvectors are different. There is no state that is simultaneously an eigenvector of both. So one cannot measure both without spoiling the information carried by the other.

The lecture then writes the general theorem on the board.

Theorem 4.4. *Suppose A and B share the same complete set of eigenvectors $|\alpha_i\rangle$:*

$$A|\alpha_i\rangle = \lambda_i|\alpha_i\rangle, \qquad B|\alpha_i\rangle = \mu_i|\alpha_i\rangle. \tag{4.28}$$

Then

$$AB = BA. \tag{4.29}$$

Equivalently,

$$[A, B] \equiv AB - BA = 0. \tag{4.30}$$

Proof. On any common eigenvector,

$$AB|\alpha_i\rangle = A(\mu_i|\alpha_i\rangle) = \lambda_i\mu_i|\alpha_i\rangle, \tag{4.31}$$

while

$$BA|\alpha_i\rangle = B(\lambda_i|\alpha_i\rangle) = \mu_i\lambda_i|\alpha_i\rangle. \tag{4.32}$$

Thus AB and BA act the same way on every basis vector $|\alpha_i\rangle$, so $AB = BA$. □

The lecture states the converse as well in the present finite-dimensional setting: commuting operators can be simultaneously diagonalized. That is the standard criterion for simultaneous measurability. The proof of the converse is not developed in detail here, but the theorem itself is part of the lecture's mathematical payload.

4.6 Any One-Spin State Is Up Along Some Axis

There is one more single-spin theorem before the lecture will allow itself to move on to entanglement. Up to now we have been given a direction $\hat{n}$ and have found the corresponding eigenvector. The lecture now reverses the problem.

Take an arbitrary normalized spinor

$$\begin{pmatrix} \alpha \\ \beta \end{pmatrix}, \qquad |\alpha|^2 + |\beta|^2 = 1. \tag{4.33}$$

Can we find a direction $\hat{n}$ such that

$$(\sigma \cdot \hat{n}) \begin{pmatrix} \alpha \\ \beta \end{pmatrix} = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}? \tag{4.34}$$

4.6.1 Question & Answer

Question. Given an arbitrary normalized one-spin state, is there always some axis along which the spin is definitely up?

Answer. Yes. The lecture proves it by counting degrees of freedom.

A unit vector $\hat{n}$ has only two independent real parameters, since $n_1^2 + n_2^2 + n_3^2 = 1$. On the other hand, the state-vector $\binom{\alpha}{\beta}$ begins with four real parameters, but normalization removes one and overall phase removes one more. Thus a physical one-spin state also depends on only two real parameters.

So the reverse eigenvalue problem has the right number of meaningful unknowns. In the lecture's language, there are two equations and two variables. Therefore there is always some direction $\hat{n}$ for which the given spinor is the $+1$ eigenstate of $\sigma \cdot \hat{n}$.

This is a concrete completion of the one-spin story. No matter how a pure one-spin state is prepared, it can always be thought of as "spin up" along some axis.

The lecture even gives an operational picture. Prepare a large number of identical copies of the same electron state. Measure one batch along one axis, another batch along a nearby axis, and keep adjusting the direction of the magnet until you find the axis along which all test electrons come out up. That is the axis corresponding to the state.

At this point the one-spin machinery is complete: arbitrary-axis observables, probability amplitudes, the measurement-as-filter rule, the commutator criterion, and the reverse theorem that identifies every pure one-spin state with spin-up along some direction. Only now does the lecture return to the topic it promised at the start.

4.7 Two Spins, Product States, and the First Entangled State

We now begin the two-spin system. The lecture simplifies the situation brutally: forget the motion of the electrons; imagine them nailed down in space so that only their spins are free to vary. That is enough to reach entanglement.

Just before switching to two spins, Susskind writes a compact action table for the Pauli matrices on the σ_3-basis states $|u\rangle$ and $|d\rangle$:

$$\sigma_1|u\rangle = |d\rangle, \qquad \sigma_1|d\rangle = |u\rangle, \tag{4.35}$$
$$\sigma_2|u\rangle = i|d\rangle, \qquad \sigma_2|d\rangle = -i|u\rangle, \tag{4.36}$$
$$\sigma_3|u\rangle = |u\rangle, \qquad \sigma_3|d\rangle = -|d\rangle. \tag{4.37}$$

This is the same content as the matrix representation, now repackaged in the language of kets. The scalar factors i and $-i$ multiply the vector from outside; they do not alter the label inside the ket.

For two electrons there are four basis states:

$$|uu\rangle, \qquad |du\rangle, \qquad |ud\rangle, \qquad |dd\rangle. \tag{4.38}$$

The first slot refers to electron 1, the second to electron 2. These four states correspond to distinct measurement outcomes of the third components of the two spins, and the lecture treats them as mutually orthogonal basis vectors.

The first electron's spin operators continue to be called $\sigma_1, \sigma_2, \sigma_3$. The second electron's are renamed τ_1, τ_2, τ_3. Their action is simple:

- σ_i acts only on the first slot and ignores the second;
- τ_i acts only on the second slot and ignores the first.

For example,

$$\sigma_1|ud\rangle = |dd\rangle, \qquad \sigma_1|dd\rangle = |ud\rangle, \qquad \sigma_3|dd\rangle = -|dd\rangle, \qquad \tau_2|ud\rangle = -i|uu\rangle. \tag{4.39}$$

That is the lecture's rule: the untouched electron simply goes along for the ride.

4.7.1 Question & Answer

Question. Why does the two-spin system require six real parameters rather than four?

Answer. Because the general two-spin state is richer than "electron one in its own state" plus "electron two in its own state."

A general state of two spins is

$$|\Psi\rangle = c_{uu}|uu\rangle + c_{du}|du\rangle + c_{ud}|ud\rangle + c_{dd}|dd\rangle, \tag{4.40}$$

with four complex coefficients. That is eight real numbers. Normalization removes one real parameter and overall phase removes another, leaving

$$8 - 1 - 1 = 6 \tag{4.41}$$

independent real parameters.

If we had merely added the two physical parameters of the first electron to the two physical parameters of the second, we would have guessed only 4. The fact that the actual answer is 6 is exactly the pressure that drives the lecture to entanglement: there are more states than product states.

Now we describe the product states first, because the lecture wants the contrast to be sharp. If electron 1 is in the state

$$\alpha_1|u\rangle + \beta_1|d\rangle, \tag{4.42}$$

and electron 2 is in the state

$$\alpha_2|u\rangle + \beta_2|d\rangle, \tag{4.43}$$

then the combined two-electron state is their product:

$$(\alpha_1|u\rangle + \beta_1|d\rangle)(\alpha_2|u\rangle + \beta_2|d\rangle). \tag{4.44}$$

Expanding it in the two-spin basis gives

$$(\alpha_1|u\rangle + \beta_1|d\rangle)(\alpha_2|u\rangle + \beta_2|d\rangle) = \alpha_1\alpha_2|uu\rangle + \alpha_1\beta_2|ud\rangle \tag{4.45}$$

$$+ \beta_1\alpha_2|du\rangle + \beta_1\beta_2|dd\rangle. \tag{4.46}$$

Such a state is a *product state*. Each electron still has its own definite polarization axis, just as in the one-spin story. Product states are the natural states of two independently prepared electrons.

The lecture then constructs a state which manifestly does *not* have that property:

$$|\psi_+\rangle = \frac{1}{\sqrt{2}}(|ud\rangle + |du\rangle). \tag{4.47}$$

This is the first entangled example in the lecture, and it is important to keep the plus sign here. Only later does Susskind remark that the minus-sign version is the better state and will matter next time.

The first property of this state is immediate. If we measure electron 2 and find it down, then electron 1 must be up. If we find electron 2 up, then electron 1 must be down. So the state carries definite anti-correlation in the 3-direction.

But the lecture does not stop with that observation. It proves something more startling: every one-spin expectation value vanishes in this state.

For σ_1,

$$\sigma_1|\psi_+\rangle = \frac{1}{\sqrt{2}}(|dd\rangle + |uu\rangle), \tag{4.48}$$

$$\langle\psi_+|\sigma_1|\psi_+\rangle = \frac{1}{2}(\langle ud| + \langle du|)(|dd\rangle + |uu\rangle) = 0. \tag{4.49}$$

No term survives because the basis states are orthogonal: the bra contains only $\langle ud|$ and $\langle du|$, while the ket contains only $|dd\rangle$ and $|uu\rangle$.

The same orthogonality argument gives

$$\langle\psi_+|\sigma_2|\psi_+\rangle = 0. \tag{4.50}$$

For σ_3, the cancellation is slightly different:

$$\sigma_3|\psi_+\rangle = \frac{1}{\sqrt{2}}(|ud\rangle - |du\rangle), \tag{4.51}$$

$$\langle\psi_+|\sigma_3|\psi_+\rangle = \frac{1}{2}(\langle ud| + \langle du|)(|ud\rangle - |du\rangle) = 0. \tag{4.52}$$

Thus

$$\langle\psi_+|\sigma_i|\psi_+\rangle = 0, \qquad i = 1, 2, 3. \tag{4.53}$$

By exactly the same reasoning,

$$\langle\psi_+|\tau_i|\psi_+\rangle = 0, \qquad i = 1, 2, 3. \tag{4.54}$$

Hence for any direction $\hat{n}$,

$$\langle\psi_+|\sigma\cdot\hat{n}|\psi_+\rangle = n_1\langle\sigma_1\rangle + n_2\langle\sigma_2\rangle + n_3\langle\sigma_3\rangle = 0, \tag{4.55}$$

and likewise for the second electron.

This is the genuinely new phenomenon. For a single pure spin state, there is always some axis along which the spin is definitely up. But for this two-electron state, every one-spin direction gives a half-half distribution. No axis is preferred for either electron by itself, even though the pair carries definite correlation information.

The lecture now asks how odd this really is. At first sight, not very. If we mix up a penny and a dime, separate them without looking, and inspect only one, then the result tells us what the other must be. So correlation by itself is not yet the deepest mystery. The penny-dime analogy is used very carefully: it shows that simple anti-correlation is not enough to distinguish the classical from the quantum. The real difference is Bell's theorem, and the lecture explicitly postpones that difference to the next time.

There is one final turn. Near the end, Susskind says that the minus-sign combination

$$\frac{1}{\sqrt{2}}(|ud\rangle - |du\rangle) \tag{4.56}$$

is the better state. He sketches a physical reason why two nearby electron magnets tend to anti-align and even names the later state as the singlet. But the reason for the minus sign is also declared to be non-obvious and is deferred. So the present chapter should end exactly where the lecture ends: with the first entangled state understood, with the better state visible on the horizon, and with Bell still waiting.

4.8 Summary

The lecture began by promising entanglement and then refusing to rush to it. That refusal was mathematically productive. We first completed the one-spin machinery: the operator $\sigma\cdot\hat{n}$, its explicit matrix form, its eigenvalues and eigenvectors, the distinction between amplitude and probability, and the formula

$$P(\sigma\cdot\hat{m} = 1 \mid \sigma\cdot\hat{n} = 1) = \frac{1 + \hat{n}\cdot\hat{m}}{2}. \tag{4.57}$$

Along the way we clarified two general principles: a measurement leaves the system in an eigenstate, and simultaneous measurability is tied to commuting operators.

Only after that could two spins reveal something new. The two-electron state space is richer than the space of simple product states, and the symmetric state $\frac{1}{\sqrt{2}}(|ud\rangle + |du\rangle)$ shows the first unmistakable sign of that richness. Each electron by itself looks completely unbiased in every direction, yet measuring one teaches us something about the other. That is the lecture's first real encounter with quantum entanglement, and it is left poised just before Bell's theorem and just before the full importance of the minus-sign singlet state.

Chapter 5

Quantum Entanglements, Part 1

This lecture begins in a very controlled way. We are not asked, at first, to define entanglement in general. We are asked something more pointed: what difference can a sign make in a two-spin state? Susskind writes down two superpositions that look almost the same, insists that they are in fact quite different, and then unfolds the answer step by step. The route matters. We first rebuild the elementary spin algebra, then discover that one-spin observables do not distinguish the states at all, then find the right observable, and only after that widen the discussion to EPR correlation, Bell's inequality, projection operators, and no-cloning. These notes follow Leonard Susskind's lecture closely, with curation by LazyingArt LLC.

5.1 Sign Choice in the Two-Spin State

Let $|u\rangle$ and $|d\rangle$ denote the eigenstates of the third component of spin. For two spins we write

$$|ud\rangle = |u\rangle_1 \otimes |d\rangle_2, \qquad |du\rangle = |d\rangle_1 \otimes |u\rangle_2,$$

and from now on we suppress the tensor-product symbol in the lecture's style.

The opening puzzle is built from the two normalized states

$$|\psi_\pm\rangle = \frac{|ud\rangle \pm |du\rangle}{\sqrt{2}}. \tag{5.1}$$

In both of them the third components are opposite. If one spin is up along the 3-axis, the other is down; if one is down, the other is up. That much is common to both states. The normalization by $\sqrt{2}$ is inserted immediately, because the discussion is about physical probabilities, not formal vectors only.

At this stage the two states sound nearly interchangeable. Both have the same up-down correlation in the third direction. But this is exactly Susskind's point: a sign that looks small in the formula can encode a large physical difference. We therefore do not begin by classifying the states abstractly; we begin by asking which calculations actually see that sign.

5.2 Recalling the Pauli Actions and the Vanishing of One-Spin Averages

Before answering the sign question, the lecture deliberately reinstalls the basic action table for the Pauli matrices. The pause is pedagogically important. The later manipulations will lean on these elementary actions over and over again.

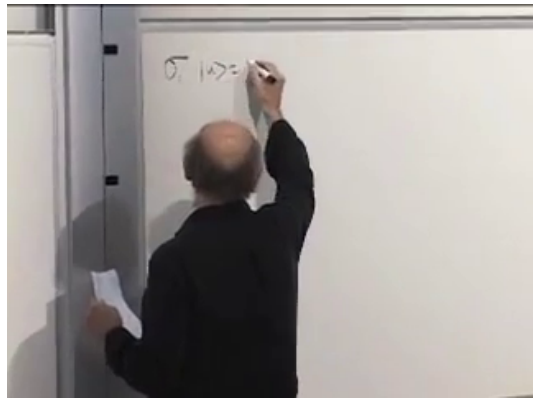


Figure 5.1: The Pauli-action table being restarted on the board from the upper left corner.

We use

$$\sigma_1|u\rangle = |d\rangle, \qquad \sigma_1|d\rangle = |u\rangle, \tag{5.2}$$

$$\sigma_2|u\rangle = i|d\rangle, \qquad \sigma_2|d\rangle = -i|u\rangle, \tag{5.3}$$

$$\sigma_3|u\rangle = |u\rangle, \qquad \sigma_3|d\rangle = -|d\rangle. \tag{5.4}$$

For the second electron we introduce another family τ_1, τ_2, τ_3, identical in algebra but acting on the second slot instead of the first.

Now comes the first important claim. In either of the states $|\psi_+\rangle$ or $|\psi_-\rangle$, every one-spin expectation value vanishes:

$$\langle\sigma_i\rangle = 0, \qquad \langle\tau_i\rangle = 0. \tag{5.5}$$

So there is no direction in which either constituent spin is definitely polarized. This is already one of the characteristic signatures of the entangled state: whatever the total state may know, neither spin by itself carries a preferred direction.

A worked check. Susskind chooses the plus state for the explicit verification and checks σ_3. We write

$$\langle\psi_+|\sigma_3|\psi_+\rangle = \frac{1}{2}(\langle ud| + \langle du|)\sigma_3(|ud\rangle + |du\rangle). \tag{5.6}$$

Since σ_3 acts on the first spin,

$$\sigma_3|ud\rangle = |ud\rangle, \qquad \sigma_3|du\rangle = -|du\rangle.$$

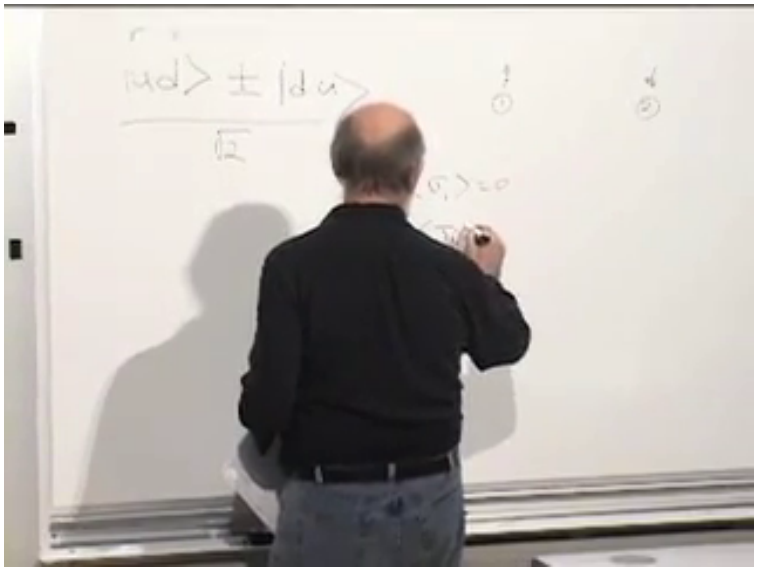


Figure 5.2: The board layout for the entangled state and the claim that the one-spin expectation vanishes. The sign in the superposition is not fixed by the frame alone; the transcript shows that the explicit check here is done for the plus state.

Therefore

$$\langle\psi_+|\sigma_3|\psi_+\rangle = \frac{1}{2}(\langle ud| + \langle du|)(|ud\rangle - |du\rangle) \tag{5.7}$$

$$= \frac{1}{2}(\langle ud|ud\rangle - \langle du|du\rangle) \tag{5.8}$$

$$= \frac{1}{2}(1-1) \tag{5.9}$$

$$= 0. \tag{5.10}$$

The cross terms vanish because $\langle ud|du\rangle = \langle du|ud\rangle = 0$. This is exactly the sort of explicit check the lecture insists on before moving on.

5.2.1 Question & Answer

Question. Does $\langle\sigma_i\rangle = 0$ mean that the spin has eigenvalue 0?

Answer. No. For a single spin component the only possible outcomes are ± 1. There is no eigenstate of σ_i with eigenvalue 0. What vanishes is the expectation value. A measurement of one spin component in these states is a fair coin toss: up and down occur with equal probability.

That warning is not incidental. The lecture is already training us not to confuse an average statement with an outcome statement. If we flatten that distinction away, we lose the logic of what comes next.

5.3 Total Spin, Singlet, and EPR Correlation

Having shown that one-spin observables do not distinguish the two states, Susskind changes the question. We should not look at one little magnetic moment at a time; we should look at the sum of the two. The natural operators are

$$\sigma_1 + \tau_1, \qquad \sigma_2 + \tau_2, \qquad \sigma_3 + \tau_3.$$

We begin with the third component. On the two basis states,

$$(\sigma_3 + \tau_3)|ud\rangle = (1-1)|ud\rangle = 0, \tag{5.11}$$
$$(\sigma_3 + \tau_3)|du\rangle = (-1+1)|du\rangle = 0. \tag{5.12}$$

Therefore

$$(\sigma_3 + \tau_3)(|ud\rangle \pm |du\rangle) = 0. \tag{5.13}$$

Both sign choices are eigenvectors of $\sigma_3 + \tau_3$ with eigenvalue 0. So the third component of the total spin still does not distinguish them.

The real difference appears when we move to the first component. Using the action table,

$$(\sigma_1 + \tau_1)|ud\rangle = |dd\rangle + |uu\rangle, \tag{5.14}$$
$$(\sigma_1 + \tau_1)|du\rangle = |uu\rangle + |dd\rangle. \tag{5.15}$$

Hence

$$(\sigma_1 + \tau_1)(|ud\rangle + |du\rangle) = 2|uu\rangle + 2|dd\rangle, \tag{5.16}$$

while

$$(\sigma_1 + \tau_1)(|ud\rangle - |du\rangle) = 0. \tag{5.17}$$

The plus state is not an eigenvector of $\sigma_1 + \tau_1$ with eigenvalue zero. The minus state is.

One should pause over that. The expectation value of the total first component may still vanish in the plus state, but the state is not annihilated by the operator. Consequently a measurement of the total first component need not return zero every time. Since each individual measurement gives ± 1, the sum can yield 2, 0, or -2.

Now check the second component. Using

$$\sigma_2|u\rangle = i|d\rangle, \qquad \sigma_2|d\rangle = -i|u\rangle,$$

and the same for τ_2, we find

$$(\sigma_2 + \tau_2)(|ud\rangle - |du\rangle) = 0. \tag{5.18}$$

Thus the minus-sign state is annihilated by all three components of the total spin. In compact notation,

$$(\boldsymbol{\sigma} + \boldsymbol{\tau}) \cdot \hat{n}\, |\psi_-\rangle = 0 \qquad \text{for every unit vector } \hat{n}. \tag{5.19}$$

This is the special state of the lecture. The minus-sign state is the singlet. The plus-sign combination in this sector is the triplet. The singlet is rotationally invariant in the precise sense relevant here: no matter which component of the total spin we measure, the result is always zero.

The EPR correlation now falls straight out of the calculation. If Bob measures a component of one spin, he knows what Alice will get if she measures the same component of the other spin, because the sum of the two answers must vanish.

5.3.1 Question & Answer

Question. Is that statement only true on average?

Answer. No. It is true every single time for the singlet, because the singlet is an eigenvector of the total spin component in every direction, with eigenvalue zero. The certainty comes from the eigenvector-eigenvalue relation, not merely from a statistical average.

5.3.2 Question & Answer

Question. Do these correlations violate causality, or allow signalling faster than light?

Answer. No. Bob's measurement gives Bob information about what Alice would get if she measures the same component, but Alice's local statistics do not change. She still sees a fifty-fifty distribution. Only after Bob sends an ordinary classical message can she combine her result with his and recover the correlation. The correlation is immediate; the usable information is not.

This is why the lecture treats EPR correlation as profound but not as a direct violation of relativistic causality. It is deeper than the simple penny-and-dime example, because it works for every direction, but it does not by itself send messages.

5.4 Bell's Inequality as a Classical Statement About Sets

Only after the singlet and its correlation have been established does the lecture pivot to Bell. The pace changes intentionally. Susskind says he wants to go slowly, and the reason is clear: Bell's inequality must first be understood as an entirely classical theorem.

One may feel, as Susskind says Bell and Feynman sometimes did, that the theorem is either profoundly deep or almost trivial. But whatever attitude one takes toward its philosophy, the mathematical statement itself is classical. Classical logic here means set theory.

Suppose we have a set of objects, each of which may or may not possess three properties A, B, and C. Then Bell's inequality is the counting statement

$$N(A, \neg B) + N(B, \neg C) \geq N(A, \neg C). \tag{5.20}$$

Here $N(A, \neg B)$ means the number of objects that have property A and do not have property B, and similarly for the other terms.

Susskind wants a picture, not a ceremonial proof. We should see the theorem in the geometry of subsets.

One clean way to express the content is

$$A \cap \neg C \subset (A \cap \neg B) \cup (B \cap \neg C).$$

Indeed, if an object lies in $A \cap \neg C$, then either it has B or it does not. If it does not have B, it lies in $A \cap \neg B$. If it does have B, then together with $\neg C$ it lies in $B \cap \neg C$. Taking cardinalities gives the inequality.

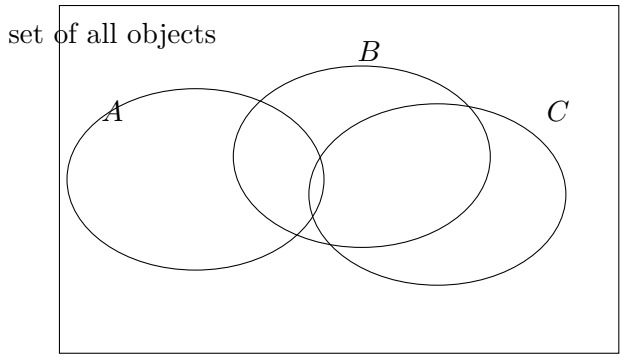


Figure 5.3: A classical picture of the properties A, B, and C as subsets of a set.

This is classical from start to finish. No noncommuting observables, no amplitudes, no Hilbert space. The lecture is careful about this because the later violation should be seen as a failure of classical set-theoretic reasoning in the quantum context, not as a cheat in the derivation.

5.5 From Spin Propositions to Projection Operators

We now specialize the abstract properties A, B, and C to propositions about the singlet pair. Following the lecture, we define

$$A : \text{spin 1 is up at } 0^\circ, \text{ that is, along the 3-axis,} \tag{5.21}$$

$$B : \text{spin 1 is up at } 45^\circ \text{ in the } zx\text{-plane,} \tag{5.22}$$

$$C : \text{spin 1 is up at } 90^\circ \text{ in the same plane.} \tag{5.23}$$

The transcript hesitates briefly about the plane, but the subsequent algebra fixes it: the intended plane is the zx-plane.

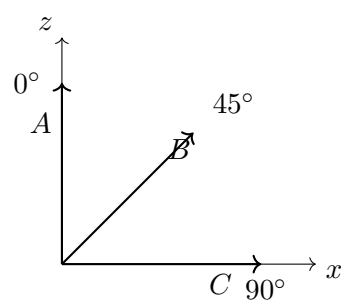


Figure 5.4: The three directions used in the Bell discussion.

Because the state is the singlet, the negation of a proposition about spin 1 becomes a positive proposition about spin 2. Thus $\neg B$ may be read as: spin 2 is up at 45°. Likewise $\neg C$ becomes: spin 2 is up at 90°.

At this point the lecture asks the practical question that forces the next construction: how do we actually compute the relevant probabilities? This is the moment where projection operators enter.

The generic board equation is only partly legible, so we write it in cautious standard form:

$$P|\alpha\rangle = |\alpha_p\rangle. \tag{5.24}$$

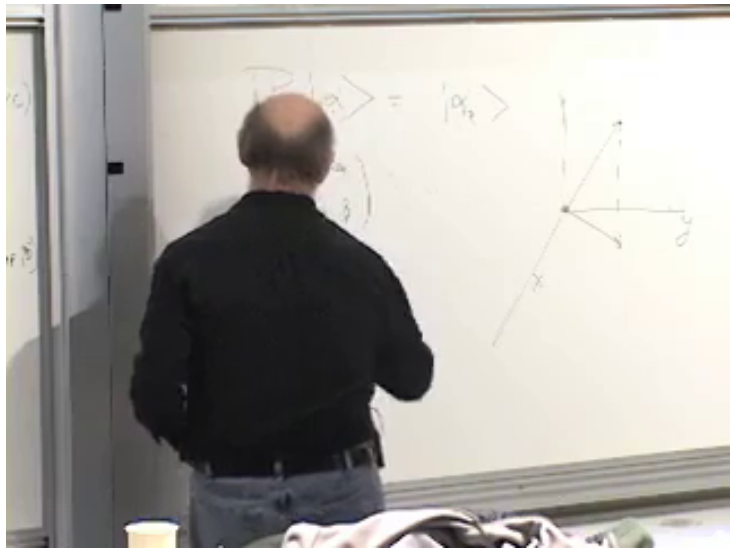


Figure 5.5: The lecture's geometric and algebraic introduction to projection operators.

The meaning is plain from the diagram: P projects the vector $|\alpha\rangle$ onto a chosen subspace and removes the component orthogonal to that subspace.

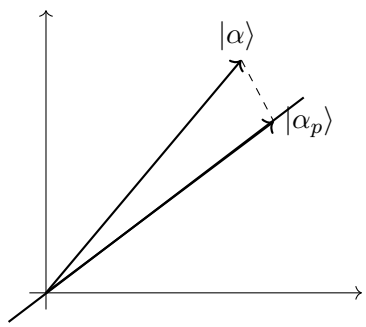


Figure 5.6: A clean redraw of the projection geometry used in the lecture.

Susskind then specializes to the one-spin space. If we project onto the subspace where the third component is $+1$, then the down component is simply discarded:

$$P_{\sigma_3=+1}\begin{pmatrix}\alpha\\ \beta\end{pmatrix}=\begin{pmatrix}\alpha\\ 0\end{pmatrix},\qquad P_{\sigma_3=-1}\begin{pmatrix}\alpha\\ \beta\end{pmatrix}=\begin{pmatrix}0\\ \beta\end{pmatrix}. \tag{5.25}$$

The operator itself is

$$P_{\sigma_3=+1}=\begin{pmatrix}1 & 0\\ 0 & 0\end{pmatrix}=\frac{1+\sigma_3}{2}. \tag{5.26}$$

And more generally, for any unit vector $\hat{n}$,

$$P_{\sigma\cdot\hat{n}=+1}=\frac{1+\sigma\cdot\hat{n}}{2}. \tag{5.27}$$

The probability postulate can now be stated in the form used by the lecture:

$$\text{Prob}(\alpha \text{ true in the state } |\psi\rangle)=\langle\psi|P_\alpha|\psi\rangle. \tag{5.28}$$

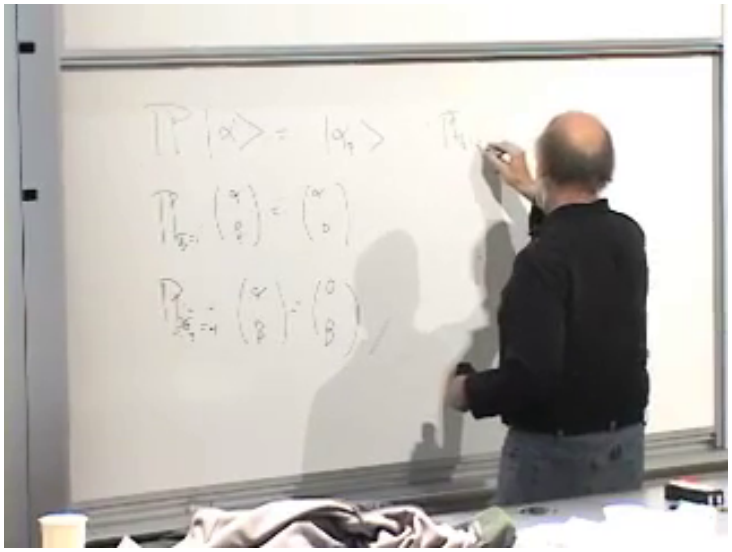


Figure 5.7: Projection onto the $\sigma_3 = \pm 1$ subspaces and the move toward matrix form.

For the Bell quantity $A \wedge \neg B$, the first statement is "spin 1 is up at 0°," and the second, after the singlet reinterpretation, is "spin 2 is up at 45°." Therefore

$$P_{A\wedge\neg B} = \frac{1+\sigma_3}{2}\,\frac{1+\tau\cdot\hat{n}_{45}}{2}, \qquad \hat{n}_{45} = \frac{\hat{x}+\hat{z}}{\sqrt{2}}, \qquad \tau\cdot\hat{n}_{45} = \frac{\tau_1+\tau_3}{\sqrt{2}}. \tag{5.29}$$

5.5.1 Question & Answer

Question. Why does the product of projectors represent the logical word "and"?

Answer. Only when the propositions are compatible, meaning that their projectors commute. For one electron, the statements "spin up along the 3-axis" and "spin up along the 1-axis" are not jointly meaningful in this way, because the corresponding projectors do not commute. Here, however, one projector acts on spin 1 and the other on spin 2. The σ's commute with the τ's, so the product does define the conjunction.

This clarification comes late in the lecture, but it is structurally important. Without it the Bell computation looks more classical than it really is.

5.6 Quantum Violation of Bell's Inequality

Now the lecture turns the machinery on. Let

$$|\psi_s\rangle = \frac{|ud\rangle - |du\rangle}{\sqrt{2}} \tag{5.30}$$

be the singlet state.

Susskind first notes the qualitative result: the quantum value of the left-hand side of Bell's inequality will come out smaller than the right-hand side. Then he says the only thing to do is to hold one's breath and compute.

The probability $P(A, \neg B)$. We evaluate

$$P(A, \neg B) = \langle \psi_s | P_{A \wedge \neg B} | \psi_s \rangle.$$

Act first with $(1 + \sigma_3)/2$. This projects onto the component where the first spin is up, so it kills the $|du\rangle$ term and leaves

$$\frac{1}{\sqrt{2}} |ud\rangle.$$

Now act with the second projector:

$$\frac{1 + \tau \cdot \hat{n}_{45}}{2} |ud\rangle = \frac{1}{2} \left(|ud\rangle + \frac{\tau_1 + \tau_3}{\sqrt{2}} |ud\rangle \right) \tag{5.31}$$

$$= \frac{1}{2} \left(|ud\rangle + \frac{|uu\rangle - |ud\rangle}{\sqrt{2}} \right). \tag{5.32}$$

So

$$\frac{1 + \tau \cdot \hat{n}_{45}}{2} |ud\rangle = \frac{1}{2} \left(\left(1 - \frac{1}{\sqrt{2}} \right) |ud\rangle + \frac{1}{\sqrt{2}} |uu\rangle \right). \tag{5.33}$$

When we now take the inner product with the singlet bra, the $|uu\rangle$ term does not contribute, because it is orthogonal to both $\langle ud|$ and $\langle du|$. The surviving piece is therefore

$$P(A, \neg B) = \frac{1}{4} \left(1 - \frac{1}{\sqrt{2}} \right). \tag{5.34}$$

The lecture does not recompute $P(B, \neg C)$ from scratch. Instead it uses rotational invariance of the singlet. The two situations differ only by a 45° rotation, so

$$P(B, \neg C) = \frac{1}{4} \left(1 - \frac{1}{\sqrt{2}} \right). \tag{5.35}$$

The probability $P(A, \neg C)$. Now $\neg C$ becomes the statement that spin 2 is up at 90°, that is, along the 1-axis. Thus the projector is

$$\frac{1 + \sigma_3}{2} \frac{1 + \tau_1}{2}.$$

As before, the first factor kills $|du\rangle$ and leaves $|ud\rangle/\sqrt{2}$. Then

$$\frac{1 + \tau_1}{2} |ud\rangle = \frac{1}{2} (|ud\rangle + |uu\rangle). \tag{5.36}$$

The $|uu\rangle$ term again has zero overlap with the singlet bra, so

$$P(A, \neg C) = \frac{1}{4}. \tag{5.37}$$

Putting the three numbers together,

$$P(A, \neg B) + P(B, \neg C) = \frac{1}{2} \left(1 - \frac{1}{\sqrt{2}} \right) < \frac{1}{4} = P(A, \neg C). \tag{5.38}$$

Numerically,

$$\frac{1}{2} \left(1 - \frac{1}{\sqrt{2}} \right) \approx 0.146, \qquad \frac{1}{4} = 0.25.$$

The decimal estimates in the lecture are informal, but the exact algebraic comparison is clean.

What is the meaning of the violation? The lecture's answer is blunt: there is no underlying classical description in which quantum propositions are governed by ordinary set theory. Quantum mechanics is not the statistical mechanics of some unseen classical system whose properties obey classical logic all the way down.

This is where Aspect's experiment enters. Susskind is careful about what it did and did not establish. It did not decide whether quantum mechanics works in general; that was already known. What it sharpened was the locality issue. By arranging the measurements so that no light-speed signal could pass from one wing of the experiment to the other in time, it blocked the obvious classical escape route in which hidden information travels from left to right. The achievement was as much one of timing and experimental control as of principle.

5.7 Linearity and the No-Cloning Theorem

The final turn of the lecture is abrupt but natural. After Bell and the discussion of compatible projectors, Susskind decides to postpone the measurement problem and instead prove a short theorem. The new input is the linearity of quantum evolution:

$$|a\rangle \to |a'\rangle, \qquad |b\rangle \to |b'\rangle \quad \Longrightarrow \quad \alpha|a\rangle + \beta|b\rangle \to \alpha|a'\rangle + \beta|b'\rangle. \tag{5.39}$$

Superpositions evolve into superpositions with the same coefficients.

Now imagine a machine that clones quantum states. At minimum it would have to work on the u/d basis:

$$|u\rangle \to |uu\rangle, \qquad |d\rangle \to |dd\rangle. \tag{5.40}$$

Next define the state pointing to the right,

$$|r\rangle = \frac{|u\rangle + |d\rangle}{\sqrt{2}}. \tag{5.41}$$

If the same machine were universal, then by linearity its output on $|r\rangle$ would be forced to be

$$|r\rangle \to \frac{|uu\rangle + |dd\rangle}{\sqrt{2}}. \tag{5.42}$$

But exact cloning would require two independent copies of $|r\rangle$:

$$|rr\rangle = |r\rangle \otimes |r\rangle \tag{5.43}$$

$$= \frac{1}{2}(|uu\rangle + |ud\rangle + |du\rangle + |dd\rangle). \tag{5.44}$$

These are different states. The linear output has only the two parallel terms $|uu\rangle$ and $|dd\rangle$. The genuinely cloned state necessarily contains the mixed terms $|ud\rangle$ and $|du\rangle$ as well.

Theorem 5.1 (No-cloning)**.** *A single machine cannot clone an arbitrary unknown quantum state.*

Proof. Assume one machine clones $|u\rangle$ and $|d\rangle$, so that

$$|u\rangle \to |uu\rangle, \qquad |d\rangle \to |dd\rangle.$$

By linearity, the same machine must send

$$\frac{|u\rangle + |d\rangle}{\sqrt{2}} \to \frac{|uu\rangle + |dd\rangle}{\sqrt{2}}.$$

But exact cloning of the input state would require

$$\left(\frac{|u\rangle + |d\rangle}{\sqrt{2}}\right) \otimes \left(\frac{|u\rangle + |d\rangle}{\sqrt{2}}\right) = \frac{1}{2}(|uu\rangle + |ud\rangle + |du\rangle + |dd\rangle).$$

Since these outputs are different, the supposed universal cloning machine cannot exist. □

The lecture does not state the moral sloppily. If the state is known in advance, one can design a device adapted to that state or to a chosen basis. What cannot be done is to build a single machine that clones everything. Cloning is effectively quadratic in the state, while quantum evolution is linear. That mismatch is the whole theorem in a nutshell.

5.8 Summary

The lecture opens with a sign and spends its first half earning the right to say why the sign matters. For the states

$$\frac{|ud\rangle \pm |du\rangle}{\sqrt{2}},$$

every one-spin expectation value vanishes, so no difference appears at the level of either spin separately. The distinction emerges only when we look at the total spin. The minus-sign state is the singlet, annihilated by every component of $\sigma + \tau$, and therefore perfectly anti-correlated in every direction.

From there the lecture broadens without losing the thread. Bell's inequality is introduced as a classical counting theorem, then translated into the singlet experiment, and then violated by an explicit projector calculation. The meaning of that violation is not that quantum mechanics is inconsistent, but that its propositions do not live inside ordinary set theory. The closing theorem makes the same Hilbert-space structure felt in another way: because quantum evolution is linear, no universal machine can clone an arbitrary unknown state. The lecture has unfolded one theme through several faces. A sign, a correlation, a violated inequality, and a no-go theorem are all part of the same quantum geometry.

Chapter 6

Subspaces, Projection Operators, and Measurement as Entanglement

This chapter follows Leonard Susskind's sixth lecture in the Quantum Entanglement series, curated by LazyingArt LLC. The order matters. The lecture begins in the afterpressure of Bell's theorem, where locality and entanglement are still the live issue in the room; then it deliberately backs up and repairs the mathematics of subspaces and projection operators; only after that does it return to the promised subject, namely the relation among measurement, collapse of the wave packet, and entanglement. The payoff is not a slogan but a mechanism: a measurement is the establishment of entanglement between a system and an apparatus, and it is precisely the recording of that information that destroys interference.

6.1 Opening Motivation: Entanglement, Locality, and Measurement

The lecture does not open with a neat new beginning. It opens with a question left over from the Bell discussion: if one member of an entangled pair is measured, is there some exchange, some message, some detectable event at the other side? Susskind's answer is operational and uncompromising. Nothing detectable happens at the far side. No usable signal arrives there. One may act on one member of an entangled pair, even rotate its spin locally with a magnetic field, and the detailed form of the joint state may change, but the distant system remains entangled with whatever system lives here.

That already forces a broader notion of what counts as the system. If one member of an entangled pair falls into a black hole, the other does not cease to be entangled; rather, it becomes entangled with the larger system containing the black hole degrees of freedom and, later, the emitted radiation. Susskind immediately domesticates the same point with the bathtub example. Drop one member of an entangled pair into a tub and let it mix violently with the water. The distant particle is no longer neatly entangled with one identifiable partner. It is entangled with the state of the bathtub.

This is not only a story about persistence. It is also a story about reversibility. If interaction can create entanglement, then interaction can, in principle, undo it. Disentanglement is possible, but only when the systems come back into contact and interact again. Anything that can happen one way can happen the other way.

The room then asks exactly the question that drives the rest of the lecture. Suppose the two particles are entangled and we measure them. Are they still entangled afterwards, or does measurement kick them out? Susskind answers that measurement does kick them out, or at least can. But he refuses to answer too quickly. Instead he states the real target of the evening: the relation among observation, collapse of the wave packet, and entanglement. Then he says, in effect, that before we can get there he must first clean up the mathematics of projection operators and subspaces.

6.2 Dimension, Linear Dependence, and Orthonormal Bases

We therefore stay, for the moment, with finite-dimensional vector spaces equipped with bras, kets, and an inner product. Susskind begins as elementarily as possible. He does not define dimension by counting components in a column vector. He defines it abstractly: the dimension of a vector space is the maximum number of linearly independent vectors that can be found in it.

Thus a collection $\{|v_i\rangle\}$ is linearly independent when the relation

$$\sum_i c_i|v_i\rangle = 0 \tag{6.1}$$

forces all the coefficients c_i to vanish. Equivalently, none of the vectors can be written as a linear combination of the others.

It is crucial here that orthogonality is not the same thing as independence. In three dimensions one may have three vectors that are not orthogonal at all and yet are linearly independent. What matters is whether one of them can be built from the other two. On the other hand, in a two-dimensional plane any third vector in that plane must be linearly dependent on the first two.

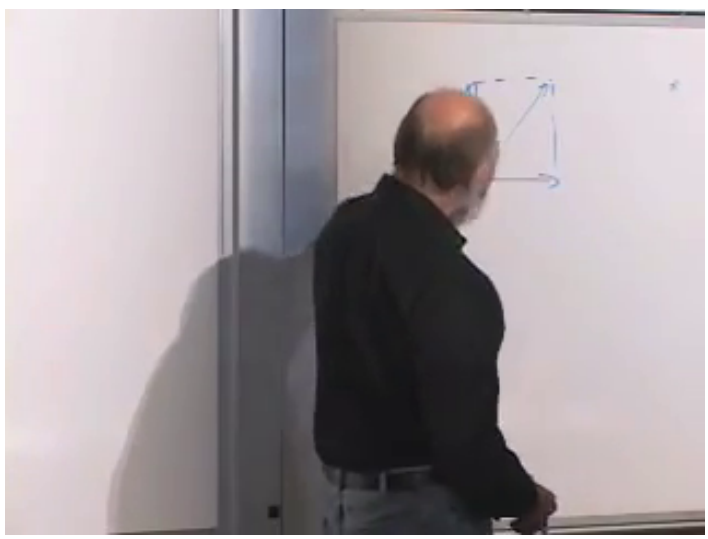


Figure 6.1: Three vectors drawn on the board. The lecture uses this sketch to separate the idea of orthogonality from the idea of linear independence, and then to emphasize that in a plane a third vector is dependent on the first two.

This is the point of the sketch. If the space is two-dimensional, the maximum number of linearly independent vectors is two. If the space is three-dimensional, it is three. More generally, if the space has dimension d, then, given an inner product, one can find d mutually orthogonal unit vectors

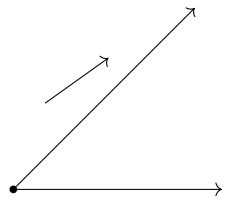


Figure 6.2: A minimal reconstruction of the board sketch. The exact coefficient relation is not fixed by the frame alone; the geometric point is that in a two-dimensional plane a third vector is not independent.

spanning it. Those are basis vectors. Susskind explicitly says he will not prove that theorem here. He only wants the conclusion: once such a basis exists, every vector in the space can be expanded in it.

6.2.1 Question & Answer: Are spatial degrees of freedom the same thing as the dimension of the state space?

No. A particle moving in ordinary three-dimensional space has three spatial degrees of freedom in the mechanical sense, but that fact does not determine the dimension of its space of states. Those are different notions. A particle moving in a room has three spatial directions available to it, but an enormous number of orthogonal possible states. The lecture stays in finite-dimensional toy models only because that is the cleanest place to see the logic of subspaces and projectors.

6.3 Basis Expansion, Dyads, and the Resolution of Identity

Now choose an orthonormal basis $\{|n\rangle\}$, with $n = 1, \ldots, d$. The blackboard at this point is especially clear and compact:

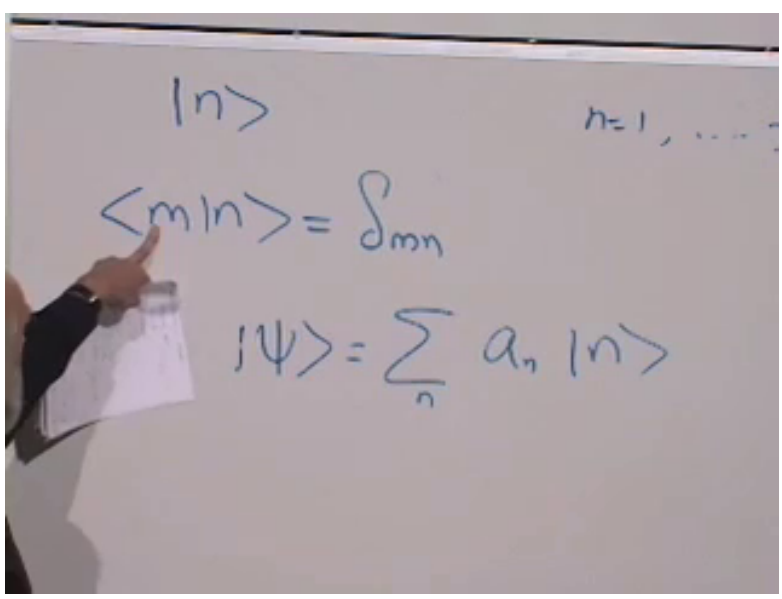


Figure 6.3: Orthonormal basis and expansion of an arbitrary state.

We write

$$\langle m|n\rangle = \delta_{mn}, \tag{6.2}$$

$$|\psi\rangle = \sum_n a_n|n\rangle. \tag{6.3}$$

Here δ_{mn} is the Kronecker delta: it is 1 when $m = n$ and 0 otherwise.

Worked derivation. The coefficients are recovered by projecting onto the basis. Multiply by $\langle m|$ on the left:

$$\langle m|\psi\rangle = \sum_n a_n\langle m|n\rangle \tag{6.4}$$

$$= \sum_n a_n\delta_{mn} \tag{6.5}$$

$$= a_m. \tag{6.6}$$

So the coefficients are simply

$$a_m = \langle m|\psi\rangle. \tag{6.7}$$

At this point the lecture pauses to note that the coefficient is just an ordinary complex number, so one may place it on either side of the ket without changing anything.

If we substitute the coefficient back into the expansion, we get

$$|\psi\rangle = \sum_n |n\rangle\langle n|\psi\rangle. \tag{6.8}$$

This suggests that the object $|n\rangle\langle n|$, and more generally $|a\rangle\langle b|$, ought to be read as an operator. That is exactly the next move. A dyad acts by

$$\big(|a\rangle\langle b|\big)|\psi\rangle = \langle b|\psi\rangle\,|a\rangle. \tag{6.9}$$

The bra extracts a complex number; the ket supplies the resulting direction. This is Susskind's low-tech route from vectors to operators.

Now read the previous expansion operatorially. Since

$$|\psi\rangle = \left(\sum_n |n\rangle\langle n|\right)|\psi\rangle \tag{6.10}$$

for every $|\psi\rangle$, the operator in parentheses must be the identity:

$$\sum_n |n\rangle\langle n| = I. \tag{6.11}$$

This is the resolution of the identity.

The lecture briefly lingers on a useful observation. The identity operator is itself a projection operator, namely the projection onto the whole space. Every vector is an eigenvector of I, and every eigenvalue is $+1$. That is not a deep theorem; it is just a clean way of seeing what the identity operator means.

6.4 Properties as Subspaces and Projection Operators

With that machinery in hand, the lecture makes its central conceptual correction. A property such as $k = \lambda$ is not, in general, one vector. It is a whole subspace.

Let k be a Hermitian observable and suppose

$$k|A\rangle = \lambda|A\rangle, \qquad k|B\rangle = \lambda|B\rangle, \tag{6.12}$$

with $|A\rangle$ and $|B\rangle$ linearly independent. Then any linear combination has the same eigenvalue:

$$\begin{aligned} k\big(\alpha|A\rangle + \beta|B\rangle\big) &= \alpha k|A\rangle + \beta k|B\rangle & (6.13)\\ &= \alpha\lambda|A\rangle + \beta\lambda|B\rangle & (6.14)\\ &= \lambda\big(\alpha|A\rangle + \beta|B\rangle\big). & (6.15) \end{aligned}$$

So the eigenspace for λ is closed under addition and scalar multiplication. It is a vector subspace.

The lecture's first concrete example is a two-spin system. The states

$$|ud\rangle, \qquad |uu\rangle \tag{6.16}$$

have different values of the second spin, but the same value of the first: the first spin is up in both cases. So they span the subspace in which $\sigma_3 = +1$ for the first spin.

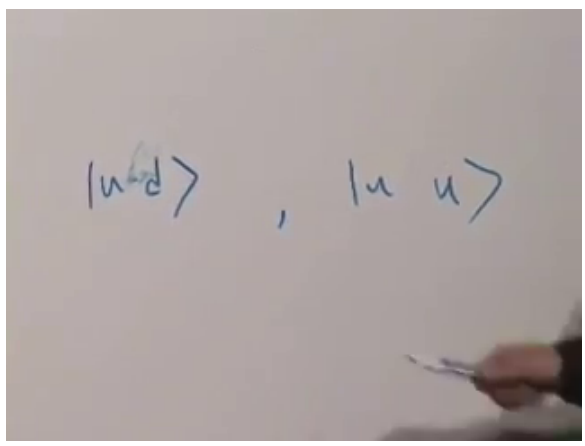


Figure 6.4: Two two-spin states with first spin up.

That is the point worth preserving: a property is not necessarily one state. It is a whole collection of states with the same eigenvalue.

To avoid the notational confusion that appears briefly on the blackboard, we shall keep capital letters $A, B, \ldots$ for particular eigenvectors and use lower-case $a, b, \ldots$ as summation indices running over an orthonormal basis of the eigensubspace. If $\{|a\rangle\}$ is such a basis for the subspace $k = \lambda$, then the projection operator onto that subspace is

$$\Pi_{k=\lambda} = \sum_a |a\rangle\langle a|. \tag{6.17}$$

This is just the identity operator restricted to the subspace. If $|\psi\rangle$ already lies in that subspace, the projector returns it unchanged. If $|\psi\rangle$ lies partly outside, the projector discards the orthogonal remainder and keeps only the component inside.

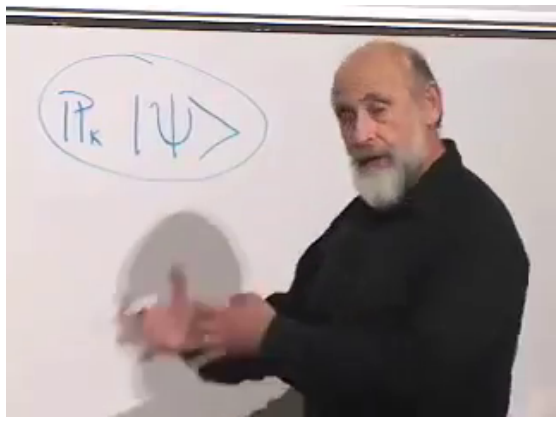


Figure 6.5: The projected state as a vector in the eigensubspace.

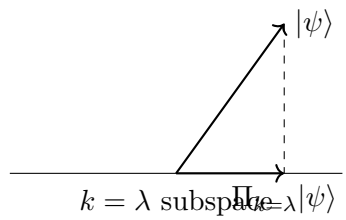


Figure 6.6: A geometric reconstruction of the lecture's point: $\Pi_{k=\lambda}|\psi\rangle$ lies in the $k = \lambda$ eigensubspace.

The blackboard emphasis is not only geometric. Algebraically, the projected vector satisfies

$$k(\Pi_{k=\lambda}|\psi\rangle) = \lambda\, \Pi_{k=\lambda}|\psi\rangle. \tag{6.18}$$

So $\Pi_{k=\lambda}|\psi\rangle$ really is a vector with the property $k = \lambda$. That is what the circled board expression is trying to make visually undeniable.

6.5 Probability Postulate, Compatible Properties, and the Bell Recap

Once properties have been reinterpreted as subspaces, the probability postulate takes its cleanest form. If $|\psi\rangle$ is normalized, then the probability that it has the property $k = \lambda$ is

$$P(k = \lambda) = \langle\psi|\Pi_{k=\lambda}|\psi\rangle. \tag{6.19}$$

Insert the dyadic form of the projector:

$$P(k = \lambda) = \left\langle \psi \left| \sum_a |a\rangle\langle a| \right| \psi \right\rangle \tag{6.20}$$

$$= \sum_a \langle\psi|a\rangle\langle a|\psi\rangle \tag{6.21}$$

$$= \sum_a |\langle a|\psi\rangle|^2. \tag{6.22}$$

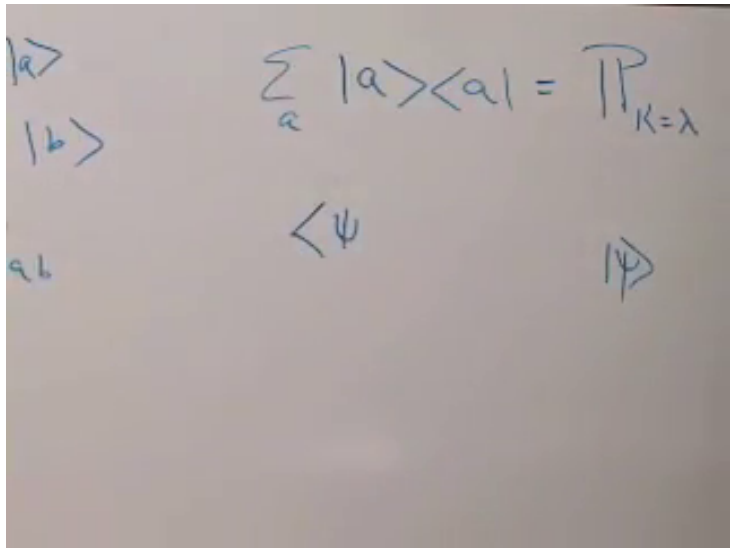


Figure 6.7: Projector onto the $k = \lambda$ eigensubspace. The blackboard layout already prepares the next move: sandwich the projector between $\langle\psi|$ and $|\psi\rangle$.

This is exactly the lecture's interpretation: the total probability for the property is the sum of the probabilities of all the orthogonal ways in which the property can occur.

For the first spin being up in a two-spin system, the relevant subspace is spanned by $|uu\rangle$ and $|ud\rangle$, so

$$\Pi_{\sigma_3=+1} = |uu\rangle\langle uu| + |ud\rangle\langle ud|. \tag{6.23}$$

In the lecture's shorthand this is also written as

$$\Pi_{\sigma_3=+1} = \frac{1+\sigma_3}{2}, \tag{6.24}$$

with the identity on the second spin suppressed.

6.5.1 Question & Answer: What is the difference between an observable, a property, and a projector?

The observable is the Hermitian operator k. The property is the statement $k = \lambda$, namely the statement that a measurement of k would certainly yield the value λ. That property is represented not by a single vector in general, but by the eigensubspace for λ. The projector $\Pi_{k=\lambda}$ is then the operator that isolates that subspace, and its expectation value in the state $|\psi\rangle$ gives the probability that the property is present.

The lecture then asks how to combine properties. Suppose we have two compatible properties, represented by commuting projectors P_k and P_ℓ. In the lecture's language, the AND statement is represented by the product:

$$\Pi_{k=\lambda \wedge \ell=\mu} = P_k P_\ell = P_\ell P_k. \tag{6.25}$$

The logic is simple. $P_k|\psi\rangle$ lies in the $k = \lambda$ subspace. Because the operators commute, $P_\ell P_k|\psi\rangle = P_k P_\ell|\psi\rangle$, so the same vector lies in the $\ell = \mu$ subspace as well. Thus the product projects onto the simultaneous property. The corresponding probability is

$$P(k = \lambda, \ell = \mu) = \langle\psi|P_k P_\ell|\psi\rangle. \tag{6.26}$$

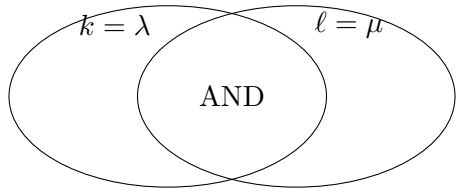


Figure 6.8: A reconstruction of the intersecting-subspace picture behind the AND statement for commuting projectors.

Remark 6.1. The lecture also treats the OR statement, in this same compatible setting, as the sum of the projectors. We keep that claim in precisely this local context and do not silently upgrade it into a more general theorem.

Compatibility itself is then tied back to physical examples. In a two-spin system, any component σ_i of the first spin commutes with any component τ_j of the second:

$$[\sigma_i, \tau_j] = 0. \tag{6.27}$$

By contrast, different components of the same spin do not commute:

$$[\sigma_1, \sigma_3] \neq 0. \tag{6.28}$$

That is why one may measure one component of each of two separated spins, but not simultaneously measure two different components of the same spin.

Now Susskind revisits Bell's theorem. He does not redo last lecture's arithmetic. He rebuilds its logic in the new language. The classical inequality, in probabilistic form, is

$$P(A \wedge \neg B) + P(B \wedge \neg C) \geq P(A \wedge \neg C). \tag{6.29}$$

This is a theorem of classical set logic, not of quantum mechanics. In the singlet state, spins are anti-aligned along any common axis, so a NOT statement for the first spin can be traded for the corresponding positive statement for the second. That lets one rewrite the inequality as a statement about joint properties of a two-spin system.

Suppressing identity operators on untouched factors, the single-property projectors appearing in the lecture are

$$\frac{1+\sigma_3}{2}, \qquad \frac{1+\frac{\tau_1+\tau_3}{\sqrt{2}}}{2}, \qquad \frac{1+\frac{\sigma_1+\sigma_3}{\sqrt{2}}}{2}, \qquad \frac{1+\tau_1}{2}. \tag{6.30}$$

Therefore the three relevant joint projectors are

$$\Pi_1 = \frac{1+\sigma_3}{2}\,\frac{1+\frac{\tau_1+\tau_3}{\sqrt{2}}}{2}, \tag{6.31}$$

$$\Pi_2 = \frac{1+\frac{\sigma_1+\sigma_3}{\sqrt{2}}}{2}\,\frac{1+\tau_1}{2}, \tag{6.32}$$

$$\Pi_3 = \frac{1+\sigma_3}{2}\,\frac{1+\tau_1}{2}. \tag{6.33}$$

The lecture speaks of the singlet in the familiar unnormalized blackboard form

$$|\Psi_{\text{singlet}}\rangle \propto |ud\rangle - |du\rangle. \tag{6.34}$$

The proportionality sign is enough here. What matters for the logic is the state itself and, in particular, the relative sign, not the overall normalization. Evaluating $\langle \Pi_1 \rangle + \langle \Pi_2 \rangle$ against $\langle \Pi_3 \rangle$ in the singlet state yields the violation obtained in the previous lecture. The important conclusion is the one Susskind emphasizes verbally: quantum mechanics is not classical set theory in disguise.

6.6 Locality After Bell

That Bell recap leads straight into the physical unease still hanging over the room. If the correlations are real, why can they not be used to send a signal?

6.6.1 Question & Answer: Why can't entanglement be used to send a faster-than-light signal?

The answer remains stubbornly operational. If the second measurement is performed only after enough time has elapsed for an ordinary light signal to travel from the first apparatus to the second, then there is no contradiction with causality. One may still wonder what mechanism carried the information, but one has not yet outrun the light cone.

The sharp Bell experiment is therefore arranged in the opposite way: the settings and outcomes are chosen so that no light signal could have crossed from one wing of the experiment to the other in time. That is what makes the result strange without turning it into a signaling device. The correlations remain, but no usable message has been transmitted. When one says that any hidden-variable account reproducing these facts must be nonlocal, one means precisely that it would need some influence not bounded by the speed of light.

This discussion does not close the lecture; it clears the ground for the next pivot. Bell has taught us that quantum mechanics is not classical logic, but the next question is different. What is a measurement, operationally and mathematically, and how does it enter the story of interference?

6.7 Finite-State Two-Slit, Interference, and Measurement as Entanglement

The lecture now changes model. Susskind turns to the two-slit experiment, but he immediately simplifies it into a finite-state problem so that the same vector-space technology can be used without bringing in the full machinery of continuum wave mechanics. He even says, in effect, that this is still the two-slit experiment, only stripped down to a small set of basis states.

There is a source state $|0\rangle$, two slit states $|1\rangle$ and $|-1\rangle$, and a set of screen states $|n\rangle$. These $|n\rangle$ are orthonormal because distinct screen positions are distinct configurations. The lecture explicitly chooses not to track normalization carefully here. Unitarity is postponed. What matters at this stage is linearity.

By symmetry the source evolves to an equal-amplitude superposition of the two slit states:

$$|0\rangle \longrightarrow |1\rangle + |-1\rangle. \tag{6.35}$$

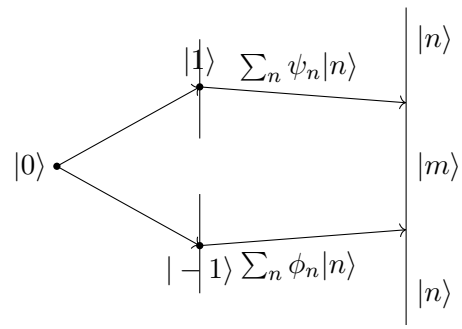


Figure 6.9: A finite-state version of the two-slit experiment: the source state $|0\rangle$, the two slit states $|1\rangle$ and $|-1\rangle$, and the screen basis $|n\rangle$.

Now let the upper slit state evolve to

$$|1\rangle \longrightarrow \sum_n \psi_n|n\rangle, \tag{6.36}$$

and the lower slit state to

$$|-1\rangle \longrightarrow \sum_n \phi_n|n\rangle. \tag{6.37}$$

The arrows here mean time evolution, not equality of states at the same instant. By linearity,

$$|1\rangle + |-1\rangle \longrightarrow \sum_n (\psi_n + \phi_n)|n\rangle. \tag{6.38}$$

That is the key rule. Amplitudes add before probabilities are formed.

The probability of arrival at the screen site m is therefore

$$P_m = |\psi_m + \phi_m|^2 \tag{6.39}$$

$$= |\psi_m|^2 + |\phi_m|^2 + \psi_m^*\phi_m + \phi_m^*\psi_m. \tag{6.40}$$

The first two terms are what classical reasoning would have produced. The last two are the interference terms.

At the symmetric point on the screen, call it $m = 0$, symmetry suggests $\psi_0 = \phi_0$. Then

$$P_0 = |\psi_0 + \phi_0|^2 = |2\psi_0|^2 = 4|\psi_0|^2. \tag{6.41}$$

Both slits open produce four times the one-slit probability at that point, not twice. This is constructive interference.

Now insert a phase shift so that $\phi_0 = -\psi_0$. Then

$$P_0 = |\psi_0 - \psi_0|^2 = 0. \tag{6.42}$$

Each slit by itself permits arrival at that point, but opening both together can make arrival there impossible. This is destructive interference. Susskind explicitly dwells on how odd this is: for his money it is as strange as Bell.

One electron, of course, produces only one hit. The interference pattern is the accumulated distribution obtained from many repetitions of the experiment, one particle at a time. This too is part of the lecture's rhythm: a single event is only a blip; the pattern is statistical.

Now we introduce the apparatus. There is one additional two-state system, a spin initially in the state $|d\rangle$. If the electron goes through the upper slit, it flips that spin to $|u\rangle$. If it goes through the lower slit, it leaves it unchanged. The initial composite state is $|0, d\rangle$, and the first update rule becomes

$$|0, d\rangle \longrightarrow |1, u\rangle + |-1, d\rangle. \tag{6.43}$$

That one line is the whole meaning of measurement in the lecture. The position of the electron has become entangled with the record stored in the apparatus.

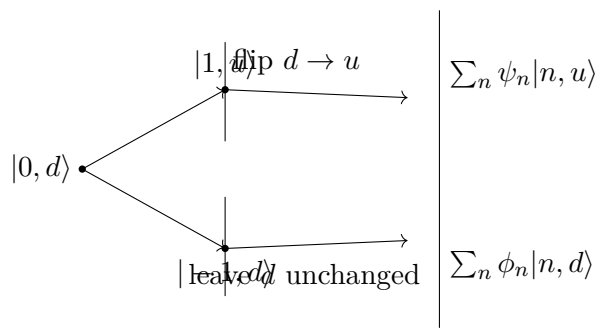


Figure 6.10: Measurement as entanglement with a detector spin. The upper path flips the apparatus spin; the lower path leaves it untouched.

Assume, as the lecture does, that nothing further happens to the spin during the journey to the screen. Then

$$|1, u\rangle \longrightarrow \sum_n \psi_n |n, u\rangle, \tag{6.44}$$

$$|-1, d\rangle \longrightarrow \sum_n \phi_n |n, d\rangle. \tag{6.45}$$

Therefore the final state with both slits open is

$$|\Psi_{\text{final}}\rangle = \sum_n \psi_n |n, u\rangle + \sum_n \phi_n |n, d\rangle. \tag{6.46}$$

This is not a superposition of states of one electron alone. It is an entangled state of electron and apparatus.

The cleanest way to compute the arrival probability at the site m is to use the projector onto that site on the electron subsystem:

$$\Pi_m = |m\rangle\langle m| \otimes I_{\text{spin}}. \tag{6.47}$$

Applying Π_m to the final state gives

$$\Pi_m |\Psi_{\text{final}}\rangle = |m\rangle\big(\psi_m |u\rangle + \phi_m |d\rangle\big). \tag{6.48}$$

So the probability of arrival at m is

$$\begin{aligned} P_m &= \langle \Psi_{\text{final}} | \Pi_m | \Psi_{\text{final}} \rangle && (6.49)\\ &= \big(\psi_m^* \langle u| + \phi_m^* \langle d|\big)\big(\psi_m |u\rangle + \phi_m |d\rangle\big) && (6.50)\\ &= |\psi_m|^2 + |\phi_m|^2 + \psi_m^* \phi_m \langle u|d\rangle + \phi_m^* \psi_m \langle d|u\rangle. && (6.51) \end{aligned}$$

But the record states are orthogonal:

$$\langle u|d\rangle = 0. \tag{6.52}$$

Hence the cross terms vanish:

$$P_m = |\psi_m|^2 + |\phi_m|^2. \tag{6.53}$$

That is the lecture's mathematical punchline. Once which-path information is recorded, interference disappears and probabilities add in the classical way.

6.7.1 Question & Answer: What is the difference between a one-system superposition and a two-system entangled state?

A state such as

$$|1\rangle + |-1\rangle \tag{6.54}$$

is a superposition of states of one system. There is only the electron, and it is in a linear combination of two position states.

A state such as

$$|1, u\rangle + |-1, d\rangle \tag{6.55}$$

is different in kind. Now there are two systems, the electron and the detector spin, and the state is not a statement about either subsystem separately. It correlates them. Looking at the spin tells us something about the path, and that is exactly why Susskind says that a measurement is the establishment of entanglement between system and apparatus.

The lecture closes by widening that claim. A deliberately built detector is not the only recorder. An atom near a slit, a bit of gas, a thermal environment, any additional degree of freedom that becomes sufficiently entangled with the electron can destroy the interference pattern. If the environmental record states are fully distinguishable, the interference is fully destroyed. If they overlap only partially, the interference is only partially destroyed. Susskind points toward that intermediate regime, but he does not develop its quantitative formalism here.

6.8 Summary

The lecture's mathematical spine is now plain. A property is represented, in general, not by a single state but by a subspace. The projector onto that subspace is built from an orthonormal basis of it, and the probability of the property is the expectation value of that projector. Compatible properties are combined, in the lecture's language, by multiplying commuting projectors. The same vector-space technology then reappears in the two-slit problem: without a record, amplitudes add and interference appears; with a record, the electron becomes entangled with an apparatus, the record states are orthogonal, the cross terms vanish, and probabilities add instead. This is the lecture's version of collapse: not an extra principle floating outside the formalism, but the disappearance of interference when information about the alternatives has been deposited somewhere in the world.

Chapter 7

Quantum Entanglements, Part 1

These notes follow Leonard Susskind's seventh lecture on quantum entanglement, curated by LazyingArt LLC. The lecture begins in repair mode. Susskind does not yet launch into entanglement entropy; instead he goes back and finishes the two-slit discussion in the simplest possible discrete language. That is the right starting point, because the real issue is not the geometry of the slits but the fate of interference once some apparatus records which route the electron took. Only after that mechanism is made precise does he turn to entropy, trace, and density matrices, which are the tools he will need next time.

7.1 Finishing the Discrete Two-Slit Setup

We begin with a stripped-down model. There is a source at a site labeled 0, two openings labeled A and B, and a screen whose possible arrival sites are denoted by $|m\rangle$. Everything else in the barrier is regarded as blocked off. For the moment the electron's own spin is irrelevant and is set aside; the states are characterized only by position.

The governing principle is linearity. If a state begins as a linear combination, then under time evolution it becomes the corresponding linear combination of the separately evolved pieces. In the symmetric situation Susskind writes, schematically,

$$|0\rangle \longrightarrow |A\rangle + |B\rangle. \tag{7.1}$$

This line is deliberately schematic. The lecture suppresses normalization on the board. A fully normalized version would carry an overall factor of $1/\sqrt{2}$, but that factor is not the point here; the point is that the source feeds both slits with equal amplitude.

Once the electron reaches one slit or the other, it propagates to the screen. The two route amplitudes are written

$$|A\rangle \longrightarrow \sum_m \psi_m |m\rangle, \tag{7.2}$$

$$|B\rangle \longrightarrow \sum_m \phi_m |m\rangle. \tag{7.3}$$

Here ψ_m is the amplitude for an electron that started at A to arrive at the m-th site on the screen, and ϕ_m is the corresponding amplitude from B.

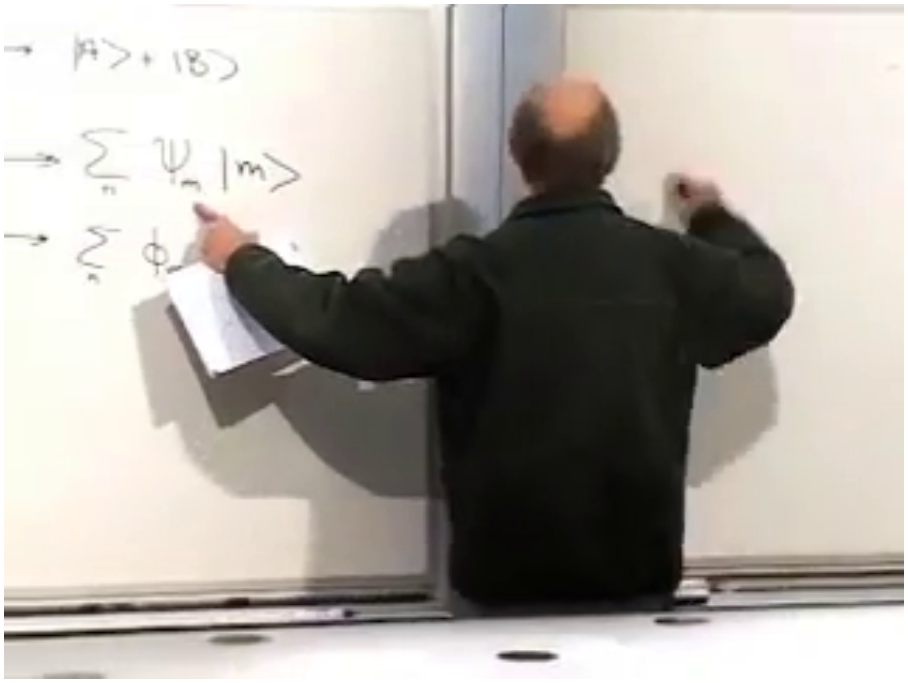


Figure 7.1: Lecture board: screen-basis amplitude expansion.

The lecture insists on the order of construction. First we understand the propagation from A alone and from B alone. Only then do we reopen both holes and ask what quantum mechanics really says.

7.2 From Path Amplitudes to the Interference Term

Suppose first that hole B is closed. Then the amplitude to arrive at site m is simply ψ_m, so the probability is

$$P_A(m) = \psi_m^* \psi_m. \tag{7.4}$$

If instead hole A is closed and only B is open, then

$$P_B(m) = \phi_m^* \phi_m. \tag{7.5}$$

At this point classical instinct suggests the wrong answer. One is tempted to say that if both holes are open, then the probability at m should be the sum of these two probabilities. Susskind pauses here to mark the pivot: not so in quantum mechanics. What we add are amplitudes.

With both slits open, linearity gives

$$|A\rangle + |B\rangle \longrightarrow \sum_n (\psi_n + \phi_n)|n\rangle. \tag{7.6}$$

So at a fixed screen site m, the relevant amplitude is $\psi_m + \phi_m$. The probability is therefore

$$P_m = |\psi_m + \phi_m|^2 \tag{7.7}$$

$$= \psi_m^* \psi_m + \phi_m^* \phi_m + \psi_m^* \phi_m + \phi_m^* \psi_m. \tag{7.8}$$

The first two terms are the direct route contributions. The last two are the interference terms. They are the genuinely quantum part of the story.

The lecture then dramatizes this with the simplest symmetry arguments. At the symmetry point on the screen we expect

$$\psi_m = \phi_m, \tag{7.9}$$

and then the probability is enhanced. In the lecture's shorthand one gets four times the single-route contribution. On the other hand, if the geometry and phases are such that

$$\psi_m = -\phi_m, \tag{7.10}$$

then the amplitudes cancel and

$$P_m = 0. \tag{7.11}$$

So despite the fact that the electron can get to m through A and it can get to m through B, with both slits open it may fail to arrive there altogether. That is the weirdness Susskind wants sitting in plain view before he asks the next question.

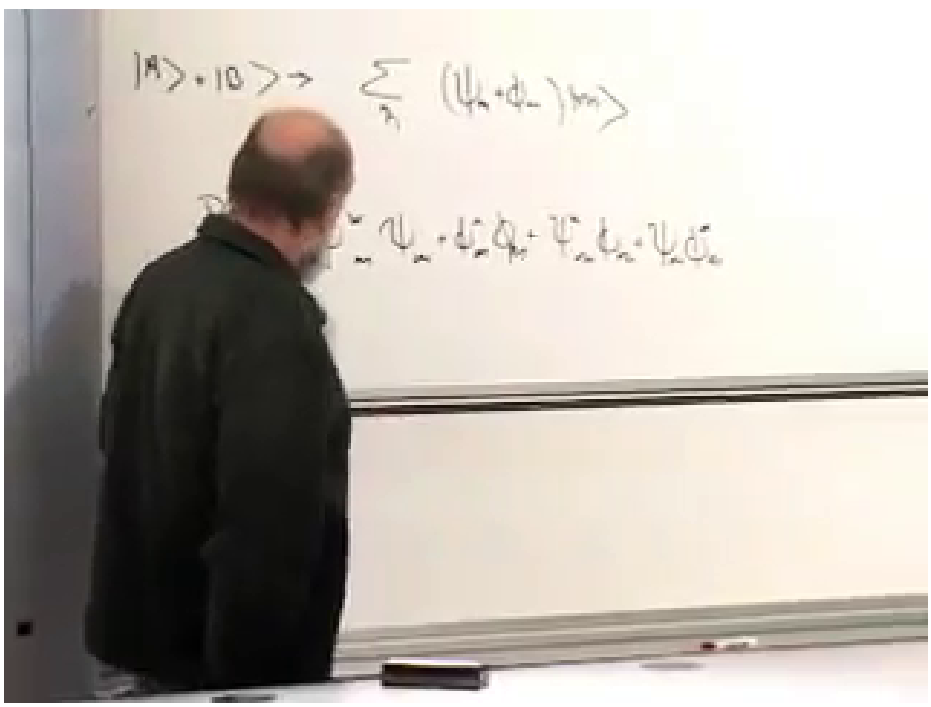


Figure 7.2: Lecture board: interference probability and cross terms.

Now the lecture can make the real turn: what happens if the experiment is arranged so that the route is recorded?

7.3 Which-Path Recording as Entanglement with a Spin

Susskind introduces the simplest recording apparatus he can think of: an auxiliary spin with two states, u and d. This could in principle be the electron's own spin, but he explicitly asks us not to think of it that way. The point is to model an external detector, however simple, that can register passage through one slit and not the other.

The initial state is therefore not just the electron at the source. It is the electron at the source together with the detector in its untriggered state:

$$|0, d\rangle. \tag{7.12}$$

The rules of the model are simple. If the electron passes through slit A, it flips the detector from down to up. If it passes through B, the detector remains down. So the post-slit state is

$$|0, d\rangle \longrightarrow |A, u\rangle + |B, d\rangle. \tag{7.13}$$

From there the electron propagates to the screen just as before, except that the detector state comes along as a route label:

$$|A, u\rangle \longrightarrow \sum_n \psi_n |n, u\rangle, \tag{7.14}$$

$$|B, d\rangle \longrightarrow \sum_n \phi_n |n, d\rangle. \tag{7.15}$$

The final state is thus

$$|\Psi_{\text{final}}\rangle = \sum_n \big(\psi_n |n, u\rangle + \phi_n |n, d\rangle\big). \tag{7.16}$$

The important change is now visible. We do not have a single electron wave function over the screen. We have a superposition of two branches, and each branch carries a different detector record.

7.3.1 Question & Answer

Question. Did the wave packet collapse, or did the electron become entangled with the detector?

Answer. Susskind's answer is that both ways of speaking occur, but they are not equally sharp. If we insist on describing only the electron and forgetting the detector, then one may use the old language of collapse: the interference terms are gone, and one says the wave packet has collapsed to one route or the other. But if we include the apparatus in the description, then that is not what has happened. The composite system has not become only one branch. It has become

$$|A, u\rangle + |B, d\rangle,$$

an entangled state of electron and detector. In that fuller description the correct statement is not "collapse occurred" but "entanglement was established."

This correction matters because the lecture is trying to replace vague talk of measurement with a precise mechanism. If we keep the apparatus in view, we can see exactly where the interference goes.

7.4 Projector onto a Screen Position and Loss of Cross Terms

Having set up the entangled state, Susskind deliberately slows down and does the formal calculation. We want the probability that the electron is found at a particular screen site m.

In the enlarged Hilbert space, the property "the electron is at m" defines a two-dimensional subspace. The basis states of that subspace are

$$|m, u\rangle \qquad \text{and} \qquad |m, d\rangle.$$

Therefore the projector onto that subspace is

$$\Pi_m = |m, u\rangle\langle m, u| + |m, d\rangle\langle m, d|. \tag{7.17}$$

This is exactly the object the board snapshot preserves most clearly.

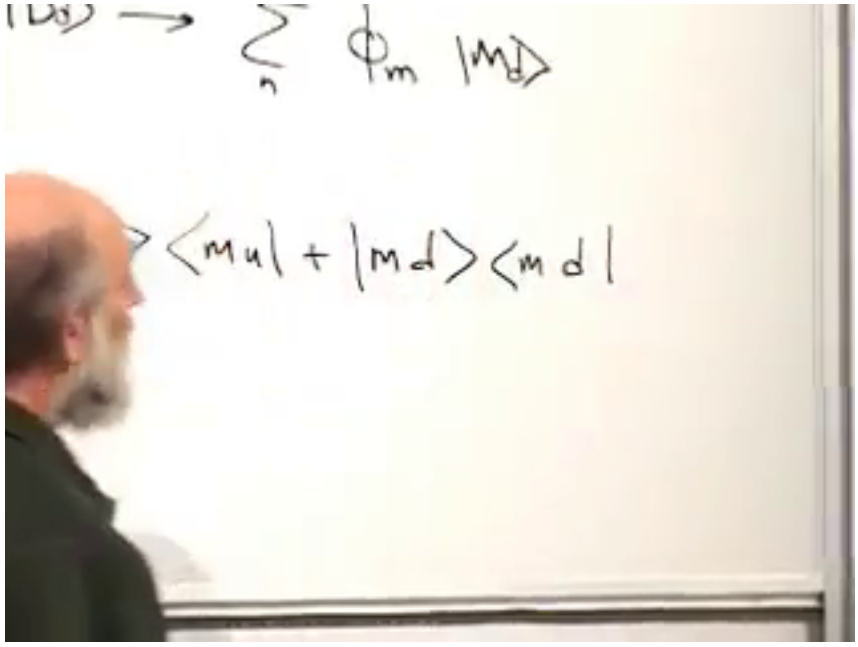


Figure 7.3: Lecture board: projector onto a fixed screen position.

The probability of arrival at m is the expectation value of this projector in the final state:

$$P_m = \langle \Psi_{\text{final}} | \Pi_m | \Psi_{\text{final}} \rangle. \tag{7.18}$$

Substituting the final state gives

$$P_m = \sum_{n,n'} (\psi_n^* \langle n, u| + \phi_n^* \langle n, d|) \Pi_m (\psi_{n'} |n', u\rangle + \phi_{n'} |n', d\rangle). \tag{7.19}$$

Now one sees why Susskind kept saying that ψ_m and ϕ_m multiply different ket vectors. The projector can pair $|n, u\rangle$ only with $|m, u\rangle$, and $|n, d\rangle$ only with $|m, d\rangle$. In particular,

$$\langle u|d\rangle = 0. \tag{7.20}$$

So the mixed terms vanish. The only surviving contributions are the direct ones, with matching screen label and matching detector label. Hence

$$P_m = |\psi_m|^2 + |\phi_m|^2. \tag{7.21}$$

This is the formal centerpiece of the lecture. The interference terms do not disappear because we announced a collapse postulate. They disappear because a measurement has correlated the route alternatives with orthogonal detector states. One must match not only the endpoint on the screen, but also the detector label. Once that extra label is there, the two amplitudes no longer add into a single coherent amplitude at m.

7.5 What Counts as a Measurement?

Once the clean spin model is on the board, the lecture opens out into a long Socratic clarification. The point is not to leave the spin as an artificial toy. It is to identify the general rule behind it.

The rule is this: a measurement has occurred if some mark has been left on the world from which one can, in principle, infer which alternative took place. No conscious observer is required at that stage. A human being may later read the record, but the decisive event is that the record exists.

7.5.1 Question & Answer

Question. What counts as a measurement?

Answer. Any interaction that leaves a distinguishable route record counts as a measurement. If the record states are orthogonal, interference is destroyed completely. If they overlap, the interference is only partially washed out.

The lecture first tests the symmetric placement of the detector. Suppose the auxiliary spin is placed so that it is kicked in exactly the same way whether the electron passes through A or through B. Then the record state is the same on both branches, and the post-slit state is effectively

$$|A, u\rangle + |B, u\rangle. \tag{7.22}$$

Because the detector no longer distinguishes the routes, the interference pattern returns.

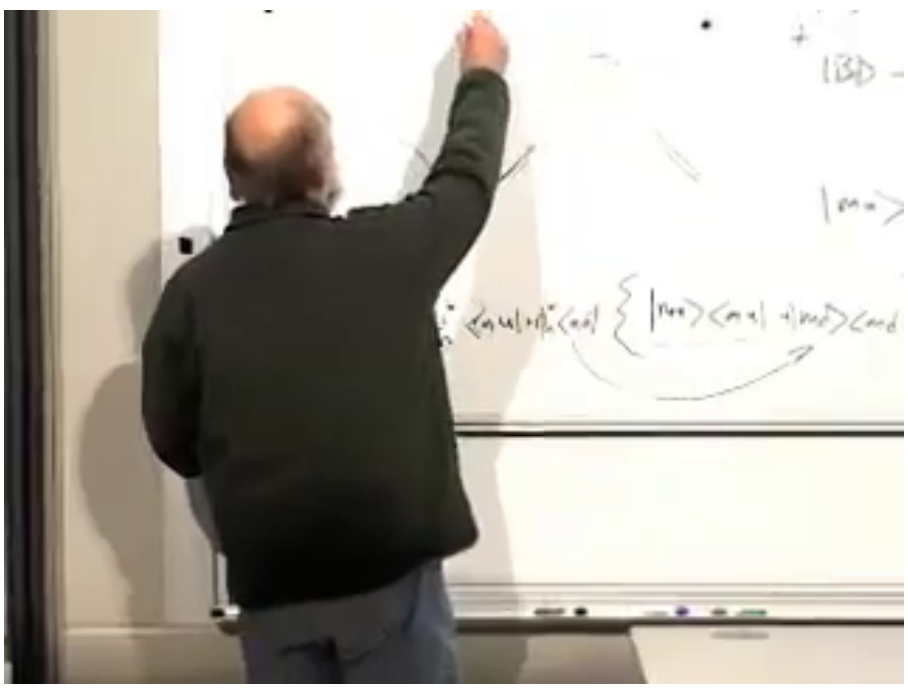


Figure 7.4: Lecture board: symmetric placement of the spin detector.

The recoil example makes the same point less idealized. If the slit plate receives an upward kick when the electron goes through the upper hole and a downward kick when it goes through the lower

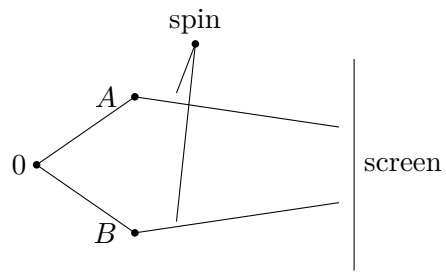


Figure 7.5: Clean schematic of the symmetric detector placement.

hole, then the plate itself may store route information. But because the plate is also localized in position, its momentum must already have some quantum uncertainty. If the kick is small compared with that spread, the two recoil states overlap strongly and are not orthogonal. This is precisely where the uncertainty principle enters the discussion. The mark exists only to the extent that the recoil states can be told apart.

The lecture does not write a general environment-overlap formula on the board, but its content is well captured by the cautious reconstruction

$$P_m = |\psi_m|^2 + |\phi_m|^2 + \psi_m^* \phi_m \langle E_A | E_B \rangle + \phi_m^* \psi_m \langle E_B | E_A \rangle, \tag{7.23}$$

where $|E_A\rangle$ and $|E_B\rangle$ are the record states left by the two routes. If $\langle E_A|E_B\rangle = 0$, the record is perfectly distinguishing and the interference is gone. If $\langle E_A|E_B\rangle$ is merely small rather than zero, the fringes are only blurred.

That is how Susskind organizes the later examples. A partial spin kick leaves the detector mostly down with only a little admixture of up, and the pattern is only partially destroyed. Emission of a photon carries route information into the radiation field, but for a slow electron the probability of emission is small and the record is incomplete. A bubble chamber or any track-forming device leaves a far more robust route record, and then the interference is destroyed outright.

One of the lecture's most important clarifications appears here. We do not have to look at the detector. The electron has already done the measurement by becoming entangled with it. It is enough that the information is recorded in principle. We send electrons through one at a time, reset the apparatus, gather statistics on arrival positions, and we see the loss of fringes whether or not anyone ever reads the auxiliary spin.

A second clarification is that the time at which one reads the record is not the essential issue. If a later measurement can reveal which route was taken, then the route information had already been stored in some correlated degree of freedom. That is why Susskind keeps returning to the same rule: ask whether something remembers.

Before leaving this part of the lecture, he briefly generalizes once more. Why do macroscopic objects not visibly spread into fuzzy superpositions? Part of the reason is their mass, but part of it is environmental bombardment. Air, light, and surrounding matter are continually leaving records. The environment is performing measurements all the time.

7.6 Measurement Chains, Schrödinger's Cat, and the Movable Cut

From here the lecture scales up the same mechanism. Schrödinger's cat is not introduced as a different mystery but as the same entanglement logic with a larger apparatus.

Suppose a gun may or may not fire, and the cat's fate is correlated with the gun. Then the composite system evolves to

$$|\Psi_{\text{cat}}\rangle = \alpha\,|\text{live}, \text{unfired}\rangle + \beta\,|\text{dead}, \text{fired}\rangle. \tag{7.24}$$

What matters is what this is not. It is not an isolated cat state

$$\alpha\,|\text{live}\rangle + \beta\,|\text{dead}\rangle.$$

The cat is entangled with the gun. The gun has already left a mark, and therefore there are no interference terms between "live" and "dead" in the ordinary description of the cat. This is, as Susskind emphasizes, the same mathematics as the slit detector.

The same structure persists when Schrödinger looks into the box, and when Wigner later asks Schrödinger what he saw. Each new observer can be included as another system with a pair of record states. Thus the chain

$$\text{system} \longrightarrow \text{detector} \longrightarrow \text{observer} \longrightarrow \text{observer of observer}$$

can be extended as far as we like.

What becomes ambiguous is not the probabilities but the placement of the cut. We may choose to say that the electron's wave function collapsed. Or we may include the detector and say that the combined wave function collapsed. Or we may include Schrödinger and postpone the collapse until Wigner's observation. Susskind's point is not that the ambiguity disappears, but that the theory remains consistent. If Wigner asks Schrödinger what he saw, the correlations line up.

This also answers an important question about the screen. Why do we talk about the probability that the electron arrives at a given point on the screen? Because the screen is itself another measuring device. A photographic plate blackens, a scintillation screen emits light, or a row of local detectors flips. In each case the arrival position becomes entangled with some macroscopic record. The probability postulate is then applied by constructing the projector onto the relevant subspace and taking its expectation value.

The lecture also sharpens the contrast with classical measurement. In classical physics one imagines that, at least in principle, an experiment can be done gently enough that it does not appreciably disturb the system. In the quantum case that picture fails exactly when incompatible alternatives are at stake. To determine among them, the apparatus must become entangled with the system, and that is a real change of state. Measurement is therefore not just passive observation. It alters whether interference exists.

7.7 Entropy, Trace, Density Matrix, and the Next-Lecture Setup

At this point Susskind says, more or less explicitly, that the lecture has not gone as far as he hoped. He wanted a measure of how entangled two systems are. So he changes gears. The remaining material is preparatory, but it is not an afterthought. It is the exact machinery that will be needed next time.

He begins with the simplest non-entangled example: two spins both up. That is a product state. Looking at one spin tells us nothing new about the other. By contrast, entangled states do correlate subsystems, and the quantity he wants to use to diagnose that correlation is entanglement entropy.

But first we go back to the classical situation.

Suppose a classical system has a finite set of possible states. Entropy, in this discussion, is not merely a property of the system; it is a property of the system together with our knowledge of it. If all we know is that the system lies in one of a set of n equally likely states, then the entropy is

$$S = \log n. \tag{7.25}$$

The logarithm measures, roughly, how many yes-no questions would be needed to pin the state down.

More generally, if the state space carries a probability distribution p_i, then the entropy is

$$S = -\sum_i p_i \log p_i. \tag{7.26}$$

The minus sign is needed because $\log p_i < 0$ whenever $0 < p_i < 1$.

The lecture checks that this reproduces the equal-probability case. If

$$p_i = \frac{1}{n} \qquad \text{for } i \in \mathcal{N}, \qquad p_i = 0 \qquad \text{for } i \notin \mathcal{N}, \tag{7.27}$$

with $|\mathcal{N}| = n$, then

$$S = -\sum_i p_i \log p_i \tag{7.28}$$

$$= -n\left(\frac{1}{n}\right)\log\left(\frac{1}{n}\right) \tag{7.29}$$

$$= \log n. \tag{7.30}$$

The states outside $\mathcal{N}$ contribute nothing, because $p \log p \to 0$ as $p \to 0$. So the Shannon formula reproduces the simpler $\log n$ rule exactly where it should.

Susskind then introduces the trace. This is the operation that will play the role of summing over configurations in quantum mechanics.

Definition 7.1. For an operator M and any complete orthonormal basis $\{|i\rangle\}$, the trace is

$$\operatorname{Tr} M = \sum_i \langle i|M|i\rangle. \tag{7.31}$$

A crucial theorem, quoted rather than proved in the lecture, is that the trace is basis independent. If M happens to be diagonal in some basis, then its diagonal entries are its eigenvalues, and therefore

$$\operatorname{Tr} M = \sum_i \lambda_i. \tag{7.32}$$

This is the sense in which the trace is an invariant.

Now Susskind makes the next step. Suppose someone prepares a system in one of the basis states $|i\rangle$, but does not tell us which one. We are told only the probabilities ρ_i for the various preparation

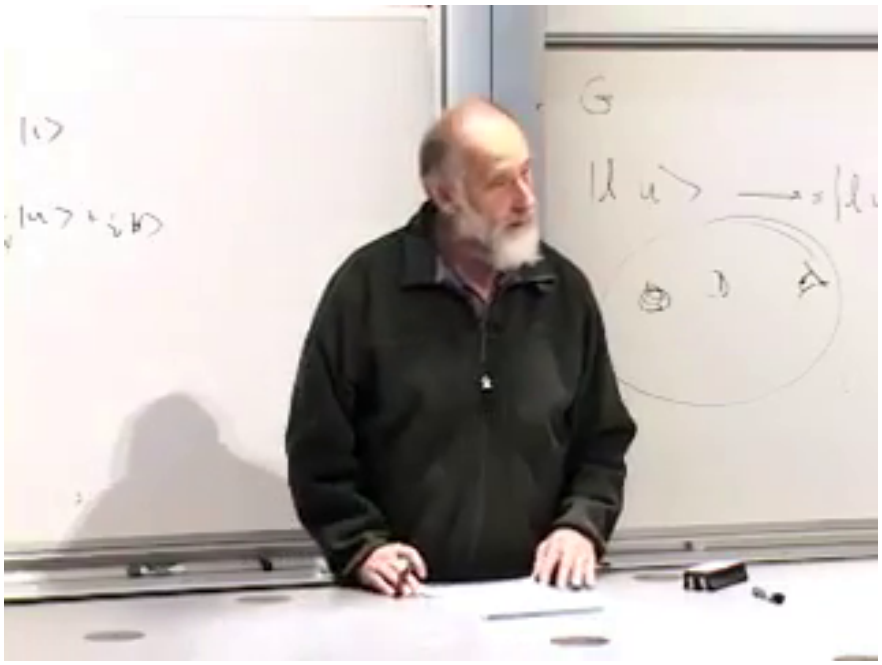


Figure 7.6: Lecture board: schematic transition to a density-matrix description.

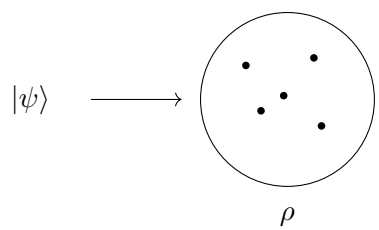


Figure 7.7: Clean schematic of the move from a state-vector description to a statistical description.

outcomes. This is not a coherent superposition. The system is not in $\rho_1|1\rangle + \rho_2|2\rangle + \cdots$. It is a statistical ensemble of definite preparation states. That is why we need an object more flexible than a state vector.

In the basis of preparation states, the density matrix is diagonal:

$$\rho = \mathrm{diag}(\rho_1, \rho_2, \rho_3, \ldots). \tag{7.33}$$

Its entries are real and nonnegative, and because probabilities sum to one,

$$\mathrm{Tr}\,\rho = 1. \tag{7.34}$$

Now let F be an observable. The lecture's central formula is

$$\langle F \rangle = \mathrm{Tr}(F\rho). \tag{7.35}$$

This is the quantum analog of the classical average $\sum_i p_i f_i$. To see the analogy, we insert a resolution of the identity,

$$\sum_j |j\rangle\langle j| = I, \tag{7.36}$$

between F and ρ. Then

$$\langle F\rangle = \mathrm{Tr}(F\rho) \tag{7.37}$$
$$= \sum_i \langle i|F\rho|i\rangle \tag{7.38}$$
$$= \sum_{i,j} \langle i|F|j\rangle\langle j|\rho|i\rangle. \tag{7.39}$$

If we choose the basis in which ρ is diagonal, then $\langle j|\rho|i\rangle$ vanishes unless $i = j$. Hence

$$\langle F\rangle = \sum_i \langle i|F|i\rangle\rho_i \tag{7.40}$$
$$= \sum_i F_{ii}\rho_i. \tag{7.41}$$

This is exactly the weighted average we wanted: the expectation value of F in each possible preparation state, multiplied by the probability that the system was prepared that way.

The trace also satisfies the cyclic property

$$\mathrm{Tr}(AB) = \mathrm{Tr}(BA), \tag{7.42}$$

even when A and B do not commute. That is why it does not matter whether we write $\mathrm{Tr}(F\rho)$ or $\mathrm{Tr}(\rho F)$.

Finally, Susskind defines the quantum entropy associated with a density matrix:

$$S = -\,\mathrm{Tr}(\rho\log\rho). \tag{7.43}$$

If ρ is diagonal, then $\log\rho$ is obtained by taking the logarithm of each eigenvalue:

$$\log\rho = \mathrm{diag}(\log\rho_1, \log\rho_2, \log\rho_3, \ldots). \tag{7.44}$$

So in that basis

$$S = -\sum_i \rho_i \log\rho_i. \tag{7.45}$$

Thus the quantum formula reduces to the classical one when the density matrix is diagonal.

The two limiting cases are exactly those emphasized in the lecture. If one eigenvalue is 1 and all the rest are 0, then the entropy vanishes. That is the pure-state case. If the density matrix is proportional to the identity, so that all states are equally likely, then the entropy is maximal:

$$\rho \propto I \qquad \Longrightarrow \qquad S = \log(\dim\mathcal{H}). \tag{7.46}$$

The lecture is very careful here. The entropy of a pure state is zero not because quantum mechanics has somehow ceased to be probabilistic, but because no extra ignorance about preparation remains. The nonzero entropy of a mixed state measures ignorance about which state was prepared.

And now the lecture stops. The payoff is deferred one more step. If a composite system AB is entangled and we describe only subsystem A, then the right object is a density matrix for A, and that density matrix will carry entropy. If A and B are not entangled, it will not. Entanglement entropy itself is therefore not yet computed in this lecture, but every ingredient needed for it has now been assembled.

7.8 Summary

The lecture unfolds in two deliberate movements. First, the discrete two-slit experiment is rebuilt so that we can see exactly where interference comes from and exactly how it disappears when the route becomes entangled with a detector. Measurement is not treated as a vague act of observation but as the establishment of system-apparatus entanglement, with orthogonal record states killing the cross terms. Second, after that mechanism is made precise, Susskind backs up and introduces the mathematics needed to quantify entanglement: classical entropy, the trace, diagonal density matrices, expectation values of the form $\mathrm{Tr}(F\rho)$, and the quantum entropy $S = -\mathrm{Tr}(\rho \log \rho)$. The chapter therefore ends at the right place: not with entanglement entropy already in hand, but with the conceptual and mathematical stage set for it.

Chapter 8

Quantum Entanglements, Part 1

These notes follow Lecture 8 of Leonard Susskind's *Quantum Entanglement* course and are written as companion notes rather than as a transcript. The lecture opens by returning to density matrices in a deliberately practical spirit: not as a new formalism, but as the right language when a system has been prepared and we do not fully know how. From there the argument unfolds in the same order as the lecture itself. First we review the rules for density matrices, then the trace formula for expectation values and the entropy that measures ignorance. Only after that does the real entanglement story begin: a pure state of a combined system may induce a mixed state for one part. Two explicit spin examples make that point concrete. The lecture then changes direction, but not logic, and turns to time evolution, unitarity, the Hamiltonian, and the phase evolution of energy eigenstates. The present chapter is curated in this archive by LazyingArt LLC, but the progression and emphasis remain those of Leonard Susskind.

8.1 Density Matrices As Incomplete Knowledge

We begin with the motivation. A system may have been prepared in a perfectly definite way, and yet we may not know how it was prepared. An electron spin may have been aligned by a magnetic field, but if someone else chose the direction of that field and did not tell us what it was, then the system is not described, for us, by a known state vector. It is described by a density matrix.

This is the quantum analog of a classical probability distribution. The lecture reviews its basic properties in exactly that spirit. If ρ is a density matrix, then

$$\operatorname{Tr}\rho = 1, \qquad \rho^\dagger = \rho. \tag{8.1}$$

If its eigenvalues are λ_i, then

$$\lambda_i \in \mathbb{R}, \qquad \lambda_i \geq 0, \qquad \sum_i \lambda_i = 1. \tag{8.2}$$

Because ρ is Hermitian, its eigenvectors form an orthonormal basis. In that basis the eigenvalues are probabilities.

The lecture immediately places beside one another the two limiting cases. If we know nothing at all, then all the eigenvalues are equal, so

$$\rho = \frac{1}{n} I, \tag{8.3}$$

where n is the dimension of the state space. This is complete ignorance. At the opposite extreme, maximal knowledge means a pure state $|\psi\rangle$, and even then it is useful to write a density matrix:

$$\rho = |\psi\rangle\langle\psi|. \tag{8.4}$$

This is simply the projector onto the state $|\psi\rangle$.

Let us check its eigenvalues, since the lecture pauses here to do exactly that. Assuming $|\psi\rangle$ is normalized,

$$\rho|\psi\rangle = |\psi\rangle\langle\psi|\psi\rangle = |\psi\rangle. \tag{8.5}$$

So $|\psi\rangle$ is an eigenvector with eigenvalue 1. If $|\phi\rangle$ is orthogonal to $|\psi\rangle$, then

$$\rho|\phi\rangle = |\psi\rangle\langle\psi|\phi\rangle = 0. \tag{8.6}$$

Thus a pure-state density matrix has exactly one nonzero eigenvalue, equal to 1, and every orthogonal direction has eigenvalue 0. In the old language this is not a mixed state at all; the old German word was *gemisch*, but here we will simply say *mixed state* when more than one eigenvalue is nonzero.

8.1.1 Question & Answer

Why is a fifty-fifty mixture of two nonorthogonal states not the right final language?

The point is not that such a preparation is impossible. The point is that a density matrix, being Hermitian, comes with a basis of orthogonal eigenvectors, and only in that basis do its eigenvalues have the clean interpretation as probabilities. One may certainly begin with a preparation procedure that mentions nonorthogonal states. But once the density matrix is formed, it reorganizes the description into its own orthogonal eigenbasis. That is where the probabilities live.

So the lecture is warning us not to confuse the story of preparation with the final language of ρ. The density matrix has its own natural basis, and that basis is orthogonal.

8.2 Expectation Values And Entropy

The next question is operational. What do we do with ρ? The answer is the trace formula

$$\bar{M} = \mathrm{Tr}(\rho M), \tag{8.7}$$

where M is any observable. The lecture also notes the modest cyclicity fact needed here:

$$\mathrm{Tr}(\rho M) = \mathrm{Tr}(M\rho). \tag{8.8}$$

This is not because M and ρ commute, but because the trace of a product of two operators is insensitive to their order.

The lecture then reviews the definition of the trace itself. In any basis $\{|i\rangle\}$,

$$\mathrm{Tr}(X) = \sum_i \langle i|X|i\rangle. \tag{8.9}$$

It does not matter which basis is used; the trace is basis-independent. Let us now see how the expectation-value rule reduces to the ordinary one for a pure state. If $\rho = |\psi\rangle\langle\psi|$, then

$$\bar{M} = \mathrm{Tr}(|\psi\rangle\langle\psi|M) \tag{8.10}$$

$$= \sum_i \langle i|\psi\rangle\langle\psi|M|i\rangle \tag{8.11}$$

$$= \langle\psi|M\left(\sum_i |i\rangle\langle i|\right)|\psi\rangle \tag{8.12}$$

$$= \langle\psi|M|\psi\rangle. \tag{8.13}$$

So the trace formula really does generalize the familiar expectation value.

Now let us do what the lecture does next: choose the basis that diagonalizes ρ. If $\rho|j\rangle = \lambda_j|j\rangle$, then

$$\bar{M} = \mathrm{Tr}(\rho M) = \sum_i \langle i|\rho M|i\rangle \tag{8.14}$$

$$= \sum_{i,j} \langle i|\rho|j\rangle\langle j|M|i\rangle \tag{8.15}$$

$$= \sum_j \lambda_j \langle j|M|j\rangle. \tag{8.16}$$

This is the correct probabilistic reading of the density matrix: the expectation value is the average of $\langle j|M|j\rangle$, weighted by the eigenvalues λ_j, but only in the basis where ρ is diagonal. That last clause matters.

The lecture then adds the second tool we will need later: entropy. For a probability distribution $\{p_i\}$,

$$S = -\sum_i p_i \log p_i. \tag{8.17}$$

If one probability is 1 and the rest vanish, then $S = 0$. If all n probabilities are equal, $p_i = 1/n$, then

$$S = -n\left(\frac{1}{n}\log\frac{1}{n}\right) = \log n. \tag{8.18}$$

So entropy measures ignorance. Zero entropy means complete knowledge. Large entropy means the probability distribution is spread out over many possibilities. The lecture is still in review mode here, but the review is pointed: entropy is about to become the diagnostic of entanglement.

8.3 Composite Systems And The Reduced Density Matrix

Now the lecture turns to entanglement itself. Entanglement begins when we have more than one system. Let us therefore consider a composite system with two parts, A and B. We choose basis states $|a, b\rangle$, where a labels subsystem A and b labels subsystem B. In the lecture these indices drift a bit on the board, so let us settle them cleanly from the start.

The most general pure state of the combined system is

$$|\Psi\rangle = \sum_{a,b} \psi(a,b)\,|a,b\rangle, \qquad \sum_{a,b} |\psi(a,b)|^2 = 1. \tag{8.19}$$

Now imagine that Alice performs an experiment involving only subsystem A. Let M be an observable acting only on A. Then M changes the a-index but leaves the b-index untouched.

To compute the expectation value, we write

$$\langle M\rangle = \langle\Psi|M|\Psi\rangle \tag{8.20}$$

$$= \sum_{a',b'}\sum_{a,b}\psi^*(a',b')\psi(a,b)\,\langle a',b'|M|a,b\rangle. \tag{8.21}$$

Since M is passive on subsystem B, its matrix elements are diagonal in b:

$$\langle a',b'|M|a,b\rangle = \langle a'|M|a\rangle\,\delta_{b'b}. \tag{8.22}$$

Therefore $b' = b$, and the expression becomes

$$\langle M\rangle = \sum_{a',a,b}\psi^*(a',b)\psi(a,b)\,\langle a'|M|a\rangle. \tag{8.23}$$

Now comes the central step. Hold a and a' fixed and sum only over b. The result depends only on a and a', so we give it a name:

$$\rho_A(a,a') = \sum_b \psi(a,b)\psi^*(a',b). \tag{8.24}$$

This is the reduced density matrix of subsystem A. With this definition,

$$\langle M\rangle = \mathrm{Tr}_A(\rho_A M). \tag{8.25}$$

So all measurements performed on subsystem A alone are encoded in ρ_A. In effect, we have summed over the B-degrees of freedom and turned a pure state of the whole into a density matrix for the part.

8.4 Pure Whole, Mixed Part

This is the sharp conceptual pivot of the lecture. The composite system may be in a pure state, described by a single state vector $|\Psi\rangle$, and nevertheless the subsystem A may be described only by a mixed density matrix ρ_A.

Classically there is no analog of this. If one has complete knowledge of a classical composite system, then one has complete knowledge of each of its parts. In classical physics, a pure state of the whole implies pure states of the parts. Quantum mechanics breaks that implication. We may know everything that quantum mechanics allows us to know about the whole, and still have only a statistical description of a part.

The lecture does not present that as a detached theorem. It appears only after the reduced-density-matrix calculation has been done. That order matters. The algebra has earned the philosophical contrast.

8.4.1 Question & Answer

How is tracing out B different from an ordinary classical marginal?

There is certainly a similarity. In both cases we sum or integrate over variables we do not observe. But the difference shows up precisely when the state of the whole is pure. In a classical system, complete knowledge of the whole implies complete knowledge of the parts. In a quantum system, complete knowledge of the whole may force us to use a mixed state for the part.

So the reduced density matrix is not just a classical marginal in new notation. It reflects the genuinely quantum fact that the state of a subsystem need not exist as an independent pure state at all.

The lecture also fields a related question here: is $\mathrm{Tr}(M\rho)$ the product $\mathrm{Tr}\, M\ \mathrm{Tr}\, \rho$? No. The trace of a product is not the product of traces.

8.5 Factorized States And The Entropy Test

At this point the lecture asks the natural question: when does the subsystem *remain* pure? In other words, what condition on $\psi(a, b)$ guarantees that the reduced density matrix of A is itself a pure-state projector?

The answer is that the wave function factorizes:

$$\psi(a, b) = \phi(a)\chi(b). \tag{8.26}$$

This is the special case in which the combined system is already a product of a pure state for A and a pure state for B.

Let us substitute this into the reduced density matrix:

$$\rho_A(a, a') = \sum_b \psi(a, b)\psi^*(a', b) \tag{8.27}$$

$$= \sum_b \phi(a)\chi(b)\, \phi^*(a')\chi^*(b) \tag{8.28}$$

$$= \phi(a)\phi^*(a') \sum_b \chi(b)\chi^*(b). \tag{8.29}$$

If χ is normalized, then the sum over b is 1, and therefore

$$\rho_A(a, a') = \phi(a)\phi^*(a'). \tag{8.30}$$

So

$$\rho_A = |\phi\rangle\langle\phi|. \tag{8.31}$$

The reduced density matrix has one nonzero eigenvalue, equal to 1, and all the rest vanish. Its entropy is therefore zero.

This gives the lecture a practical criterion. Product structure means zero subsystem entropy. More generally, the smaller the entropy, the closer the state is to a product; the larger the entropy, the farther it is from one. The lecture is careful not to overstate this. The diagnostic is practical rather than mystical: calculate the reduced density matrix, calculate its eigenvalues, then calculate its entropy.

8.6 Two Spin Examples: Singlet And Product State

The lecture now puts the machinery to work on two familiar two-spin states.

8.6.1 The Singlet State

The first example is the singlet,

$$|\text{sing}\rangle = \frac{1}{\sqrt{2}}\Big(|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle\Big). \tag{8.32}$$

In component form,

$$\psi(\uparrow,\uparrow) = 0, \qquad \psi(\uparrow,\downarrow) = \frac{1}{\sqrt{2}}, \qquad \psi(\downarrow,\uparrow) = -\frac{1}{\sqrt{2}}, \qquad \psi(\downarrow,\downarrow) = 0. \tag{8.33}$$

Let us compute the reduced density matrix for spin 1. The diagonal entries are

$$\rho_A(\uparrow,\uparrow) = \sum_b \psi(\uparrow,b)\psi^*(\uparrow,b) = |\psi(\uparrow,\uparrow)|^2 + |\psi(\uparrow,\downarrow)|^2 = \frac{1}{2}, \tag{8.34}$$

$$\rho_A(\downarrow,\downarrow) = \sum_b \psi(\downarrow,b)\psi^*(\downarrow,b) = |\psi(\downarrow,\uparrow)|^2 + |\psi(\downarrow,\downarrow)|^2 = \frac{1}{2}. \tag{8.35}$$

For the off-diagonal term,

$$\rho_A(\uparrow,\downarrow) = \sum_b \psi(\uparrow,b)\psi^*(\downarrow,b) \tag{8.36}$$

$$= \psi(\uparrow,\uparrow)\psi^*(\downarrow,\uparrow) + \psi(\uparrow,\downarrow)\psi^*(\downarrow,\downarrow) = 0, \tag{8.37}$$

and similarly $\rho_A(\downarrow,\uparrow) = 0$. Hence

$$\rho_A = \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & \frac{1}{2} \end{pmatrix} = \frac{1}{2}I. \tag{8.38}$$

This is the maximally mixed density matrix for a single spin. Its entropy is

$$S_A = -2\left(\frac{1}{2}\log\frac{1}{2}\right) = \log 2. \tag{8.39}$$

Now let us use the reduced density matrix the way the lecture uses it. For spin along any axis $\hat{n}$,

$$\langle\sigma\cdot\hat{n}\rangle = \mathrm{Tr}\big(\rho_A\,\sigma\cdot\hat{n}\big) = \frac{1}{2}\,\mathrm{Tr}(\sigma\cdot\hat{n}) = 0, \tag{8.40}$$

because every Pauli matrix has zero trace. So the spin is equally likely to point along $\hat{n}$ or opposite to $\hat{n}$, no matter what direction $\hat{n}$ is. This is the familiar content of the singlet state, now recast in density-matrix language.

8.6.2 A Product State In Disguise

The second example is the normalized state with all four amplitudes equal:

$$|\Psi\rangle = \frac{1}{2}\Big(|\uparrow\uparrow\rangle + |\uparrow\downarrow\rangle + |\downarrow\uparrow\rangle + |\downarrow\downarrow\rangle\Big). \tag{8.41}$$

When we trace over spin 2, we obtain

$$\rho_A = \begin{pmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{pmatrix}. \tag{8.42}$$

Its trace is still 1, as it must be, but now the determinant is

$$\det \rho_A = \frac{1}{2} \cdot \frac{1}{2} - \frac{1}{2} \cdot \frac{1}{2} = 0. \tag{8.43}$$

Since the eigenvalues of a 2×2 matrix have sum equal to the trace and product equal to the determinant, the two eigenvalues are 1 and 0. Therefore the entropy is zero. So despite appearances, this is not an entangled state at all; it is a product state.

8.6.3 Question & Answer

What product state is it?

The lecture answers this by testing the expectation values of the Pauli matrices. First,

$$\sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \qquad \mathrm{Tr}(\rho_A \sigma_3) = 0. \tag{8.44}$$

So the spins are not aligned along the z-axis.

Next the lecture turns to σ_1. The spoken line in the transcript is garbled, but the intended matrix is supported by the board frame and by standard notation:

$$\sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}. \tag{8.45}$$

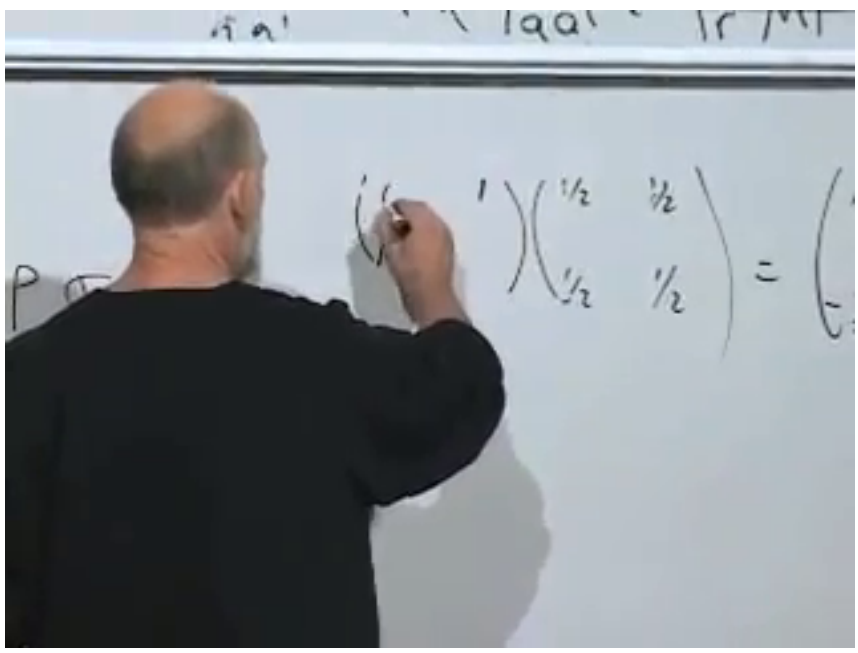


Figure 8.1: σ_1 acting on the reduced density matrix. The frame is kept as evidence for the board's matrix-multiplication layout, while the clean algebra is reconstructed nearby.

Now compute

$$\rho_A \sigma_1 = \begin{pmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{pmatrix}, \tag{8.46}$$

$$\mathrm{Tr}(\rho_A \sigma_1) = 1. \tag{8.47}$$

So the expectation value of σ_1 is 1, while the expectation values of the other components vanish. The conclusion is that the two spins are unentangled and both point along the x-axis.

The lecture then makes the broader interpretive point. If we perturb such a zero-entropy density matrix slightly, its entropy becomes small but nonzero. That small entropy measures a small amount of entanglement. Entropy and entanglement go together here.

Before leaving the subject, the lecture pauses to separate two entropies that are easy to confuse. In the examples above, the *combined* system is in a pure state and therefore has entropy zero. The entropy we have been computing is the entropy of the *subsystem*. It arises because the subsystem is entangled with what we traced over. That is why it is called entanglement entropy. It is also why the entropy is not additive in the naive classical way: the whole may have entropy zero while each part has nonzero entropy.

A student then asks whether entanglement has units. The lecture's answer is crisp. Entropy is a pure number. One may speak loosely of bits, but that is not a physical dimension. It is best to think of entropy here as dimensionless.

Finally, the lecture raises a suggestive idea without forcing it into dogma: perhaps every density matrix can ultimately be viewed as coming from entanglement with something else. Sometimes that "something else" is merely hidden in the preparation apparatus, the environment, or the observer's ignorance. The lecture leaves that as a suspicion rather than as a theorem.

8.7 From Unitarity To The Schrödinger Equation

Now the lecture changes direction. We leave entanglement for a while and ask how states change with time. The transition is deliberate. The same concern remains in the background: distinctions must be preserved.

In a discrete classical system, reversible evolution is just a permutation of states. The law tells us where each state goes next, and because the evolution is reversible, two different states never collapse into one. Information is not lost. In quantum mechanics the corresponding principle is unitarity.

8.7.1 Question & Answer

How can a system with only finitely many orthonormal states evolve continuously in time?

The lecture answers by returning to the spin. A spin-$\frac{1}{2}$ system has only two orthogonal outcomes for a measurement along a fixed axis, but its state vector is not confined to those two basis vectors. It may point continuously through an entire family of superpositions. That is why a finite-dimensional quantum system can still evolve continuously in time. The discreteness lies in the dimension of the vector space, not in the set of all possible state vectors.

8.7.2 Linear Evolution And Unitarity

The first principle is that time evolution is linear. If the state at time 0 is $|\psi(0)\rangle$, then at time t,

$$|\psi(t)\rangle = U(t)|\psi(0)\rangle, \qquad U(0) = I. \tag{8.48}$$

The second principle is that inner products are preserved:

$$\langle\phi(t)|\psi(t)\rangle = \langle\phi(0)|\psi(0)\rangle. \tag{8.49}$$

Taking the Hermitian conjugate of the ket evolution,

$$\langle\phi(t)| = \langle\phi(0)|U^\dagger(t). \tag{8.50}$$

Therefore

$$\langle\phi(t)|\psi(t)\rangle = \langle\phi(0)|U^\dagger(t)U(t)|\psi(0)\rangle. \tag{8.51}$$

Since this must equal $\langle\phi(0)|\psi(0)\rangle$ for every pair of initial states, the only possible conclusion is

$$U^\dagger(t)U(t) = I. \tag{8.52}$$

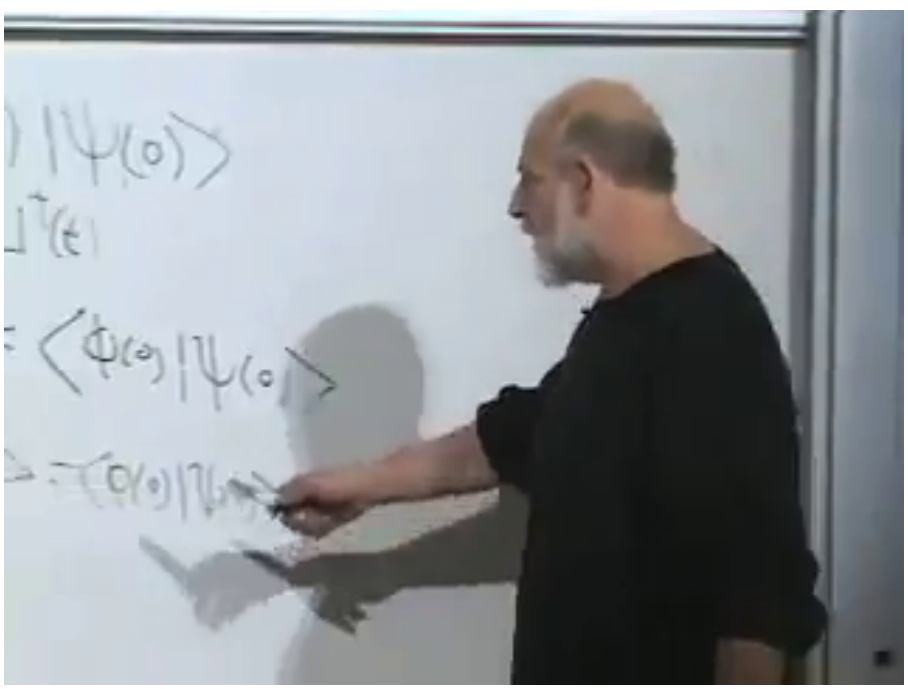


Figure 8.2: Bra-ket form of the unitarity argument. The screenshot is preserved because it records the lecturer's stacked board derivation from time evolution to $U^\dagger U = I$.

That is unitarity. It is the quantum version of the statement that distinctions are conserved.

The lecture then fields a useful caution. Preservation of inner products does *not* mean that unentangled systems must remain unentangled forever. If that were true, systems could never become entangled, and measurement itself would be impossible. Unitarity preserves inner products, not separability.

8.7.3 Infinitesimal Evolution And The Hamiltonian

Now take a very small time interval ϵ. Because $U(0) = I$, the lecture writes

$$U(\epsilon) = I - i\epsilon h, \tag{8.53}$$

to first order in ϵ. At this stage the lecture temporarily suppresses $\hbar$; that is part of the historical order of the derivation, and we will restore $\hbar$ in the final cleaned equation.

Taking the Hermitian conjugate,

$$U^\dagger(\epsilon) = I + i\epsilon h^\dagger. \tag{8.54}$$

Now impose unitarity:

$$\begin{aligned} I &= U^\dagger(\epsilon)U(\epsilon) && (8.55)\\ &= (I + i\epsilon h^\dagger)(I - i\epsilon h) && (8.56)\\ &= I + i\epsilon h^\dagger - i\epsilon h + O(\epsilon^2). && (8.57) \end{aligned}$$

Ignoring $O(\epsilon^2)$, we obtain

$$h^\dagger = h. \tag{8.58}$$

So the infinitesimal generator is Hermitian. The lecture then gives it its name: the Hamiltonian. Its eigenvalues are the energy levels.

To pass from the infinitesimal evolution operator to a differential equation, write

$$|\psi(\epsilon)\rangle = (I - i\epsilon h)|\psi(0)\rangle. \tag{8.59}$$

Subtract $|\psi(0)\rangle$ and divide by ϵ:

$$\frac{|\psi(\epsilon)\rangle - |\psi(0)\rangle}{\epsilon} = -ih|\psi(0)\rangle. \tag{8.60}$$

Now let $\epsilon \to 0$. This becomes

$$\frac{\mathrm{d}}{\mathrm{d}t}|\psi(t)\rangle = -ih|\psi(t)\rangle. \tag{8.61}$$

Restoring $\hbar$ and switching to the standard capital H, we arrive at the final form

$$i\hbar\frac{\mathrm{d}}{\mathrm{d}t}|\psi(t)\rangle = H|\psi(t)\rangle. \tag{8.62}$$

This is the general time-dependent Schrödinger equation.

8.7.4 Question & Answer

If a state remains an energy eigenvector for all time, does that not mean it does not change?

This is exactly the objection the lecture raises near the end, and it is worth preserving. The answer is no. An eigenvector may be multiplied by a complex number and remain the same eigenvector. So "remaining an eigenvector" does not mean "remaining numerically identical."

8.7.5 Energy Eigenstates And Phase Evolution

Now suppose

$$H|\psi\rangle = E|\psi\rangle. \tag{8.63}$$

We try the lecture's ansatz

$$|\psi(t)\rangle = f(t)|\psi(0)\rangle, \tag{8.64}$$

with $f(t)$ a scalar function of time.

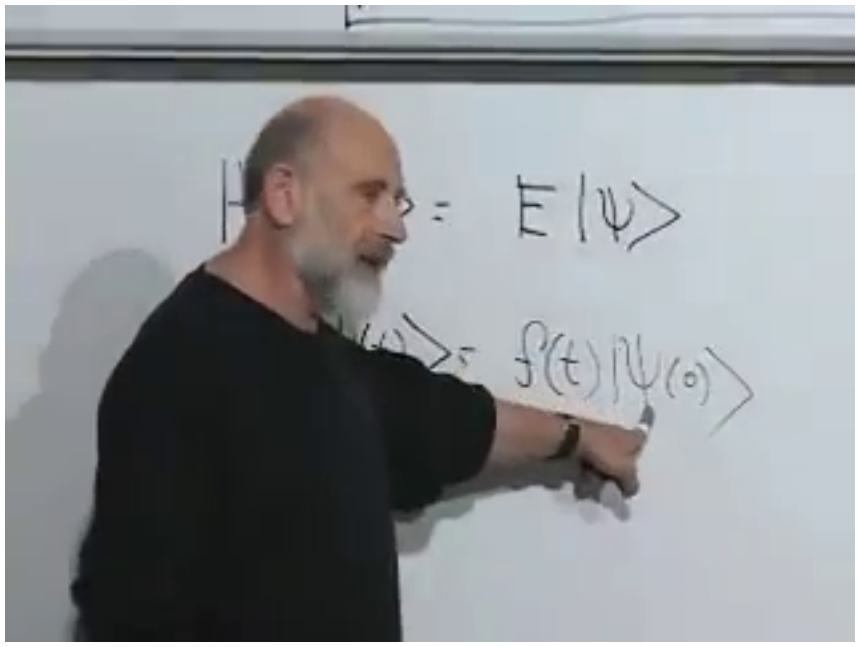


Figure 8.3: Energy eigenstate and scalar time evolution. The board shows the two-line setup that leads from $H|\psi\rangle = E|\psi\rangle$ to the time-dependent phase factor.

Substitute into the Schrödinger equation:

$$i\hbar \frac{\mathrm{d}}{\mathrm{d}t}\big(f(t)|\psi(0)\rangle\big) = Hf(t)|\psi(0)\rangle \tag{8.65}$$

$$i\hbar \frac{\mathrm{d}f}{\mathrm{d}t}|\psi(0)\rangle = f(t)E|\psi(0)\rangle. \tag{8.66}$$

Cancel the common eigenvector:

$$\frac{\mathrm{d}f}{\mathrm{d}t} = -\frac{iE}{\hbar}f(t). \tag{8.67}$$

Hence

$$f(t) = e^{-iEt/\hbar}, \tag{8.68}$$

and therefore

$$|\psi(t)\rangle = e^{-iEt/\hbar}|\psi(0)\rangle. \tag{8.69}$$

So an energy eigenstate evolves only by a phase.

The lecture closes by interpreting the coefficient of t in that phase as an angular frequency:

$$\omega = \frac{E}{\hbar}, \qquad \text{or equivalently} \qquad E = \hbar\omega. \tag{8.70}$$

That is where the lecture stops. The general equation of motion is in hand, but the concrete choice of Hamiltonian is postponed.

8.8 Summary

The lecture begins with density matrices as the language of incomplete preparation and ends with the Schrödinger equation as the law of time evolution. In between, one central quantum fact is established carefully and then illustrated: a composite system may be in a pure state while each subsystem is mixed. The reduced density matrix is obtained by summing over unobserved degrees of freedom, and its entropy measures the entanglement of the subsystem with what has been traced out. Product states have one eigenvalue equal to 1 and zero entropy; the singlet has a reduced density matrix proportional to the identity and entropy $\log 2$. After that, the lecture pivots to time evolution, argues that inner products must be preserved, derives unitarity, identifies a Hermitian Hamiltonian as the infinitesimal generator of motion, and finally shows that an energy eigenstate evolves only by a time-dependent phase with angular frequency $E/\hbar$.

Chapter 9

Quantum Entanglements, Part 1

This lecture opens by deliberately going backward. We return to last week's discussion of time dependence not because there is no new material, but because the logic is important enough to rebuild from the state-vector point of view. Only after that logic is made solid do we move on to examples, and only after the examples are in hand do two coupled spins enter and entanglement begin to evolve in time. The chapter therefore unfolds in the same order as the lecture itself: linear evolution, unitarity, infinitesimal time steps, the Schrödinger equation, stationary states, superposition, commutators, precession, and finally the first genuinely entangled dynamics.

9.1 Time Dependence, Inner Products, and Unitarity

We begin with the state vector. Up to now one could think of a ket as something timeless, but the lecture now insists that we put time into the notation and ask how the state changes. The most general thing we are willing to write, under the postulate that evolution is linear, is

$$|\psi(t)\rangle = U(t)|\psi(0)\rangle. \tag{9.1}$$

Nothing detailed has yet been assumed about $U(t)$ except linearity. The system moves around in its vector space under the influence of a linear operator.

The corresponding rule for bras is the Hermitian-conjugate one:

$$\langle\phi(t)| = \langle\phi(0)|U^\dagger(t). \tag{9.2}$$

The transcript is noisy at this point, but the intended content is standard and clear: bras evolve by the conjugate transpose of the operator that evolves kets.

Now comes the first real restriction. It is not enough that states evolve linearly; probabilities must remain well defined. In particular, if a physical state is normalized at one time, it must remain normalized:

$$\langle\psi(t)|\psi(t)\rangle = \langle\psi(0)|\psi(0)\rangle. \tag{9.3}$$

The lecture then broadens the point. If the length of every vector is preserved by a linear transformation, then the inner product of any pair of vectors is preserved as well. Thus

$$\langle\phi(t)|\psi(t)\rangle = \langle\phi(0)|\psi(0)\rangle. \tag{9.4}$$

Substituting the evolution rules gives

$$\langle\phi(0)|\, U^\dagger(t)U(t)\,|\psi(0)\rangle = \langle\phi(0)|\psi(0)\rangle. \tag{9.5}$$

And because this holds for arbitrary vectors $|\phi(0)\rangle$ and $|\psi(0)\rangle$, the operator in the middle must be the identity:

$$U^\dagger(t)U(t) = 1. \tag{9.6}$$

So the time-evolution operator is unitary. In the lecture this is not an isolated axiom. It is the consequence of demanding that distinctions and probabilities not be erased by time evolution.

9.2 Infinitesimal Evolution and the Schrödinger Equation

Having established unitarity, the lecture immediately changes scale. Instead of asking what happens over a finite interval, it asks what happens over a very short one. This is the usual move in mechanics: break time into tiny pieces, understand one such piece, and then build finite evolution from infinitesimal evolution.

Let $t = \epsilon$, with ϵ so small that ϵ^2 can be ignored. Since zero elapsed time should mean no change at all, we must have $U(0) = 1$. The next term must therefore be linear in ϵ. In the lecture's convention one writes

$$U(\epsilon) = 1 - \frac{i\epsilon}{\hbar}H. \tag{9.7}$$

The lecture pauses to say that the factors of i and $\hbar$ are conventional. They are inserted for convenience and for comparison with established notation. In particular, the i is chosen so that the operator that survives in the end will be Hermitian rather than anti-Hermitian.

Taking the Hermitian conjugate gives

$$U^\dagger(\epsilon) = 1 + \frac{i\epsilon}{\hbar}H^\dagger. \tag{9.8}$$

Now impose unitarity to first order:

$$\begin{aligned} 1 &= U^\dagger(\epsilon)U(\epsilon) && (9.9)\\ &= \left(1 + \frac{i\epsilon}{\hbar}H^\dagger\right)\left(1 - \frac{i\epsilon}{\hbar}H\right) && (9.10)\\ &= 1 + \frac{i\epsilon}{\hbar}(H^\dagger - H) + O(\epsilon^2). && (9.11) \end{aligned}$$

Neglecting $O(\epsilon^2)$, we conclude that

$$H^\dagger = H. \tag{9.12}$$

So the generator of infinitesimal time evolution is Hermitian. This is already enough to identify it as an observable, and in the lecture it is now given its name: the Hamiltonian.

The next pivot is the crucial one. We no longer want merely an update rule over a small interval; we want a differential equation. The board writes the derivative at the chosen origin of time as

$$\frac{d}{dt}|\psi\rangle = \frac{|\psi(\epsilon)\rangle - |\psi(0)\rangle}{\epsilon}. \tag{9.13}$$

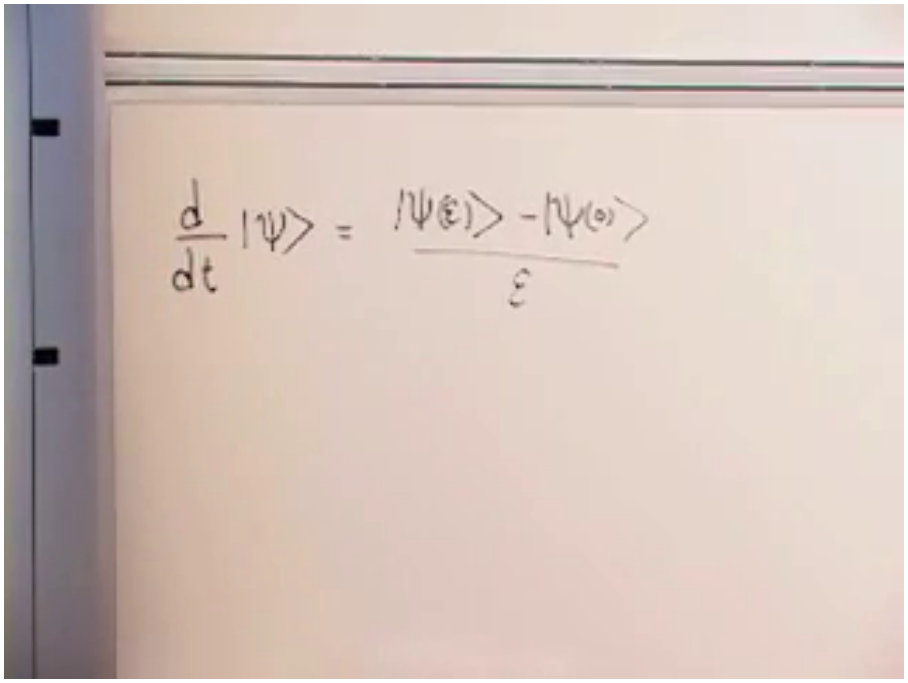


Figure 9.1: Difference quotient for the state vector

The displayed equation in the screenshot suppresses the explicit time label on the ket at the left. In the notes it is safer to say what the lecture also says in words: the choice of the origin is arbitrary. Any instant can be called 0. So the cleaned version is

$$\frac{d}{dt}|\psi(t)\rangle = \frac{|\psi(t+\epsilon)\rangle - |\psi(t)\rangle}{\epsilon}. \tag{9.14}$$

Now substitute the infinitesimal evolution law,

$$|\psi(t+\epsilon)\rangle = \left(1 - \frac{i\epsilon}{\hbar}H\right)|\psi(t)\rangle. \tag{9.15}$$

Then

$$\frac{d}{dt}|\psi(t)\rangle = \frac{\left(1 - \frac{i\epsilon}{\hbar}H\right)|\psi(t)\rangle - |\psi(t)\rangle}{\epsilon} \tag{9.16}$$

$$= -\frac{i}{\hbar}H|\psi(t)\rangle. \tag{9.17}$$

Equivalently,

$$i\hbar\frac{d}{dt}|\psi(t)\rangle = H|\psi(t)\rangle. \tag{9.18}$$

This is the Schrödinger equation. The lecture is quite explicit here: this is the whole engine of quantum time evolution. Give the state at one instant and the Hamiltonian, and the equation determines the state at later times.

9.3 Energy Eigenvectors and Stationary Time Dependence

The lecture does not solve the general Schrödinger equation immediately. It first chooses the simplest solvable case: a state that is already an eigenvector of the Hamiltonian.

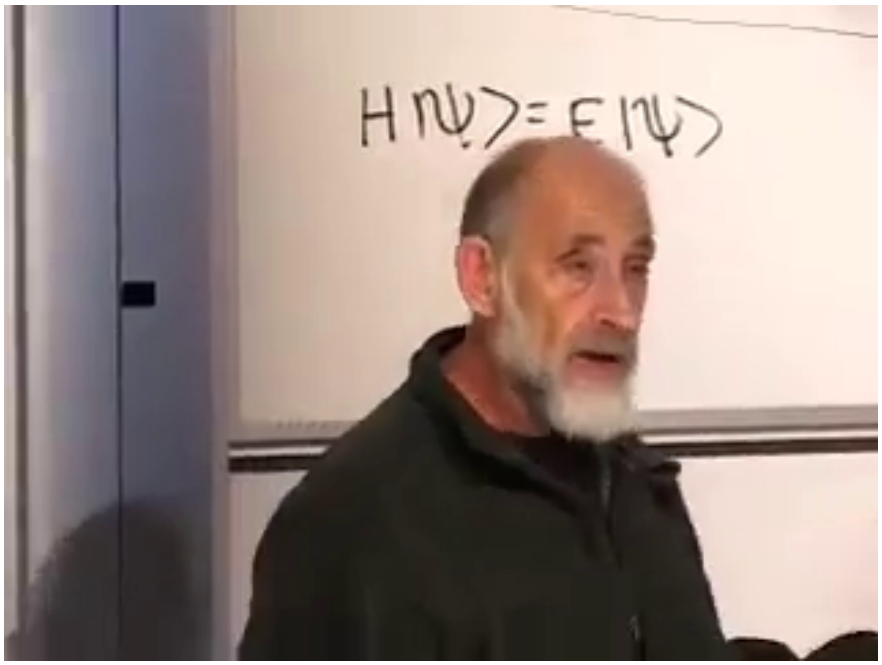


Figure 9.2: Energy eigenvalue equation

Suppose

$$H|\psi_E\rangle = E|\psi_E\rangle. \tag{9.19}$$

If the Hamiltonian acting on the state merely returns the same state multiplied by a number, then the time evolution cannot bend the state into a new direction of the vector space. It can only multiply it by a time-dependent scalar. So we try

$$|\psi_E(t)\rangle = f(t)|\psi_E\rangle. \tag{9.20}$$

Insert this ansatz into the Schrödinger equation:

$$i\hbar \frac{d}{dt}(f(t)|\psi_E\rangle) = H(f(t)|\psi_E\rangle), \tag{9.21}$$

$$i\hbar\, \dot{f}(t)|\psi_E\rangle = f(t)E|\psi_E\rangle. \tag{9.22}$$

Since $|\psi_E\rangle$ is fixed, we reduce the vector equation to the scalar one

$$\dot{f}(t) = -\frac{iE}{\hbar} f(t). \tag{9.23}$$

That equation says that the derivative of f is proportional to f itself, so its solution is exponential:

$$f(t) = e^{-iEt/\hbar}. \tag{9.24}$$

Therefore

$$|\psi_E(t)\rangle = e^{-iEt/\hbar}|\psi_E(0)\rangle. \tag{9.25}$$

This is why the lecture treats Hamiltonian eigenvectors as stationary states. Nothing dynamically interesting happens to a single energy eigenvector except the accumulation of phase.

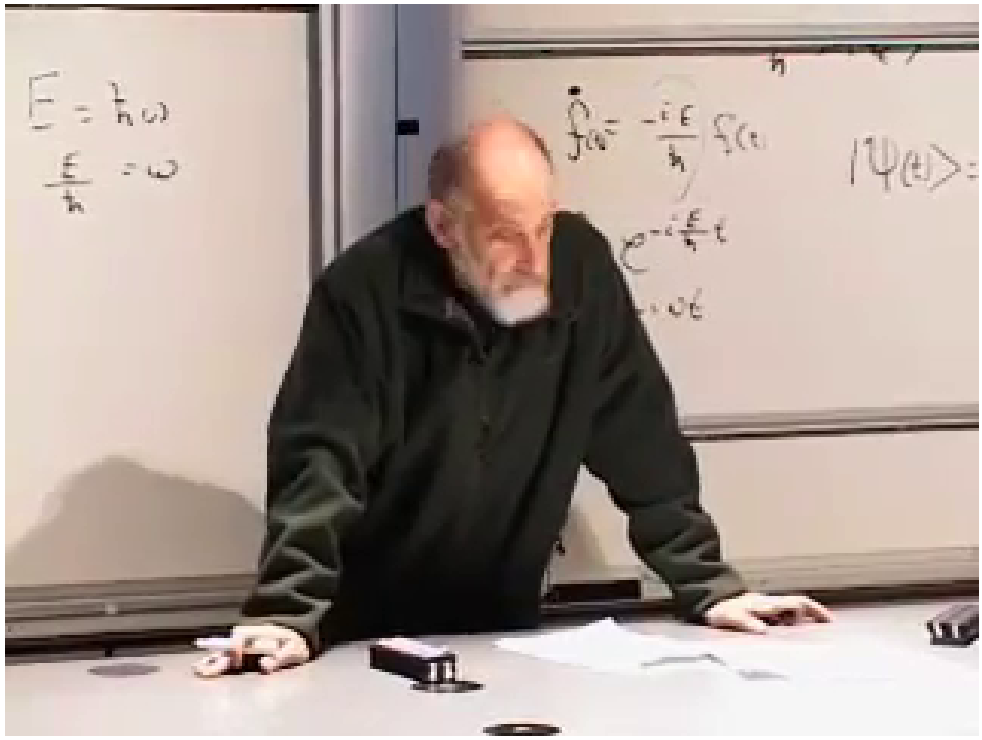


Figure 9.3: Energy-frequency relation beside phase evolution

The lecture then turns the phase into a frequency statement. After an on-the-spot correction of signs, one has

$$e^{-iEt/\hbar} = \cos\left(\frac{Et}{\hbar}\right) - i\sin\left(\frac{Et}{\hbar}\right). \tag{9.26}$$

So the phase rotates with angular frequency

$$\omega = \frac{E}{\hbar}, \tag{9.27}$$

or, written the other way,

$$E = \hbar\omega. \tag{9.28}$$

The board image supports this connection but does not preserve every algebraic step cleanly, so it is better to typeset the cleaned chain beside the screenshot. Still, the lecture's intent is unmistakable: the quantity E appears precisely in the role that Einstein and Planck taught us to identify as energy.

9.3.1 Question & Answer

Why do we call H the energy?

The lecture answers this in two stages. The first answer comes from the stationary-state calculation itself. If a state evolves as $e^{-i\omega t}$, and if the coefficient in the exponent is $E/\hbar$, then E is exactly the quantity that enters the Einstein–Planck relation $E = \hbar\omega$. In that sense the eigenvalue of the Hamiltonian sits where energy belongs.

But that is not the only answer, and the lecture is careful not to stop there. A more structural answer is about conservation. In an abstract system, before we know any classical formula such

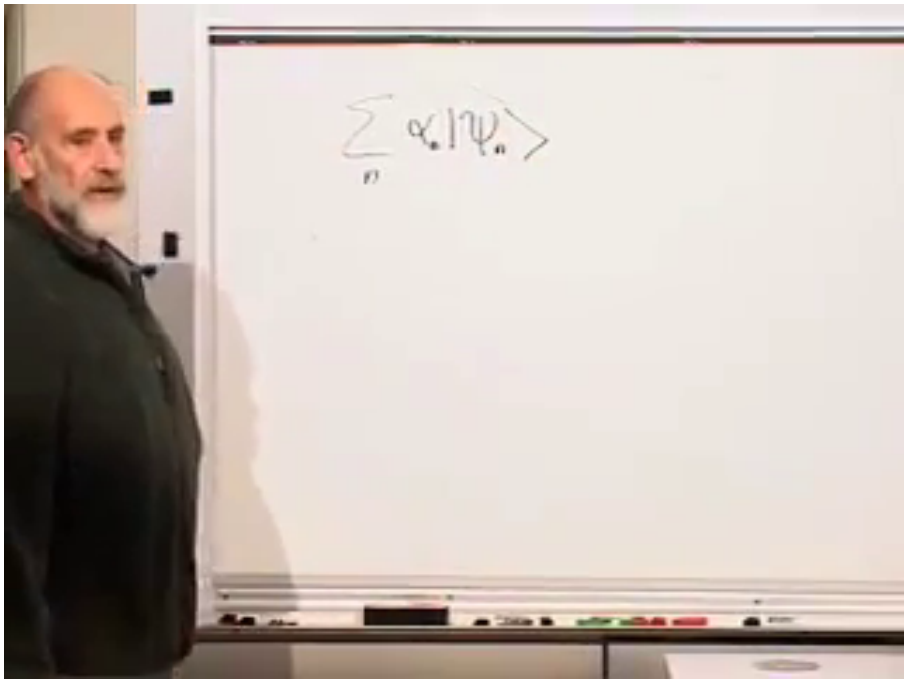


Figure 9.4: Expansion in Hamiltonian eigenvectors

as $p^2/2m$, the first thing we would look for in a definition of energy is that it be conserved. The next section shows precisely that: $\langle H \rangle$ does not change with time. So H earns its name twice, once through frequency and once through conservation.

There is also a small but useful remark about convention. The lecture returns to the inserted factor of i and explains why it was chosen. Without that i, the infinitesimal generator would have come out anti-Hermitian. Multiplying by i converts an anti-Hermitian operator into a Hermitian one. That choice ensures real eigenvalues and, with unitarity, keeps the norm of the state fixed.

9.4 Superposition in the Energy Basis and the General Solution

At this point the lecture says, in effect, that a single energy eigenvector is too boring. Multiplying an entire state by an overall phase is not where the interesting physics lies, because an overall phase does not affect probabilities. Real time dependence begins when the state is a superposition of components with different energies.

Because H is Hermitian, it has a complete orthonormal basis of eigenvectors. Therefore any state can be written as

$$|\psi\rangle = \sum_n \alpha_n |\psi_n\rangle, \tag{9.29}$$

with

$$H|\psi_n\rangle = E_n|\psi_n\rangle. \tag{9.30}$$

Now linearity does the rest. Since each eigenvector evolves with its own phase, the whole state evolves as

$$|\psi(t)\rangle = \sum_n \alpha_n e^{-iE_n t/\hbar}|\psi_n\rangle. \tag{9.31}$$

This is the general solution in the energy basis.

The lecture emphasizes the qualitative content: the high-energy components rotate faster, the low-energy ones more slowly, and what matters physically is not the overall phase of the whole state but the changing relative phases among different components. Those relative phases are the dynamical substance of quantum mechanics.

In the energy basis the Hamiltonian is diagonal. That immediately tells us that the evolution operator is diagonal in the same basis, with entries $e^{-iE_n t/\hbar}$. In abstract operator notation one summarizes the whole discussion by writing

$$U(t) = e^{-iHt/\hbar}. \tag{9.32}$$

This should not be thought of here as a brand-new theorem dropped from nowhere. It is a compact way of repackaging the energy-basis solution we have just derived.

A short but important remark about combined systems appears at this stage. If two systems A and B are non-interacting, then

$$H_{AB} = H_A + H_B, \tag{9.33}$$

and the energy eigenvalues of the combined system are sums of the separate energies. If the systems interact, this simple additivity no longer holds in that form. The lecture flags this point now because it will matter later, when two spins are coupled to one another.

9.5 Observables, Commutators, and Conserved Quantities

Only after the discussion of state evolution is in place does the lecture pivot to observables. The observables themselves are fixed. It is the states that change with time, and therefore the expectation values of observables change.

Let A be a fixed observable. Its expectation value is

$$\langle A \rangle = \langle \psi | A | \psi \rangle. \tag{9.34}$$

Differentiate it. Since A is fixed and the bras and kets are time dependent, the product rule gives

$$\frac{d}{dt}\langle A \rangle = \langle \dot{\psi} | A | \psi \rangle + \langle \psi | A | \dot{\psi} \rangle. \tag{9.35}$$

Now use the Schrödinger equation for the ket and its Hermitian conjugate for the bra:

$$|\dot{\psi}\rangle = -\frac{i}{\hbar} H|\psi\rangle, \qquad \langle \dot{\psi}| = \frac{i}{\hbar}\langle \psi | H. \tag{9.36}$$

Substituting,

$$\frac{d}{dt}\langle A \rangle = \frac{i}{\hbar}\langle \psi | HA | \psi \rangle - \frac{i}{\hbar}\langle \psi | AH | \psi \rangle \tag{9.37}$$

$$= \frac{i}{\hbar}\langle \psi | (HA - AH) | \psi \rangle \tag{9.38}$$

$$= \frac{i}{\hbar}\langle [H, A] \rangle. \tag{9.39}$$

This is one of the central identities of the lecture. The Hamiltonian governs not only the time dependence of the state vector but also the time dependence of every expectation value through its commutator with the corresponding observable.

The first application is immediate. Put $A = H$. Then

$$\frac{d}{dt}\langle H\rangle = \frac{i}{\hbar}\langle [H, H]\rangle = 0. \tag{9.40}$$

So the expectation value of the Hamiltonian is conserved. At this point the lecture circles back to the earlier question and answers it more sharply: what better candidate could there be for energy than the observable every system possesses and every system conserves?

9.5.1 Question & Answer

What is the physical meaning of operators that commute with H?

The lecture answers this cleanly. If

$$[H, A] = 0, \tag{9.41}$$

then

$$\frac{d}{dt}\langle A\rangle = 0. \tag{9.42}$$

So the set of operators commuting with the Hamiltonian is the set of conserved quantities for that system. Energy is always in that set, because every operator commutes with itself. Beyond that, the answer depends on the details of the Hamiltonian. For a free particle, momentum is conserved. Once forces are present, momentum may fail to commute with the Hamiltonian and need no longer be conserved. The general rule is the one the lecture wants us to remember: commuting with H is the quantum criterion for conservation.

9.6 Example I: One Spin in a Magnetic Field

Only now does the lecture move to a concrete system. It is the simplest one available: a single spin in a magnetic field. The Hamiltonian is the energy of the magnetic moment in the field, and in the lecture's convention it is written as

$$H = \frac{\mu}{2}\boldsymbol{\sigma}\cdot\mathbf{B}. \tag{9.43}$$

The factor of 1/2 is treated in the lecture as a convention that may be absorbed into μ; we keep the lecture's convention. Now choose the magnetic field along the third axis, $\mathbf{B} = B\hat{z}$. Then

$$H = \frac{\mu B}{2}\sigma_3. \tag{9.44}$$

The goal is not first to solve for the full state vector, but to see how the average spin components change with time. To stay close to the lecture, we temporarily suppress the expectation-value bars and write $\sigma_1, \sigma_2, \sigma_3$ for the expectation values themselves.

Start with the third component:

$$\dot{\sigma}_3 = \frac{i}{\hbar}[H, \sigma_3]. \tag{9.45}$$

Since H is proportional to σ_3, the commutator vanishes:

$$\dot{\sigma}_3 = 0. \tag{9.46}$$

So the component along the magnetic field is conserved. The lecture immediately turns that into a picture: if the projection on the field direction is fixed, then the spin cannot wander arbitrarily; its tip must move on a cone of fixed opening angle.

Now turn to the transverse components. For the first one,

$$\dot{\sigma}_1 = \frac{i}{\hbar}[H, \sigma_1] = \frac{i\mu B}{2\hbar}[\sigma_3, \sigma_1]. \tag{9.47}$$

The lecture works out the commutator directly from the Pauli matrices and finds

$$[\sigma_3, \sigma_1] = 2i\sigma_2. \tag{9.48}$$

Hence

$$\dot{\sigma}_1 = -\frac{\mu B}{\hbar}\sigma_2. \tag{9.49}$$

Similarly,

$$\dot{\sigma}_2 = \frac{\mu B}{\hbar}\sigma_1. \tag{9.50}$$

If we define

$$\Omega = \frac{\mu B}{\hbar}, \tag{9.51}$$

then the equations become

$$\dot{\sigma}_1 = -\Omega\sigma_2, \qquad \dot{\sigma}_2 = \Omega\sigma_1, \qquad \dot{\sigma}_3 = 0. \tag{9.52}$$

This is simply planar rotation. A convenient solution is

$$\sigma_1(t) = \cos(\Omega t - \theta), \qquad \sigma_2(t) = \sin(\Omega t - \theta), \tag{9.53}$$

with σ_3 constant. The phase θ is not deep; it only tells us where in the orbit the motion starts.

The lecture then returns from the equations to the experimental meaning. If we prepare the spins so that σ_1 is sharp and positive at $t = 0$, then σ_2 and σ_3 are not sharp; each is maximally uncertain. After a quarter period, the expectation values have rotated into the 2-direction, so a measurement of σ_2 becomes definite while σ_1 becomes uncertain. After a half period, the spin points oppositely along the 1-axis. In this way the expectation values precess exactly as a classical magnetic moment would precess in a magnetic field.

A student then raises a useful objection: in practice one often measures averages over many electrons, and might different electrons carry different phases? The lecture answers that this would mean the ensemble had not been prepared identically. If the same experiment is done on each member of the ensemble, they must all begin in the same state. If they do not, we are dealing with a mixed state instead of the coherent pure-state evolution being described here. That clarification belongs to the logic of the example and should not be omitted.

9.7 Example II: Two Coupled Spins and the First Entanglement Dynamics

Entanglement finally enters not as a slogan but as a Hamiltonian problem. A student asks about two entangled electrons. The lecture refuses to answer immediately and quite rightly so: we cannot say what they will do until we know the energy of the system.

So let us take a concrete interaction. There is no external magnetic field; instead each spin provides the field for the other. Then the Hamiltonian is proportional to

$$H \propto \boldsymbol{\sigma} \cdot \boldsymbol{\tau} = \sigma_1\tau_1 + \sigma_2\tau_2 + \sigma_3\tau_3. \tag{9.54}$$

Here $\boldsymbol{\sigma}$ acts on the first spin and $\boldsymbol{\tau}$ on the second. The proportionality constant is not fixed in the lecture, so we keep the energies schematic.

The state space is four-dimensional. A convenient basis adapted to the Hamiltonian is the triplet-singlet basis:

$$|T_+\rangle = |\uparrow\uparrow\rangle, \tag{9.55}$$

$$|T_-\rangle = |\downarrow\downarrow\rangle, \tag{9.56}$$

$$|T_0\rangle = \frac{|\uparrow\downarrow\rangle + |\downarrow\uparrow\rangle}{\sqrt{2}}, \tag{9.57}$$

$$|S\rangle = \frac{|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle}{\sqrt{2}}. \tag{9.58}$$

What matters is that there are two distinct energy eigenvalues, not three. The triplet sector is threefold degenerate, with energy E_T, and the singlet is nondegenerate, with energy E_S.

If we start in any energy eigenstate, nothing very interesting happens: the state only picks up its own phase. So the lecture chooses the more interesting initial state

$$|\uparrow\downarrow\rangle, \tag{9.59}$$

which is not an energy eigenstate. Its decomposition is

$$|\uparrow\downarrow\rangle = \frac{1}{\sqrt{2}}|T_0\rangle + \frac{1}{\sqrt{2}}|S\rangle. \tag{9.60}$$

Time evolution therefore gives

$$|\psi(t)\rangle = \frac{1}{\sqrt{2}}e^{-iE_Tt/\hbar}|T_0\rangle + \frac{1}{\sqrt{2}}e^{-iE_St/\hbar}|S\rangle. \tag{9.61}$$

Now go back to the unentangled basis $|\uparrow\downarrow\rangle, |\downarrow\uparrow\rangle$:

$$|\psi(t)\rangle = \frac{1}{2}\left(e^{-iE_Tt/\hbar} + e^{-iE_St/\hbar}\right)|\uparrow\downarrow\rangle \tag{9.62}$$

$$+ \frac{1}{2}\left(e^{-iE_Tt/\hbar} - e^{-iE_St/\hbar}\right)|\downarrow\uparrow\rangle. \tag{9.63}$$

At this point the lecture isolates the average energy and then the energy difference. Writing

$$\Delta E = E_T - E_S, \tag{9.64}$$

we obtain the cleaned form

$$|\psi(t)\rangle = e^{-i(E_T+E_S)t/(2\hbar)}\left[\cos\left(\frac{\Delta E}{2\hbar}t\right)|\uparrow\downarrow\rangle - i\sin\left(\frac{\Delta E}{2\hbar}t\right)|\downarrow\uparrow\rangle\right]. \tag{9.65}$$

The overall phase in front is physically irrelevant, exactly as the lecture stresses. Therefore the probabilities depend only on the difference of energies:

$$P_{\uparrow\downarrow}(t) = \cos^2\left(\frac{\Delta E}{2\hbar}t\right), \qquad P_{\downarrow\uparrow}(t) = \sin^2\left(\frac{\Delta E}{2\hbar}t\right). \tag{9.66}$$

So the system oscillates back and forth between $|\uparrow\downarrow\rangle$ and $|\downarrow\uparrow\rangle$, with a frequency set by the singlet–triplet splitting. That is the first genuinely entangled dynamics of the lecture.

And now the chapter ends where the lecture ends: with method. The rule is always the same. Start with the basis relevant to the question, rewrite the state in the eigenbasis of the Hamiltonian, let each energy eigenvector evolve with its own phase, and then recombine in the basis in which you want probabilities. This is the ritual to which the lecture steadily reduces the problem.

9.8 Summary

The lecture is called a lecture on entanglement, but its mathematical spine is the reconstruction of time evolution. We begin with the most general linear law

$$|\psi(t)\rangle = U(t)|\psi(0)\rangle, \tag{9.67}$$

impose probability conservation, and thereby derive unitarity. We then shrink time to an infinitesimal interval, identify the Hermitian generator H, and convert the small-step update into the Schrödinger equation. After that we solve first for a single energy eigenvector and then for a general superposition, learning along the way that $E = \hbar\omega$ and that real dynamics comes from relative phases, not overall ones.

Only then do observables enter through

$$\frac{d}{dt}\langle A\rangle = \frac{i}{\hbar}\langle[H, A]\rangle, \tag{9.68}$$

which immediately yields conservation of $\langle H\rangle$ and more generally identifies operators commuting with H as conserved quantities. The examples then cash out the formalism. A single spin precesses around a magnetic field, and two coupled spins oscillate between $|\uparrow\downarrow\rangle$ and $|\downarrow\uparrow\rangle$ because triplet and singlet components accumulate different phases. In that precise sense, entanglement appears here not as an extra postulate but as ordinary quantum time evolution once the Hamiltonian and the right basis have been found.